OECD ECONOMIC SURVEYS

IRELAND

1997

SPECIAL FEATURES

- LABOUR MARKET AND HEALTH CARE

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IRELAND

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF IRELAND

THE LAND

Area (thousand sq. km)	70
Agricultural area, 1995, as per cent of total area	57

Population of major cities, 1996[1] census (thousands):	
Dublin (County and Co. Borough)	1 057
Cork, Co. Borough	127
Limerick, Co. Borough	52

THE PEOPLE

Population (census, 1996)[1]	3 621 000
No. of inhabitants per sq. km	52
Increase in population: annual average 1991-1996	19 063
Natural increase in population: annual average 1991-1996	18 163

Net emigration:	
Annual average 1991-1996	900
Annual average per thousand of population	0.2
Labour force, total, April 1996 (thousands)	1 475
Civilian employment in (thousands):	
Agriculture, forestry and fishing	136
Industry and construction	351
Other sectors	798

THE GOVERNMENT

Public current expenditure on goods and services, 1994 (as per cent of GNP)	20
Current government receipts, 1994 (as per cent of GNP)	46
National debt, 1995 (as per cent of GDP)	82

	Seats
Composition of Parliament (1994):	
Fianna Fail	67
Fine Gael	47
Labour	32
Progessive Democrats	8
Democratic Left	6
Others	6
Last general election: December 1994	

FOREIGN TRADE

Exports:	
Exports of goods and services, as per cent of GNP (1995)	82
Main exports, 1995 (per cent of total):	
Meat and meat preparation	4
Dairy products	4
Beverages	2
Organic chemicals	9
Medical and pharmaceutical products	5
Textile manufacturing, clothing and footwear	3
Machinery and transport equipment	35
of which:	
Office machines	21
Electrical machinery	8
Main customers, 1995 (per cent of total):	
United Kingdom	23
Other European Union	50
United States	8

Imports:	
Imports of goods and services, as per cent of GNP (1995)	60
Main imports, 1995 (per cent of total):	
Petroleum products	3
Chemicals and related products	13
Textile manufacturing, clothing and footwear	5
Machinery and transport equipment	42
of which:	
Office machines	17
Electrical machinery	11
Main suppliers, 1995 (per cent of total):	
United Kingdom	32
Other European Union	24
United States	18

THE CURRENCY

Monetary unit: Irish pound

Currency unit per US dollar, average of daily figures:	
Year 1996	0.63
March 1997	0.64

1. Preliminary.

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Ireland by the Economic and Development Review Committee on 19 March 1997.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 8 April 1997.

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The previous Survey of Ireland was issued in June 1995.

Assessment and recommendations

An outstanding economic performance...

The strong expansion of the Irish economy during the past three years has boosted the real level of activity by almost one-fourth, more than three times the average increase in the rest of Europe. Although incomes have grown somewhat less because of a deterioration in the terms of trade and a fall in EU transfers, private consumption has risen by over 15 per cent during the same period. Rapid output growth has been accompanied by significant gains in employment; indeed, more jobs have been created since 1993 than in the previous thirty years in Ireland. Such developments have been reflected in a significant decline in the unemployment rate, although, at an estimated 11¼ per cent, it still remains above the OECD average. The expansion in economic activity has also been accompanied by a fall in the rate of inflation to less than 2 per cent. At the same time, the government deficit has remained amongst the lowest in the OECD area, helping to reduce the gross public debt ratio by 21 percentage points to 73 per cent of GDP.

... associated with sound macroeconomic management...

Macroeconomic management has played an important role in this performance by enabling Ireland to remain relatively immune to the recessionary forces evident in much of Europe. The devaluation of the Irish pound, which was forced by the market in January 1993, avoided the need to maintain excessively high interest rates to protect the exchange rate, thereby allowing an early easing of monetary conditions. Subsequently, the monetary authorities re-

established the confidence of the markets in the stability of the currency, generating further falls in short-term rates until the end of 1994. Thereafter, the decline in interest rates has not been as large as that in continental Europe and, more recently, the Central Bank has tightened monetary policy somewhat in order to forestall possible inflationary pressures. As for fiscal policy, the budget deficit was already within the Maastricht limit of 3 per cent of GDP when the current upturn started in 1993, and, during the period 1994 to 1996, Ireland has had the second lowest budget deficit (1.4 per cent of GDP on average) in the European Union. By 1996, the estimated structural government deficit had been reduced to 1 per cent of GDP.

... rising labour supply...

The current upswing has clearly benefited from a very favourable *supply-side response.* The interaction of three factors have helped in this respect: the growth in the labour supply, continued inflow of foreign investment and an extended period of social consensus. A rapid increase in the size and quality of the *labour supply* has been the key, helped by the large number of young people with relatively high qualifications entering the labour force. Almost half of school-leavers continue into third-level education in Ireland, which produces, proportionally, the highest number of scientists and engineers in the OECD area. In addition, there has been a marked increase in the female labour force participation rate in line with the rising educational qualifications of women and the decline in the birth rate. Finally, net emigration (to the United Kingdom in particular) has virtually ceased, reflecting increased job opportunities in the Irish labour market.

... sustained foreign investment...

Foreign investment by multinational enterprises has also contributed importantly to the upswing, though precise numbers are not available on its extent. Production in the modern sector of the economy (dominated by enterprises in

the computer, chemical and soft drink industries) has expanded by over two-thirds in the past three years, while the post-tax profits of all foreign companies represented 15 per cent of Irish value-added in 1995. Multinational companies have boosted employment and stimulated the service sector of the economy through their purchases of inputs. The extension of the 10 per cent corporate tax rate from the manufacturing sector to international financial services has attracted a large number of foreign enterprises since the early 1990s.

... and supported by income policy agreements

These positive factors have been reinforced by a high degree of *social consensus*, which is reflected in the income policy agreements – covering wages, taxation and government spending – between the social partners and the government over the past nine years. The most recent agreement, the ''Programme for Competitiveness and Work'', which ended in 1996, has generated moderate private-sector wage increases during a period of rapid growth, thereby maintaining Ireland's competitive position relative to its trading partners. In addition, it has helped reduce industrial unrest. ''Partnership 2000'' – the new accord covering the period to 1999 – is intended to consolidate these developments through continued wage moderation in exchange for tax cuts and public spending increases, while allowing more flexibility in the determination of salaries at the local level.

High growth, albeit at a slower pace, is expected to continue...

Against this background, the economy should continue to expand rapidly over the next two years, though at a somewhat slower pace than in the past three years. The rebound of the export sector in the second half of 1996 is a positive sign in this perspective, as is the recovery that is underway in the computer industry. Private consumption seems likely to remain strong, especially as the scrapping scheme for old cars based on tax incentives is to be prolonged until the end

of 1997. Moreover, the boom in the housing sector may encourage some anticipatory increase in residential property investment. Finally, the recent falls in long-term interest rates may stimulate business investment.

... but any further pickup in activity could generate inflation pressure

Further acceleration in growth from that projected for the next two years might pose problems for the economy as there does not appear that much capacity is left to sustain the expansion at such a rapid pace. Indeed, according to Secretariat estimates, the economy is currently running at, or slightly above, its potential. Although this concept of potential is of less relevance in Ireland than in other countries given possible increases in the labour force and the open nature of the economy, some strains are already emerging in the residential real estate market, with the price of second-hand homes having increased by 18 per cent in the year to September 1996. These pressures might spill over into wages in the construction sector and then into the rest of the economy. Such factors give rise to concern about possible inflationary pressures, the more so since the recent increase in monetary and credit aggregates was more than double the growth of nominal income. For traded goods, the key factor will be the movement in the exchange rate. So far, despite the appreciation of sterling, the effective exchange rate of the Irish currency has risen strongly; a reversal of this development could push up import prices.

Therefore, a cautious monetary policy is needed...

In such circumstances, monetary authorities should be prepared to err on the side of caution and to tighten further before markets become concerned about possible inflationary tensions. Such a tightening would tend to dampen the demand for credit and, in turn, lessen the pressures in the real estate market. However, given ongoing significant capital inflows and the strength of the pound, which is currently well above its central rate within the European Exchange Rate Mechanism, the scope for raising interest

rates might be constrained by the need to maintain a stable exchange rate in the run-up to EMU. Consequently, the authorities may be faced, for a time, with the dilemma of either giving priority to inflation control, which might generate additional upward pressure on the pound, or focusing on the exchange rate but with the risk of tending to undermine price stability.

... as well as a tighter fiscal stance...

In view of this potential dilemma, fiscal policy should support monetary policy in keeping inflation low. The 1997 budget envisages a modest increase in the general government deficit to 1½ per cent of GDP. This reflects the fact that, despite rising public expenditure, close to half of the tax cuts promised by the Partnership 2000 programme over the three years to 1999 (see above) is included in the budget. Such tax reductions were the key to securing the new programme of wage moderation. However, they could tend to stimulate domestic demand, which, at a time the economy appears to be approaching capacity limits, might contribute to short-term inflationary pressures. To help prevent such an outcome, fiscal policy in 1998 and 1999 should aim for more demanding targets than those of keeping the budget deficit at the same level as in 1997, as envisaged in the government's current three-year fiscal plan. Moreover, reducing the deficit would help accelerate the fall in the still relatively high burden of public debt and increase the fiscal room for manoeuvre in the context of the strict budget limits required in the EMU from 1999.

... based on expenditure restraint

The key to achieving this result will be to restrain the growth of public expenditure. For the past two years, the government has overshot its target of limiting the increase in spending to 2 per cent above the rate of inflation. This objective has been extended to 1998 and 1999, despite continuing pressures on public-sector wages. Keeping spending on target would require that the contingency

reserve, included in the three-year plan and amounting to almost ¾ per cent of GDP, be used to cushion any revenue shortfalls in the event of an economic slowdown rather than to finance additional expenditure. While allowing further inroads into the budget deficit, this approach would also create room for further lowering average tax rates.

This would allow tax reductions...

Continuing to reduce taxation in Ireland is important for several reasons. *First,* in order to meet the tax promises of the Partnership 2000 programme, the government will have to make cuts in the 1998 and 1999 budgets. *Second,* corporate tax rates in the service sector should be lowered towards that currently paid in manufacturing industries. *Finally,* and as importantly, the high marginal tax rates and welfare benefit withdrawal rates faced by many low-paid workers would need to be reduced. Such changes to benefit the low paid would be essential to generate further improvements in the labour market situation.

... in order to improve work incentives

High tax rates have indeed tended to reduce work incentives, particularly for the low-skilled, which has contributed to maintaining the unemployment rate at a relatively high level. This is especially the case for the long-term unemployed who generally suffer from low levels of education; indeed, about three-fourths of those out of work for more than one year failed to complete upper secondary school. Given the high marginal tax rate on low incomes, the potential earnings of such persons are low relative to the permanent flat-rate unemployment support. Consequently, it is unlikely that poorly-educated unemployed persons can significantly boost their disposable incomes by working.

While the government has started to address this issue,...

Following a study of this problem, the government has adjusted the tax and benefit systems to boost the labour supply. Most notably, it has lowered average tax rates, reduced or eliminated social security contributions on low-

income workers and significantly improved the Family Income Supplement for working parents. At the same time, there has been an effort to broaden the secondary school curriculum in order to reduce the number of students who leave the educational system without qualifications. The government also operates or funds various training programmes to improve labour force skills. In total, more than 30 active labour market programmes have been introduced, with related government spending substantially above the OECD average. These programmes include a large public job scheme in the voluntary social sector, supplemented by training measures and several subsidies to encourage private-sector hiring of the long-term unemployed.

... significant changes are still required...

While many of these recent initiatives have been in the right direction, much more remains to be done. Indeed, a further overhaul and integration of the tax and benefit system appears necessary to address the causes of Ireland's persistent unemployment problem. One key would be to reduce the number of taxpayers on the marginal relief system in which a 47 per cent effective marginal tax rate applies to low-income workers, as, even after the 1997 budget, this rate is still paid by one-eighth of all taxpayers. A step in this direction would be to increase the personal income tax allowance over time towards the level of the income tax exemption limit, a move which would lower the marginal rate to 33 per cent for those previously affected by the marginal relief system.

... including changes in the benefit system...

On the benefit side, further reducing the bias of the current child allowance system in favour of the unemployed would be beneficial. Reducing the disincentive effect of housing support, including the withdrawal of rent and mortgage assistance, should also be addressed. Moreover, unemployment benefits should be made more clearly conditional on efforts by the unemployed to search for work and to accept

job proposals. The long-term unemployed, in particular, should be required to participate in programmes that offer counselling in job-search techniques, monitor job-seeking activities and verify eligibility for benefits. Enforcing such conditions would require closer co-operation between the Department of Social Welfare, which is responsible for the payment of unemployment benefits, and the National Training Agency (FAS), which provides services for the unemployed, such as training programmes, job referral and counselling. A greater integration of these two functions, as seen in other OECD countries, might help in this regard. A final step would be to reduce benefits as the spell of unemployment increases.

... and a rationalisation of public spending in labour market programmes

The government should also rationalise and simplify its spending on labour market programmes. The schemes currently in operation would need to be carefully evaluated to ensure that resources are concentrated on those programmes that have proved to be the most effective. For example, carefully targeted employment subsidies appear best placed to reduce the number of long-term unemployed. Finally, it is essential to reduce the number of students who leave school early. The changes introduced to broaden the secondary school curriculum need to be pursued and revised as necessary. In addition, resources should be further focused on schools in disadvantaged areas which have large numbers of early school-leavers.

Health expenditure, too, could be made more efficient...

Another category of public expenditure where there is some scope for increased efficiency is *health care,* the subject of the special chapter in this Survey. As a result of rapid economic growth and a major downsizing of the hospital network during the 1980s, the share of GNP devoted to public health care in Ireland fell to a low of 6.8 per cent in 1996, without jeopardising the high quality of the service provided. Public health costs have also been contained by

the fact that a significant proportion of the Irish population is insured privately for hospital care, with the government funding first-level medical care only for the poorest people. This has limited the government's exposure to demand-led growth for medical care. Although per capita spending on health is still markedly lower than in other European countries, the size of government expenditure in this field points to the need for better controls in a number of areas, given official targets to limit overall spending growth.

... by improving budgetary management of hospitals...

Improving the management of hospital budgets is one key area. At present, government control over hospitals seems unduly bureaucratic, with little devolution of budgetary authority and accountability to medical staffs. Hospital costs also seem to have been held down by extensive reliance on junior doctors. Finally, there has been some concern about managing the complicated interface between public and private provision of medical care. Despite several government initiatives to resolve these problems, further action is still required. In particular, more decentralisation of budgetary control is needed from the centre to the regional and local levels in order to improve efficiency. Hospital budgets should also be more closely related to standard costs of provision of health care. This is being achieved by increasing the weight given to the case mix of a given hospital when determining its budget. However, while it is reasonable to introduce this progressively to avoid major budget cuts for certain hospitals in the short term, the process should be speeded-up.

... and clarifying the split between public and private care

In the same spirit, other initiatives could be envisaged. For instance, within hospitals, value for money could be improved by making clinical units into cost centres with senior medical staff responsible for management. At the same time, resource allocation in state-funded hospitals could be improved by reforming the contracts of hospital

doctors in a way that defines the expected commitment to both the public and private sectors. Resource allocation would also be improved by putting charges for the use of public hospital facilities on a more economic basis. In addition, there may be a case for concentrating the training of doctors into a shorter period and improving their career structure in order to retain experienced staff and improve the provision of health care. Finally, the impact on health provision of the new competitors to the government-owned health insurance monopolist should be carefully monitored, so as to avoid high risk groups being pushed back into the public sector with adverse consequences on health expenditure. Such changes should enable the Irish health system to continue to deliver high quality care at a time of increasing budgetary stringency.

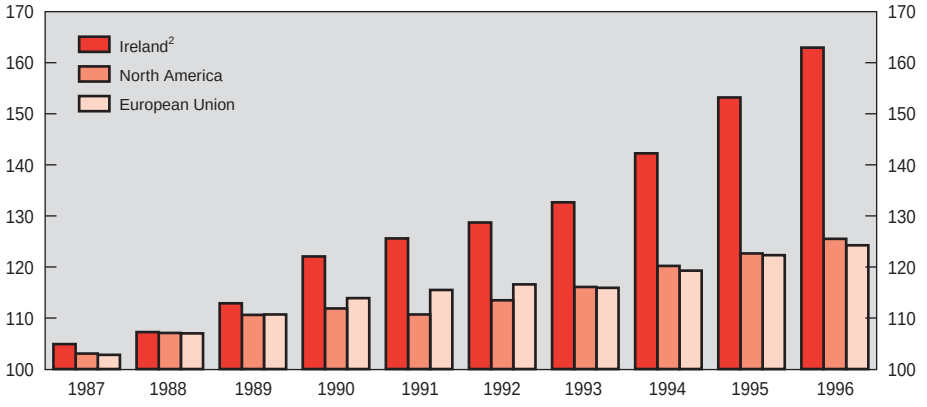
Summary

In sum, the Irish economy has performed well over the past three years. This favourable outcome has been based on sound fiscal and monetary management, supported by an effective incomes policy which allowed continued wage moderation. However, the economy could not have kept growing so rapidly without favourable supply-side factors. The key in this area has been the substantial inflow of direct investment by foreign companies, drawn by factors such as a young, well-qualified labour force and a favourable tax regime. As a result, there has been a substantial rise in employment, with the unemployment rate falling by 4 percentage points since 1992. However, unemployment remains above the OECD average. Reducing it to a more normal level would require a further overhaul of the tax and social welfare system which, together with a stricter enforcement of eligibility rules and job counselling, would give the long-term unemployed more incentive to work, thus increasing labour market flexibility. In conjunction with more ambitious fiscal consolidation, this would strengthen Ireland's economic performance and further prepare the ground for successful participation in the EMU.

I. Recent trends and prospects

Economic growth in Ireland has been the second highest in the OECD area during the past three years. The 22 per cent increase in GNP was more than three times faster than that seen in both the European Union and North America, and on par with countries in the Pacific region of the OECD such as Korea and New Zealand (Figure 1). This upswing has generated fast job creation that has significantly reduced unemployment. Growth peaked in 1995, when the rapid expansion of the computer and semiconductor industries worldwide helped boost GDP by over 10 per cent and GNP by 7 per cent (Table 1). In 1996, with the surge in the high-tech industries slowing, GNP growth is estimated to have

Figure 1. **OUTPUT GROWTH IN AN INTERNATIONAL PERSPECTIVE**[1]
1986 = 100



1. At 1990 prices and exchange rate.
2. GNP.
Source: OECD Secretariat.

Table 1. **Demand and output**

	1992	1993	1994	1995	1996 estimates
	Per cent of GDP	Percentage change on previous year			
Private consumption	60.2	1.7	6.8	3.7	6.0
Public consumption	14.8	1.2	3.4	2.0	1.9
Gross fixed investment	15.2	–2.1	8.7	10.1	10.2
Building and construction	9.5	–5.9	11.7	12.9	10.5
Machinery and equipment	5.7	4.1	4.1	5.5	9.7
Final domestic demand	90.2	0.9	6.6	4.5	6.1
Stockbuilding	–0.3	–0.2	–0.5	1.0	0.3
Total domestic demand	90.0	0.7	6.0	5.7	6.4
Exports of goods and services	66.9	9.7	13.6	17.0	10.6
Imports of goods and services	56.8	7.2	14.3	12.3	10.3
Foreign balance	10.0	2.4	1.2	5.1	1.9
Gross Domestic Product	100.0	3.1	6.5	10.1	7.2
Net factor income	–11.4	3.1	1.0	32.2	13.7
Gross National Product	88.6	3.1	7.2	7.5	6.3
Gross National Disposable Income		4.2	4.4	5.0	4.2
Memorandum item					
Output gap	–1.0	–3.8	–3.1	0.0	0.3
Household saving rate	8.3	13.8	10.6	11.7	10.0
Output by sector					
Agriculture	9.9	–8.1	–1.4	4.3	
Chemicals, computers, electrical engineering	14.4	5.6	19.8	19.3	
Other industries	23.2	1.3	2.9	12.4	
Distribution	17.3	8.3	1.8	7.4	
Other services	35.2	3.3	5.6	4.6	
GDP (factor cost)	100.0	2.9	5.8	9.3	

Source: Central Statistics Office, OECD, estimates.

slowed slightly to 6½ per cent. By then, nearly all the economy-wide spare capacity, which had been as high as 3 per cent in 1994, had been eliminated. The upswing has not resulted in any generalised pressure on resources: the rate of inflation has remained low, with some measures even falling slightly, while wage increases have remained under control.

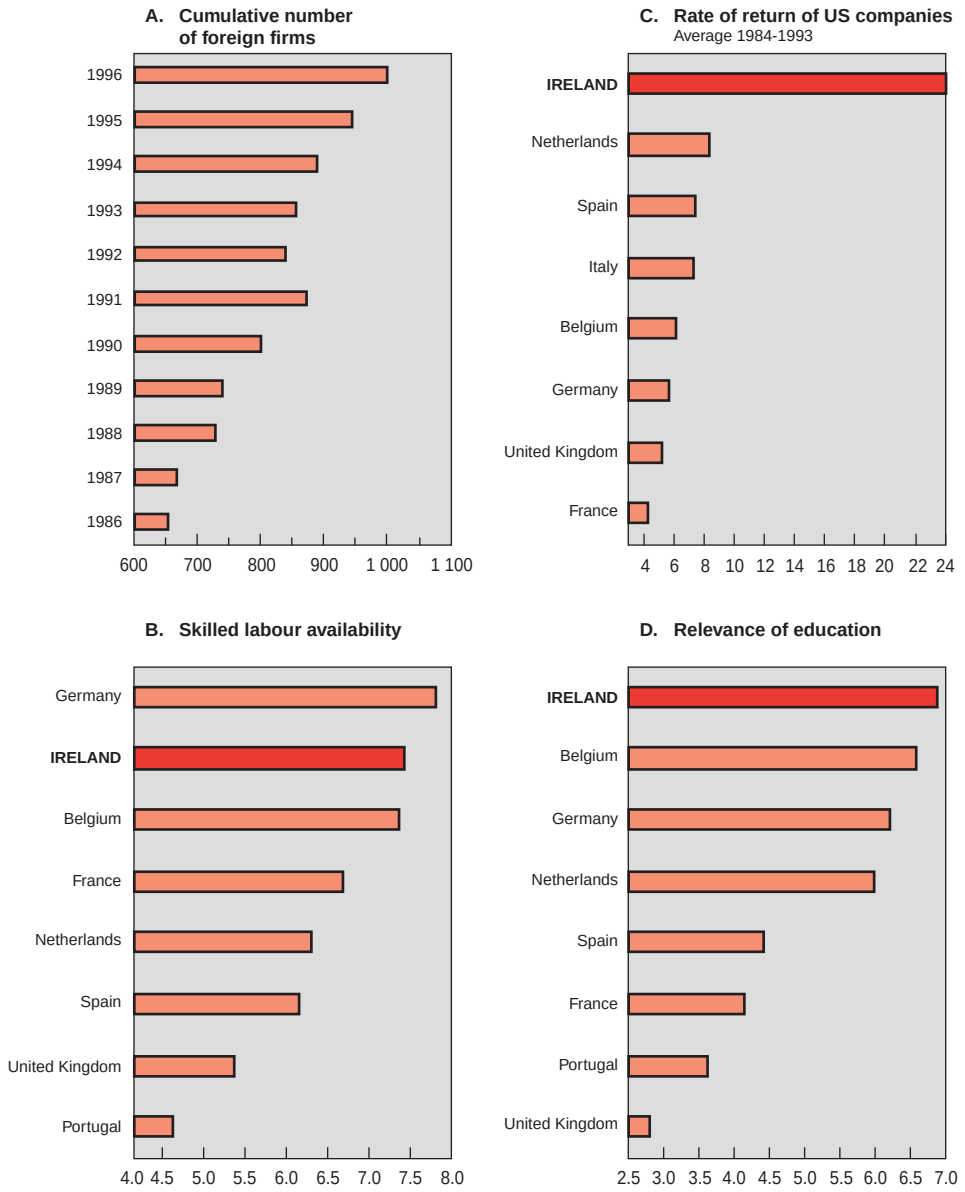
Growth sustained by favourable supply-side factors...

Although macroeconomic policies did help to ensure a recovery from the 1993 recession, it is estimated that only one-fifth of the expansion in activity represents the elimination of the cyclical slack. Instead, it is primarily the result of the growing supply potential of the economy, which has markedly improved during the 1990s. Three favourable elements, in particular, are associated with this development: *foreign direct investment* by multinational companies; *an expanding well-qualified labour force;* and *a social consensus on pay policy* that has left management free to concentrate on improving efficiency.

The number of foreign firms investing in Ireland has risen by over 50 per cent during the last ten years, led by American companies (Figure 2, Panel A). According to US data, the total stock of investment by American firms in the Irish manufacturing sector between 1992 and 1995 grew by 45 per cent to $6.9 billion (12 per cent of GNP). Manufacturing investment by US firms averaged 1½ per cent of Irish GNP annually during that period and appears to have accounted for about half of all investment in the Irish manufacturing industry. Moreover, nearly half of the foreign enterprises investing in Ireland were American in 1996, a year when a record number of new jobs were created by incoming multinational companies (Table 2). Most foreign investment has been concentrated in the ''modern sector'' of computers, semiconductors, office equipment, software, pharmaceuticals, electrical engineering and soft drink concentrates. Key parts of this sector expanded by more than 40 per cent in 1994 and 1995, bringing its overall share of net manufacturing output towards 60 per cent (Table 1). The pace of growth, though, slowed in 1996, as there was some destocking in the computer and semiconductor industries.

Government policy has played a key role in attracting foreign investment. A major factor has been the low 10 per cent corporate tax rate, which applies to the entire manufacturing sector until end-2010. In 1990, this low rate was extended to companies operating in the International Financial Services Centre and the Shannon Zone until end-2005. In addition, the standard rate for employers' social security taxes was only 12.2 per cent in 1995 and was reduced in the 1996 budget to only 12 per cent. Employers pay an even lower rate (8.5 per cent) for workers earning less than the average wage. Finally, the Industrial Development Agency

Figure 2. **FOREIGN INVESTMENT: DETERMINING FACTORS**



Source: IDA, US Department of Commerce, *IMD World Competitiveness Report,* 1996.

Table 2. **Major foreign investments in Ireland**

1996

Date	Company	Sector	Jobs created
January	Various companies		550
February	Eastman Kodak	Compact discs	250
April	Digital Equipment	Technical support	220
	Alps Electric	Car parts	250
May	Oracle	Telemarketing	400
	Gateway	Computers	1 200
	EPC	Car parts	300
	24 minor companies	Various	2 300
June	IBM	Customer centre	750
	Copeland	Compressors	300
July	Hewlett Packard	Printer cartridges	830
October	3Com and Stream	Computers	1 000
November	Dell	Computers	750
	DSC	Telecommunications	470
	Fore systems	Electronics	210
December	IBM	Service centre	3 000
	Others		1 000
Total employment creation			13 780
(Per cent labour force)			0.9

	Number	Per cent of total
Geographic origin of investors		
United States	440	45
Germany	175	18
United Kingdom	160	16
Other EU	200	21
Total	975	100

Source: Irish Development Authority.

offers a series of grants to new companies to pay for part of the costs of installation and initial training for the new labour force.

Another favourable element was the establishment of the International Financial Services Centre in 1987 to act as a base for international banking, insurance, leasing and fund management. This centre has expanded rapidly as more than 400 international banks and securities companies have set up offshore operations in Dublin, mainly dealing with administrative functions. The number of funds based in Dublin grew by 18 per cent in 1996 and their assets under management rose 65 per cent. Indeed, 40 per cent of the projects reported by the

Industrial Development Agency in 1996 were in the International Financial Services Centre. As a result, total employment in this sector rose by 34 per cent in 1996 to nearly 3 500. In addition, many companies have established service centres in Ireland to cover all world markets, a strategy encouraged by increased competition in the telecommunications market (see Chapter III). More than 3 000 people (over ¼ per cent of the labour force) now work in this area.

While the tax regime has been an important factor in drawing foreign companies to Ireland, the availability of low-cost, well-qualified labour, stemming from the government's education and training policies, has also been essential. In 1996, an international survey of businessmen reported that Ireland ranked second in Europe in the availability of skilled labour (Figure 2, Panel B). Moreover, they regarded the Irish educational system as the most relevant to the needs of business (Figure 2, Panel D). These last two factors often appear to give the Irish economy an edge over other parts of Europe.

As discussed in detail in the 1995 Survey, new entrants to the labour force have a high level of education. In particular, the proportion of school-leavers who have completed upper secondary education is the highest in the OECD area. In addition, the graduation rate from third-level education was the ninth highest in the OECD area in 1994 and has probably risen since then, given the considerable expansion of higher education during the past few years. Moreover, the new graduates have skills that are relevant for the incoming firms. The graduation rate in science, engineering and mathematics was the fifth highest in the OECD. Indeed, the share of young science graduates in the labour force, with qualifications from either universities or technical colleges, is now slightly higher than in Japan and Korea. By contrast, those who are retiring from the Irish labour force have a relatively low level of qualifications – a situation that will continue for another two decades as free universal secondary education was introduced only in 1968. As a result, the rise in the stock of human capital in Ireland over the next 25 years is projected to be the second most rapid in the OECD area.

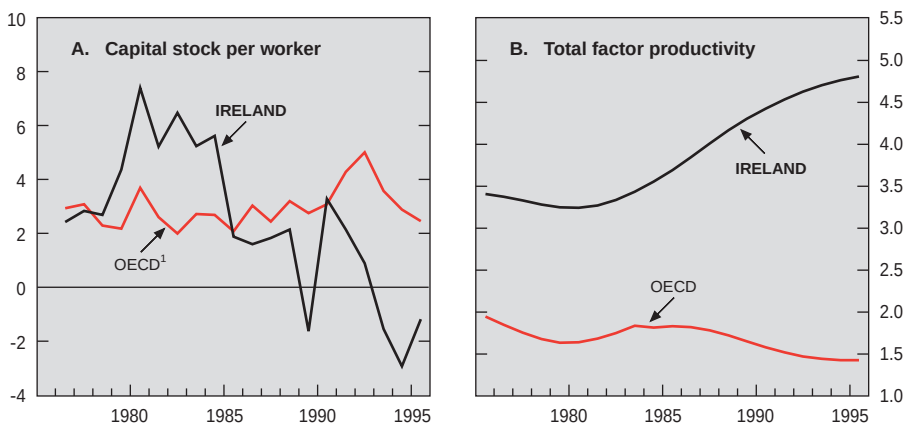
Overall, the combination of various factors such as an inexpensive, well-qualified workforce, location, language and a favourable corporate taxation regime, has made manufacturing investment profitable. In 1995, for example, US companies earned a post-tax rate of return of 23 per cent on capital employed in Ireland, up to five times higher than that of US companies elsewhere in Europe (Figure 2, Panel C).

Structural shifts resulting from the inflow of foreign capital have pushed up the growth of labour productivity. Three factors have been at work. First, labour productivity in the modern sector grew over 10 per cent per year in 1994 and 1995, on average, compared with only 3 per cent in traditional areas, where some industries, such as clothing, experienced falls. Second, the rising share of employment in the modern sector, which had a productivity level 2¾ times higher than in traditional areas in 1993, has raised the average level of productivity. Finally, overall productivity in the economy is being boosted by the ongoing decline in the number of people employed in agriculture, where output per worker was barely half that found in the manufacturing sector. The decrease in this sector accelerated in 1996.

Foreign direct investment has increased labour productivity by bringing new technology and management skills to Ireland. However, its effect on the size of the total capital stock has been relatively small. Indeed, total capital per worker in the business sector has been falling, dropping by 1¾ per cent per year between 1993 and 1996, according to Secretariat estimates (Figure 3, Panel A).

Figure 3. **CAPITAL INTENSITY AND TOTAL FACTOR PRODUCTIVITY GROWTH**

Per cent change per year



1. Countries for which data are available.
Source: OECD Secretariat.

Box 1. **The measurement of national income**

The concentration of economic activity in multinational high-tech companies and the relatively high level of external debt means that GDP, the normal indicator of output, overstates both the level and growth of Irish incomes. The level of GNP, a more appropriate measure of income, was over 12 per cent less than GDP in 1995, while its rate of increase was about ½ percentage point per year less than that of GDP in the five years ending 1995 (Table 3). Another factor working against the growth of real output has been a systematic fall in the terms of trade. The decline reflects high productivity growth in multinational companies, which are increasingly centred on computers and electronics, resulting in price reductions for their output. As nearly all of their production is shipped abroad, export prices tend to increase less rapidly than import prices. In 1995 and 1996, this gap was just over 1¾ percentage points per year, in line with the average of the previous five years. Given that exports of goods and services (at constant prices) represent over 90 per cent of GNP, this terms-of-trade loss has reduced the growth of real national disposable income to about 1 percentage point below that of GNP. Finally, the decline in transfers from the EU has further slowed the increase in real national disposable income by a further ½ percentage point. Overall, the gap between the growth of GDP and that of real national disposable income (adjusted for the movements in the terms of trade) has been 2 percentage points during the 1990s. Nevertheless, this measure of real national income increased by as much as 4½ per cent annually in 1995 and 1996.

Table 3. **GNP and real national disposable income**

	1991	1992	1993	1994	1995	estimate 1996
	Per cent change					
GDP	2.1	4.0	3.0	6.6	10.1	7.2
Profit adjustment[1]	0.4	–1.7	–0.2	–0.2	–3.4	–1.7
GNP	2.8	2.5	3.1	7.2	7.5	6.3
Terms of trade adjustment[1]	–1.7	–0.9	1.2	–2.0	–2.4	–1.8
Foreign transfers adjustment[1]	0.7	–1.4	0.0	–0.7	–0.3	–0.1
Real national disposable income	1.7	0.3	4.3	4.3	4.8	4.5

1. Contribution to growth, *i.e.* change in the adjustment factor expressed as a percentage of the previous year's GDP or GNP.

Source: Central Statistics Office and OECD Secretariat estimates.

This decline reflects the fact that the share of business sector investment in total output was the lowest in the OECD area during that period. In contrast, capital-per-worker in Ireland was rising more rapidly than the OECD average during the 1980s despite an over-supply of labour, perhaps due to the investment incentives then available.

With both capital and labour productivity rising, the trend growth of total factor productivity in the whole economy has progressively accelerated. By 1996, it is estimated to have reached an annual rate of nearly 5 per cent, more than three times the rate seen elsewhere in the OECD area (Figure 3, Panel B). The increase in productivity accounted for more than 85 per cent of the estimated growth of potential output in Ireland – much higher than the OECD average of 55 per cent.

The acceleration in the rise of total factor productivity during this cycle has been accompanied by rapid demographic growth, the absence of net emigration and the increase in the participation rate (Figure 3 and below). As a result of these factors, the potential growth rate of output is estimated to have risen to 6½ per cent on a GDP basis in 1996, up from 4 per cent at the end of the 1980s, which is equivalent to GNP growth of about 5¾ per cent (Box 1). Despite this acceleration, the strength of the upswing means that the economy is now operating close to capacity. However, given the possibility that a small proportion of the 2 million Irish living in the United Kingdom and 1 million in the United States might return to their country of origin, there is a considerable degree of uncertainty about the level of potential output.

... as well as by buoyant high-tech exports...

Overall, exports have been the leading element of demand in the current upswing (Table 4). In 1995, the value of sales abroad by the modern sector rose by 31 per cent and its share of total exports increased to 62 per cent, with computer hardware and software alone accounting for 29 per cent. Moreover, the slight slackening in the growth of such exports in 1996 was responsible for most of the slowdown in GNP growth to 6¼ per cent from 7¼ per cent the previous year. The weakness in exports, which was concentrated in the first half of 1996, was reversed by the end of the year.

Table 4. **Exports and imports of goods**

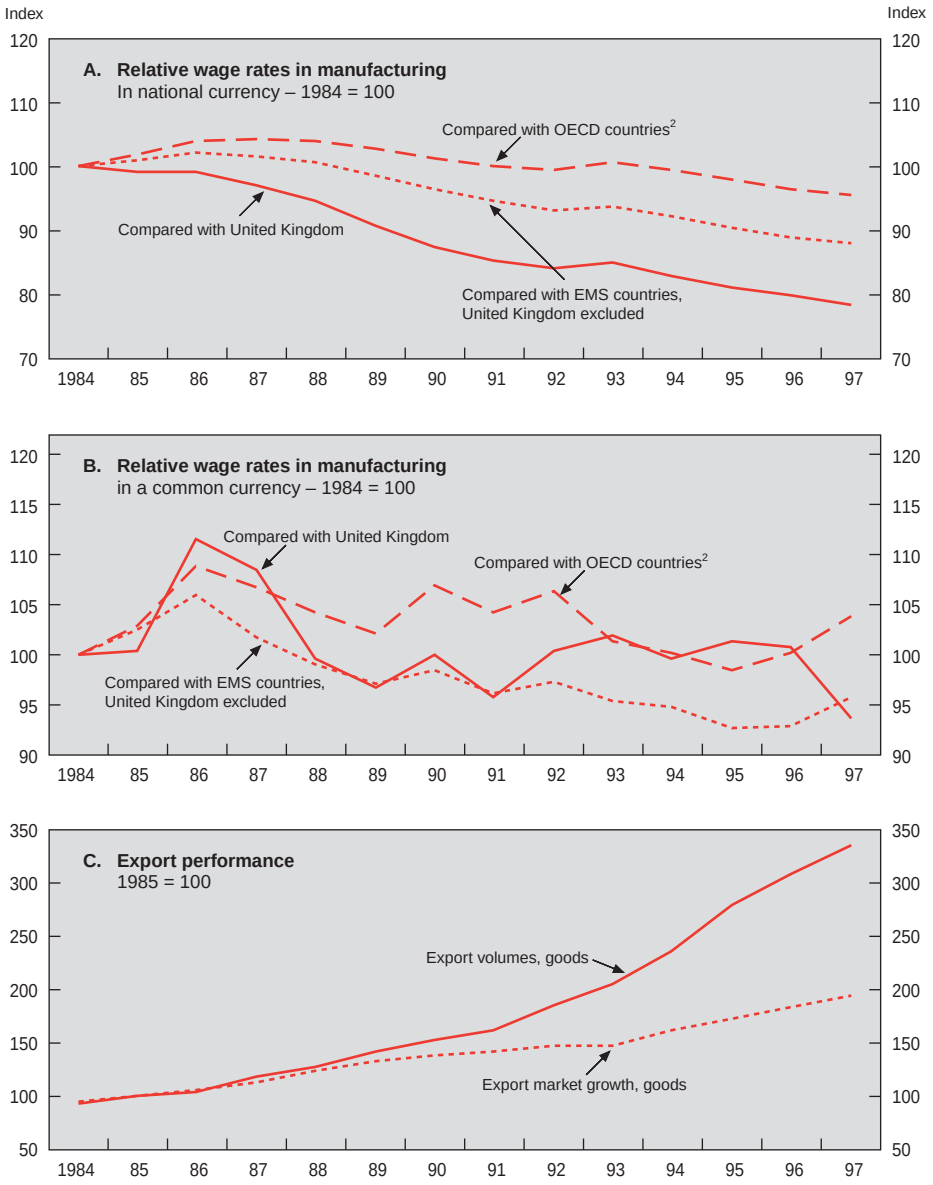
	1991	1992	1993[1]	1994	1995	1996 1st half
	Percentage change from previous year					
Volumes						
Total exports	5.6	14.7	10.2	14.8	19.8	11.4
Agricultural	6.6	28.3	–12.3	3.5	–5.0	–2.0
Industrial	5.8	11.4	13.7	17.1	19.5	13.1
Total imports	0.8	4.8	7.0	13.2	13.6	10.9
Capital goods	–9.9	–3.3	30.9	48.2	27.2	12.0
Consumer goods	4.4	4.1	–3.4	–6.5	3.0	10.5
Raw materials	1.5	7.1	5.1	5.3	17.5	11.1
Prices						
Exports	–0.7	–2.8	7.6	–0.1	1.8	0.4
Imports	2.3	–2.1	5.1	2.6	4.3	1.0
Terms of trade	–2.9	–0.7	2.4	–2.6	–2.4	–0.6
	Per cent					
Share of total export values						
Agricultural	13.6	15.6	12.8	11.5	10.5	8.1
Industrial	83.8	82.2	79.9	84.6	85.9	88.1
United Kingdom destination	32.0	31.5	28.5	28.1	25.4	

1. Sectoral level comparisons between 1992 and 1993 should be treated with particular caution because the introduction of Intrastat in 1993 led to large unclassified (by commodity) balances for both import and export statistics. The 1993 volume growth rates in this table reflect the allocation of these balances to sectors on a pro rata basis (pro rata to the shares of those sectors in total imports and exports).

Source: Central Statistics Office, ESRI, OECD.

In contrast to the modern sector, the growth of exports in the traditional industrial sector has been more modest and has decelerated from 17 per cent in value terms in 1994 to 9 per cent in 1995 and slowed further in 1996. The slowdown reflects changes in export market growth and competitiveness, which are more important as determinants of foreign sales in this sector. More importantly, the favourable impact on international competitiveness of the 1993 devaluation was reversed by the effect of a slow, but steady, appreciation during 1994 and 1995, which accelerated in 1996. As a result, competitiveness began to deteriorate from 1995 onwards despite slow wage growth relative to competitors (Figure 4, Panels A and B). Although this had a negative effect on the traditional sector, the increase in total Irish exports continued to outpace the growth of foreign markets (Figure 4, Panel C).

Figure 4. **EXPORT COMPETITIVENESS AND EXPORT MARKET SHARE**[1]



1. Relative to trade-weighted average of trading partners.
2. Mexico and Turkey excluded.

Source: OECD.

The movement of agricultural exports has been erratic during the upswing. In 1995, exports rose 13 per cent in value terms, reflecting the closing of the gap between EU and world market prices, which allowed a marked reduction in the level of intervention stocks. In 1996, however, agricultural prices fell markedly, partly as a result of a 14 per cent decline in the price of beef (33 per cent of gross agricultural output) due to the BSE beef crisis. In addition, dairy prices were also weak. As a result, the value of agricultural exports dropped some 17 per cent in the first nine months of 1996. Although the weakness of beef prices cut production values by about Ir£ 175 million in 1996 (0.5 per cent of GNP), the effect on farmers' incomes was partially offset by an Ir£ 70 million rise in EU support payments. Additional support of around Ir£ 50 million has been agreed for 1997.

The growth of exports and production in the modern sector feeds directly into the domestic economy, as well as through the links generated by the consumption of the employees in that sector. As noted in previous OECD Surveys, multinational companies' purchases of Irish inputs are only a small proportion of their total sales. However, their purchases of domestic services per employee are actually higher than those of indigenous firms. As a result, the number of jobs in the service sector indirectly linked to those in the modern manufacturing sector was estimated to be 105 per cent of direct employees in that area, compared to only 80 in traditional manufacturing industries. Furthermore, in the modern sector, this ratio has risen steadily from 93 in 1983, whereas the linkage between traditional manufacturing and the service sector has fallen or, at best, remained stable. As employment in foreign-owned firms has increased rapidly in the current upturn, this finding suggests that domestic service sector activity has been boosted considerably by the growth of high-tech exports.

... and rapid growth of domestic demand

In contrast to the fluctuations in the growth of exports, the expansion of total domestic demand has been stable during the upswing, increasing at close to 6 per cent each year (Table 1). However, private sector demand has tended to accelerate while government demand has slackened.

The rapid growth of private incomes has been driven essentially by the strong expansion in employment. The total number of employees rose by 12 per cent between 1993 and 1996 while the growth in real average earnings was less than 1 per cent, leading to a 15 per cent rise in real compensation of employees (Table 5). Real transfer incomes grew somewhat slower at 10 per cent, mainly reflecting an increased number of beneficiaries, as the real value of individual benefits rose only slightly. Overall, real personal incomes expanded by 12 per cent. With the average household tax rate (including social security contributions) remaining constant, real disposable incomes have risen by a similar amount. The increase was slower at the beginning of the upturn and peaked in 1995 when the government paid the first instalment of the court-ordered equal

Table 5. **Prices, wages and profits**

	1991	1992	1993	1994	1995	1996 estimates
	Percentage change from previous period					
Consumer prices						
All items	3.2	3.1	1.4	2.4	2.5	1.6
Except mortgage interest	3.2	2.8	2.0	3.0	2.3	1.9
Food	1.4	1.7	0.4	3.4	3.1	1.9
Energy	1.5	–1.6	0.2	0.7	0.6	3.7
Wholesale prices						
All items	1.2	0.8	4.8	1.0	2.1	0.8
Manufacturing	0.8	1.7	4.6	1.1	2.5	0.7
Agriculture	–3.6	1.5	6.5	1.5	2.3	–3.4
Price deflators						
GDP	1.8	2.2	4.2	1.2	0.5	0.7
Total domestic demand	3.5	3.2	3.1	2.8	2.2	2.1
Exports of goods and services	–0.3	–2.0	6.2	0.7	1.8	0.1
Imports of goods and services	2.3	–1.2	4.3	2.7	3.9	1.5
Hourly wages	5.6	4.6	5.8	1.8	2.4	3.5
Wages and profits						
Compensation of employees	6.6	6.9	8.3	6.5	6.4	..
Agricultural profits	–8.5	16.8	2.9	5.6	8.0	..
Non-agricultural profits	3.1	0.5	14.2	4.8	18.5	..
Memorandum items						
Real wages	2.3	1.4	3.8	–0.6	–0.3	1.6
Terms of trade	–2.5	–0.8	1.9	–2.0	–2.1	–1.4

Notes: Availability of 1996 data as follows: consumer prices: December; wholesale prices: all items until October; output prices: manufacturing until December; output prices: agriculture until December; hourly wages: first quarter.
Source: Central Statistics Office, *Economic Series*, OECD.

treatment payments to women, which totalled Ir£ 190 million (0.9 per cent of disposable incomes). The final payment, of Ir£ 60 million, was made in 1996.

Private consumption has increased somewhat faster than personal disposable incomes during this upswing. Consumers became markedly more confident at the beginning of the expansion as interest rates fell. However, the pace of consumption slackened in 1995 despite rapid income gains. The most rapidly growing component of consumer spending has been foreign travel, which rose by over 56 per cent between 1993 and 1996. Recent indicators suggest a pickup in the growth of private consumption in 1996, though the increase in personal incomes may have slackened, thereby resulting in a fall in the saving ratio. The rise in consumption is partly due to a surge in car sales (Table 6), which have been

Table 6. **Recent economic indicators**

Per cent change on previous year

		1993	1994	1995	1996 to date	1996			
						Q1	Q2	Q3	Q4
Manufacturing									
Output		5.4	12.8	20.1	8.7	14.4	10.9	2.8	5.1
Employment	Number	0.1	2.7	6.1	5.1	5.4	4.8	5.1	
Construction									
Housing permits	Number	12.8	7.1	12.1	10.6	12.5	8.8		
Dwellings completed	Number	–4.8	25.6	13.8	12.7	14.6	10.8		
Cement imports	Volume	–47.5	25.5	26.5	75.1	74.0	76.2		
Employment	Number	–6.0	2.5	4.5	8.2	6.8	8.9	8.2	8.8
Investment									
Industrial building	Value	–17.8	14.3	32.1	25.7	2.6	48.9		
Commercial building	Value	–3.1	12.7	50.8	22.3	20.1	24.5		
Capital goods imports	Value	27.4	27.9	29.5	24.3	14.7	33.9		
Vehicle registrations	Number	–15.9	24.3	8.3	20.7	18.2	29.9	23.8	21.4
Consumption									
Retail sales (non-garage)	Volume	2.2	3.5	2.9	4.7	4.5	4.8	5.2	3.3
Car licensings [1]	Number	2.2	33.5	6.8	23.5	28.0	28.0	13.3	20.5
Retail sales (total)	Volume	1.4	5.5	2.8	6.2	6.7	7.7	4.6	
Government spending	Value	8.6	6.4	7.9	7.9	14.0	3.6	6.1	
Foreign trade									
Exports	Volume	10.2	14.8	20.1	11.4	12.5	10.4		
Imports	Volume	7.0	13.2	14.4	10.8	10.6	11.1		
Net exports	Volume	20.0	19.5	36.4	12.7	16.7	8.8		

1. New cars and newly registered second-hand cars.
Source: Central Statistics Office, *Economic Series.*

boosted by the government policy of giving a Vehicle Registration Tax rebate of Ir£ 1 000 for new car purchasers who were simultaneously scrapping an old car. This was one of the factors that led to a 32 per cent increase in new car registrations, pushing up the annual growth of overall retail sales to 6 per cent by October 1996.

Housing investment has also been buoyant, reflecting the low level of interest rates and the strong rise in household formation. The upturn in new housing permits was particularly marked in 1995 before slowing in 1996, perhaps due to the 17 per cent increase in second-hand house prices in the year to December 1996 (Table 6). However, house completions continued to accelerate during 1996.

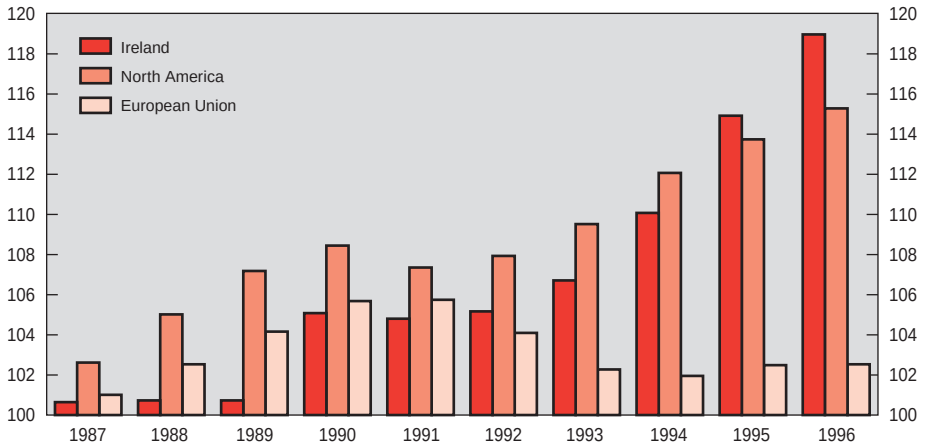
Business investment strengthened later in the upswing than either private consumption or housing. The major factor responsible for the rise in investment has been the growth in private consumption which, in turn, has led to a sharp increase in the construction of new retail outlets. Several large shopping malls have opened recently in the Dublin area, bringing a significant influx of foreign retail chains, mainly from the United Kingdom. With hotel construction also buoyant, there was a 15 per cent increase in non-dwelling construction in 1995, but recent data suggest that the scale of this boom has slackened in 1996 (Table 6). Given the rise in construction, total investment by manufacturing firms expanded in 1995 by 7 per cent despite a continued decline in investment in industrial equipment and machinery. The overall increase, though, was markedly less than the 20 per cent rise in manufacturing output. Overall, gross industrial investment in manufacturing was only 7½ per cent of gross value-added, a low rate by international standards, which confirms the picture of relatively labour-intensive growth.

As a result, the labour market has improved markedly...

The upswing of the past three years has generated a 12 per cent rise in employment (Figure 5), more than in the previous 30 years and a faster rate of job creation than in all but one OECD country. The increase was concentrated in the private non-agricultural sector where employment rose more than 12 per cent over 1995 and 1996 (Table 7). While the largest gains were in manufacturing and construction in 1994, the leading sector in 1995 and 1996 was other services

Figure 5. **EMPLOYMENT GROWTH IN AN INTERNATIONAL COMPARISON**

1986 = 100



Source: OECD Secretariat.

(*i.e.* private services excluding commerce, distribution, transport and communications), where employment increased almost 25 per cent. This reflects the expansion of the hotel industry and the financial sector. Although government employment has increased, its growth remained modest, rising by 3½ per cent between 1994 and 1996. In contrast, employment in agriculture fell by over 4 per cent in 1996, perhaps due to strong activity elsewhere in the economy.

There has been a marked fall in unemployment from 15.7 per cent of the labour force in 1993 to an estimated 11.2 per cent in 1996. The rate is now below the level at the peak of the last business cycle in 1990, which brings it almost back to the level of 1982. However, the fall in the number of unemployed since 1993 was less than one-third of the increase in employment, with only half of the decline coming from the ranks of the long-term unemployed. The natural increase in the labour force, which was not offset by net emigration as in the past, provided another third. The remainder was accounted for by a rise in the female participation rate, which jumped by over 2 percentage points in 1996.

Table 7. **Labour market developments**

	1996	1993 to 1996	1991	1992	1993	1994	1995	1996
	Number (000)		Percentage increase					
Population	3 621	1.4	0.6	0.8	0.5	0.3	0.4	0.6
Natural increase	17	1.4	0.6	0.6	0.6	0.5	0.5	0.4
Net immigration	6	–0.1	–0.1	0.2	0.0	–0.2	–0.1	0.2
Population aged 15-64	2 342	4.3	1.4	1.6	1.3	1.2	1.4	1.7
Active	1 475	7.1	2.5	1.3	1.3	1.6	2.1	3.2
Education	358	9.5	3.5	6.5	5.1	1.7	2.7	5.0
Inactive[1]	1 029	–2.0	–0.1	2.3	0.1	0.8	–0.1	–2.6
Employment	1 285	11.9	0.0	0.8	0.4	3.0	4.8	3.7
Industry and construction	351	12.9	0.9	–1.5	–2.5	5.8	4.9	1.7
Commerce and transport	353	12.8	1.0	1.7	3.6	1.3	6.3	4.7
Other services	226	30.6	3.8	6.9	7.4	4.0	11.7	12.4
Government	218	5.3	1.2	–0.7	0.5	3.4	0.0	1.9
Agriculture	136	–4.9	–8.3	0.0	–7.1	–1.4	0.0	–4.2
Non-agricultural private sector	808	15.8	2.3	1.0	2.0	2.9	6.1	6.0
Public sector	301	4.2	0.3	0.0	0.3	0.7	1.0	2.4
			Percentage of labour force					
Unemployment								
Standardised definition	178	13.6	14.7	15.3	15.1	14.7	12.2	11.9
Survey definition	190	14.6	15.6	16.0	16.7	15.6	13.4	12.9
Registered definition	279	19.9	18.9	20.7	21.0	19.8	19.2	18.9
Redundancies	13	1.2	1.5	1.6	1.6	1.3	1.1	1.0

1. Includes students and housewives.
Source: Central Statistics Office, *Labour Force Survey, Economic Series.*

Less information than usual is available about recent movements in unemployment since the monthly publication of the standardised unemployment rate was abandoned in June 1996. In the past, the ‘‘Live Register’’ of unemployed was used to estimate monthly movements in the rate. However, a September 1996 government study showed that this was a misleading indicator of the number of jobless since up to 44 per cent of the registered unemployed were either working full time or had stopped looking for a job (see Chapter III for more details). In the six months following this disclosure, the number of registered unemployed fell by over 10 per cent. However, the speed of the decline in unemployment appears to have slowed in 1996 as a result of two factors. First, by the spring of 1996, short-term unemployment had fallen to 4.7 per cent, a rate below the OECD

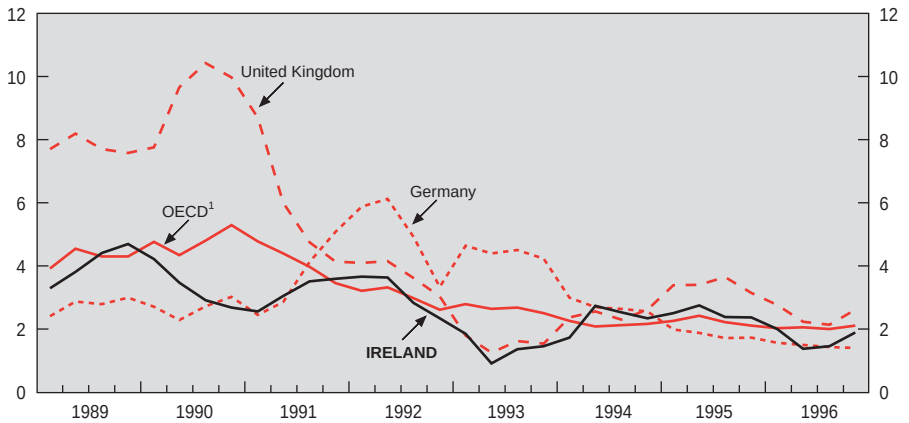
average. Secondly, the most employable of the long-term unemployed, who in total account for 60 per cent of the unemployed, may have found jobs in the first stage of the upswing.

... while inflation remained subdued...

Between 1993 and 1996, annual hourly wage increases in the manufacturing sector averaged 2½ per cent, less than half the rate recorded in the period 1990 to 1993. Wage growth did accelerate somewhat to 3.5 per cent in 1996 (Table 5), in line with the terms of the Programme for Competitiveness and Work (PCW). This trilateral agreement between the unions, employers and the government set the terms for wage increases in the three-year period ending in December 1996. It provided for an overall rise of 8.2 per cent: 2 per cent in the first year, 2.5 per cent in the second year and a total of 3.5 per cent by the end of the final year of the agreement. Wage increases in the manufacturing, financial and public sectors have remained close to the limit, ranging between 7 and 8 per cent. On the other hand, pay hikes in the construction industry, an area where labour demand has risen particularly rapidly in the past three years, have surpassed the limit with wages increasing by 13 per cent in the first two years of the period covered by the settlement.

Price inflation has also remained low, dropping to almost 1½ per cent for 1996 as a whole. Following the devaluation of the pound at the beginning of 1993, inflation (as measured by the CPI excluding mortgage interest) had been as high as 3.5 per cent at the start of 1994 (Figure 6). However, as the effective exchange rate started to appreciate, inflation decelerated progressively to 2.5 per cent by the end of 1994 and 2.2 per cent by the end of 1995. This downward movement continued until February 1997, when inflation stabilised at 1.5 per cent.[1] The deceleration was more marked in the manufacturing sector, where prices in December 1996 were 1½ per cent lower than in the same period a year earlier. The decline was concentrated in products sold on export markets, reflecting the importance of the exchange rate in the pricing of tradable goods. On the other hand, the increase in prices of certain non-tradable goods and services have tended to accelerate. This has been the most noticeable in the housing market.

Figure 6. **CONSUMER PRICE INFLATION**
12 month change in the consumer price index



1. Excluding Turkey, Mexico, Czech Republic, Hungary and Poland.
Source: OECD Secretariat.

... and the current account surplus fell

The volume of imports increased at an average annual rate of more than 13 per cent in 1994 and 1995, reflecting the combination of strong growth in domestic demand and a rapid expansion of exports. The largest increases in imports were in the areas of capital goods, raw materials and components, such as computer hardware, which alone accounted for 17 per cent of total imports in 1995. The slackening in export growth in the first half of 1996 resulted in a marked slowing of imports of these two components (Table 4). However, the pickup in consumption, especially of cars, led to a sharp increase in consumer good imports which may have also been boosted by the steady appreciation of the currency in effective terms. As a result, the gap between the growth of imports and exports narrowed in 1996. Consequently, the trade surplus, which had expanded significantly during 1995, stabilised during 1996. Nevertheless, it remained above the level of 1995 (Table 8), with net exports continuing to support growth.

Table 8. **The current account of the balance of payments**

		1991	1992	1993	1994	1995	1996 1st half
		Ir£ million					
Exports		14 674	16 504	19 459	22 424	27 504	21 940
Imports		12 609	13 003	14 633	17 028	20 232	15 949
Trade balance		2 065	3 501	4 826	5 396	7 271	5 993
Royalties	Net	–477	–654	–897	–1 218	–1 520	–1 342
Other services	Net	–193	–561	–469	–805	–1 135	–712
Factor incomes	Net	–2 797	–3 210	–3 521	–3 574	–4 815	–3 850
Company earnings	Credit	186	157	151	296	556	371
	Debit	2 538	3 042	3 355	3 786	5 435	4 280
National debt	Debit	1 030	923	1 020	1 081	1 015	705
Other	Net	437	442	532	813	898	600
Current transfers	Net	1 609	1 245	1 309	1 155	1 111	669
Current account balance		209	321	1 248	955	913	758
		Per cent of GNP					
Memorandum items							
Trade balance		8.1	13.0	16.8	17.3	21.3	
Royalties		–1.9	–2.4	–3.1	–3.9	–4.5	
Other services		–0.8	–2.1	–1.6	–2.6	–3.4	
Company earnings		–9.2	–10.7	–11.1	–11.2	–14.4	
National debt interest		–4.0	–3.4	–3.5	–3.5	–3.0	
Other factor incomes		2.3	2.2	2.4	3.2	3.2	
Current transfers		6.3	4.6	4.6	3.7	3.3	
Current balance account		0.8	1.2	4.3	3.1	2.7	

Source: Central Statistics Office.

At the same time, the strength of manufacturing output led to an increase of nearly 50 per cent in the profits and royalty payments of multinational companies, boosting these current account debits to 19 per cent of GNP[2] (Table 8). In addition, net current transfers from abroad, principally from the EU, fell 15 per cent between 1993 and 1995. As a result, the current account surplus has fallen from a peak of 4.3 per cent of GNP in 1993 to 1¼ per cent in 1996.

Strong growth projected to continue over the medium term

Given the favourable developments on the supply side, with many foreign firms announcing plans to expand their Irish operations, the economy seems

likely to maintain its rapid pace of growth. The increase in GNP may total over 12 per cent in 1997 and 1998 (Table 9). Exports should continue to lead the expansion, boosting the trade and services balance by an amount equivalent to half of the projected growth in GDP. After allowing for a rise in the profits of multinational companies, the foreign balance may account for over one-third of the increase in GNP. The expansion in domestic demand should also continue. Rising employment, the cuts in income tax and some increase in real wages, due

Table 9. **Short-term projections**

	Estimates		Projections	
	1995	1996	1997	1998
	Percentage change on previous year			
Demand and output				
Private consumption	3.7	6.0	4.5	4.8
Public consumption	2.0	1.9	2.5	1.9
Gross fixed investment	10.1	10.2	7.4	9.7
Final domestic demand	4.5	6.1	4.7	5.3
Stockbuilding	1.0	0.3	0.3	0.1
Total domestic demand	5.7	6.4	5.0	5.3
Exports of goods and services	17.0	10.6	9.4	9.7
Imports of goods and services	12.3	10.3	8.0	8.5
Foreign balance	5.1	1.9	2.5	2.6
Gross Domestic Product	10.1	7.2	6.7	7.0
Net factor income	32.2	13.7	11.3	10.9
Gross National Product	7.5	6.3	6.0	6.3
Inflation				
GDP deflator	0.5	1.0	1.2	1.3
Private consumption deflator	2.0	1.8	2.2	2.5
Export deflator	1.9	–0.2	0.4	1.0
Import deflator	3.8	1.0	1.5	2.2
Labour market				
Total employment	4.4	4.0	3.3	3.3
Unemployment rate	12.2	11.2	11.0	10.7
	Per cent of GDP			
Current account balance	2.4	1.0	0.7	0.6
General government financial balance	–2.3	–1.0	–1.5	–1.0

Source: OECD.

in part to pay hikes in the public sector, will be the main factors boosting consumption by 9½ per cent over the two-year period. The secular decline in the terms of trade associated with heavy dependence on the export of computer hardware and software will keep the growth of domestic demand below that of GNP. With output growing broadly in line with potential and with the effective exchange rate assumed to appreciate in 1997, inflation (as measured by the private consumption deflator) should rise to only 2½ per cent in 1998.

These projections are based on the following assumptions:

- a constant effective exchange rate for the pound from March 1997 onwards at a rate of 2.66 against the Deutschemark and 96.8 against the pound sterling. This would generate a 3¾ per cent appreciation in the effective exchange rate of the Irish pound between 1996 and 1997 and a further ¼ per cent appreciation in 1998;
- a pickup in export market growth from an estimated 6½ per cent increase in 1996 to 7¼ per cent in 1997 and 7¾ per cent in 1998 (these figures do not take into account the particular products exported from Ireland);
- an 8 per cent fall in oil prices in 1997, followed by a further 4 per cent decline in 1998;
- a slight fall in short-term interest rates to 5.3 per cent in the second half of 1997 and 4.9 per cent in 1998. Long-term rates are projected to remain constant at 6.4 per cent in 1997 and 1998, respectively. By 1998, the short-term differential against German rates is assumed to have fallen to 170 basis points and that for long rates to 60 basis points.

The medium-term growth outlook will depend on supply-side developments. The favourable performance of the economy over the past five years, with GNP increasing at an average rate of 5¼ per cent, has not resulted in inflationary pressures. Growth at around this rate, which is close to the Secretariat's estimate of potential output at between 5½ and 6 per cent (on a GNP basis), should be sustainable in the period to 2002 (Figure 7). However, if the expansion continues to be oriented towards high productivity areas, the growth of real national disposable incomes might be only 3½ to 4 per cent per year. Inflation should remain under control while the current account surplus may decline somewhat, reflecting increasing profits earned by foreign companies. In addition, the improvement in the quality of the labour force suggests that unemployment may

Figure 7. **MACROECONOMIC PERFORMANCE OVER THE MEDIUM TERM**

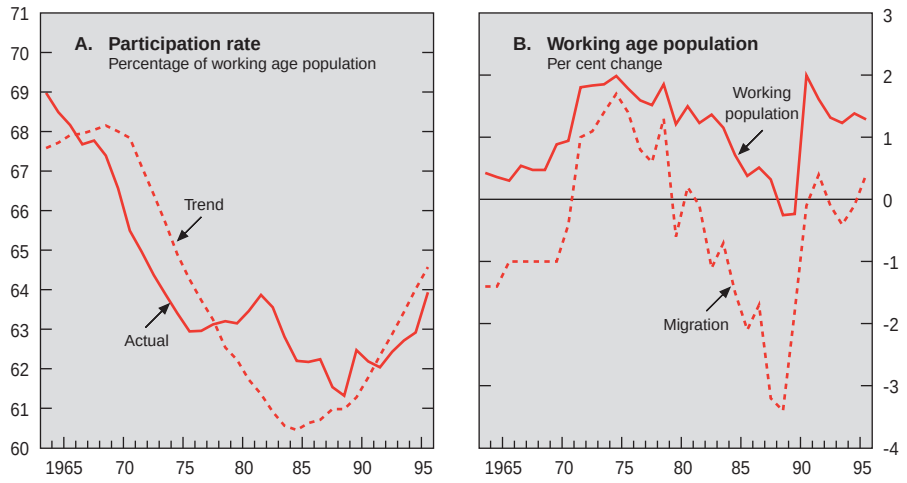
Source: OECD Secretariat.

continue to fall as those with higher levels of education have markedly lower unemployment rates (see Chapter III).

Another favourable development for growth during the next five years is the expected continued rapid expansion of the labour force, despite some slowing in the natural growth rate. The fall in the birth-rate during the 1980s from 22 to 14 per 1 000 will reduce the growth of the working-age population from about 2 per cent annually to only ¾ per cent by 2005 (Figure 8, Panel B). However, much of this slowdown will be offset in the second half of the 1990s by the ending of net outward migration. This contrasts with the situation during the 1980s when a large share of the natural increase in the labour force was absorbed by emigration, which peaked at 5½ per cent in 1989 (Figure 8, Panel B). Since then, the strength of the labour market has reduced emigration, resulting in a net population inflow in 1996.

The growth of the labour force is also likely to be boosted by the increased participation of women. The rising enrolment of women in higher education, where more females than males now graduate, and the fall in the birth rate are

Figure 8. **ELEMENTS DETERMINING LABOUR FORCE GROWTH**



Source: OECD Secretariat.

expected to sustain the ongoing increase in female participation. Since 1990, the female participation rate has steadily increased by about 1 per cent per year. This trend has reversed the fall in the overall participation between 1965 and 1990. The rise in female participation could continue for another generation before reaching the levels in other English-speaking countries.

The medium-term outlook, though, depends on whether the structural transformation of the Irish economy continues at the same pace as in the past five years, with a rapid inflow and expansion of multinational companies, especially in the areas of computers and electronics. If this occurs, and the recently-announced plans of foreign firms suggest that it will, there should be a rapid convergence of income levels in Ireland and those in the rest of Europe. By 2002, GNP per capita would be 95 per cent of the average EU level, up from 75 per cent in 1994, while the gap in the level of private consumption should also narrow by a similar amount.

There are a number of risks surrounding these projections. In the short term, the main one remains higher-than-expected inflation, as the economy is currently operating at close to potential and the output gap has been closed. Although the link between capacity utilisation and price developments appears to have weakened since 1987, there is a concern that inflationary pressures on costs, notably wages, in the non-tradable area could spill over into the traded sector. This, in turn, could reduce profitability, thus weakening the incentive for foreign companies to invest in Ireland and reducing the pace of output growth. Over the medium term, the increasing specialisation of the Irish economy in electronics-based industries could also pose problems by generating cyclical fluctuations that differ from those affecting other members of the EMU. Moreover, investment by firms in the International Financial Services Centre may slacken as the end of the current favourable tax regime in 2005 approaches.

II. The setting of macroeconomic policies

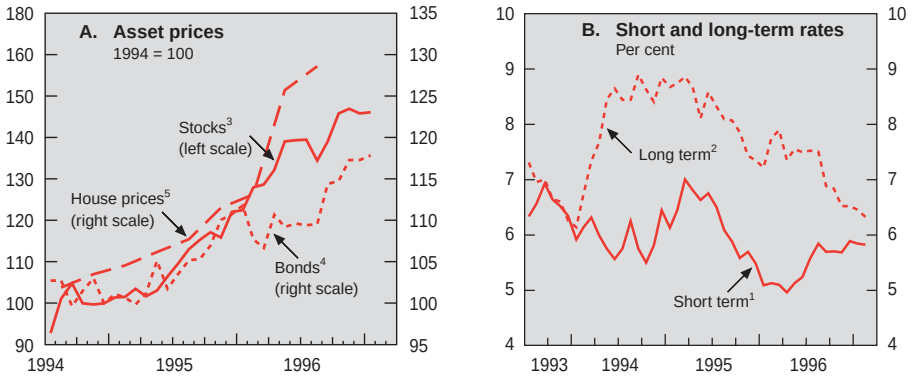
Over the past two years, macroeconomic policy in Ireland has aimed at consolidating the progress made in reducing some of the major imbalances in the economy. Monetary policy has been targeted on holding inflation at a low level, while keeping the exchange rate within the limits of the European Exchange Rate Mechanism. In doing so, it has been supported by income policy agreements which have ensured wage moderation by establishing a social consensus on pay rises. At the same time, fiscal polices have focused on maintaining the general government budget deficit below 2½ per cent of GDP, in order to further reduce the still relatively high level of public debt. Since the ratification of the Maastricht treaty, achieving such objectives has been essential for Ireland to qualify for the last phase of the European monetary unification process. As this process will limit the use of macroeconomic policy instruments for stabilisation purposes, strengthening labour market flexibility appears necessary. In this regard, the government has continued to reform the tax and welfare systems in order to improve incentives in the labour market and generate additional declines in unemployment. This chapter briefly reviews the implementation of such policies.

Monetary management

In a context of rising asset prices,...

Although there were only limited inflationary pressures in goods and services markets during 1995 and 1996, asset prices increased markedly. Housing prices were buoyant, particularly in the Dublin area (Figure 9). The 25 per cent nationwide increase in the two years to September 1996 brought the ratio of house prices to disposable incomes back to the peak level seen at the top of the

Figure 9. **INTEREST RATES AND ASSET PRICES**



1. Three-month interbank rate.
2. Ten-year central government bonds.
3. Stock prices refer to the Irish Stock Exchange index and exclude all UK-registered companies.
4. Bond prices refer to a national 10-year bond with a coupon of 7.8 per cent.
5. Second-hand houses in the Dublin area.

Source: Department of the Environment and OECD.

previous economic cycle in 1990. Stock and bond prices were also strong. Increases in business sector profitability helped push up stock market prices by nearly a half in the two years to the end of 1996. During the same period, bond prices rose nearly 18 per cent, giving a total annual return (interest plus holding gains) of over 17 per cent in local currency and more than 21 per cent in DM terms. Foreign investors, probably attracted by the prospect of Ireland's participation in the EMU, accounted for a significant share of bond purchases. Consequently, capital inflows accelerated during 1996 (Table 10). By September 1996, foreign investors held 34 per cent of the stock of domestic Irish government bonds, up from 28 per cent one year previously, though it is not known to what extent their currency exposure has been hedged. By March 1997, ten-year government bond yields had fallen to 6.6 per cent, a 2 percentage point drop over two years. Although this was about 30 basis points more than the fall in German bonds over the same period, the differential with German yields remains around 90 basis points. Consequently, the inflow of foreign capital appears to be continuing.

Table 10. **Capital account of the balance of payments**

Ir million

	1993	1994	1995	1995		1996[1]
				1st half	2nd half	1st half
Current account						
Current account surplus	1 248	954	850	252	597	200
Capital account						
Market flows						
Irish-pound government securities	1 672	–421	605	205	401	696
Credit institutions	–844	214	1 721	1 067	654	–90
Private capital flows and residual	438	99	–1 403	–904	–499	–706
Non-market flows						
Capital transfers	513	251	511	178	334	164
Exchequer foreign borrowing[2]	–78	–416	–614	–153	–461	–687
Semi-state companies	–141	–285	–260	–173	–86	0
Other transactions	–1 053	–498	33	35	–2	–2
Official reserves	–1 756	102	–1 443	–506	–937	425
Valuation adjustment	410	–135	–12	0	–12	0
Level of reserves (end period)	4 278	4 041	5 473	4 547	5 473	5 048

1. Estimate by the Central Bank of Ireland.
2. This differs from government foreign-currency borrowing in that it excludes foreign-currency borrowing from resident banks and through Section 69 bonds.

Source: Central Statistics Office and Central Bank of Ireland.

... fast growth of money and credit...

The growth of credit and money was extremely rapid in 1995 and 1996. The adjusted broad money stock rose over 25 per cent in the two years ending December 1996, significantly more than the 15 per cent growth of nominal incomes in the same period, driven essentially by strong credit demand from the domestic private sector (Table 11). In addition, lending by a large number of offshore banks in the expanding International Financial Services Centre (see Chapter I) increased rapidly, so biasing the unadjusted broad money data (M3E). Indeed, in the year to May 1996, about one-third of the overall increase in total private-sector credit was accounted for by the lending of IFSC banks and changes in the valuation of existing foreign currency credits of this type (Table 12).

The real estate and construction sector accounted for more than half of the increase in credit to the domestic private sector, reflecting the high demand for

Table 11. **Money growth and its counterparts**

	1994	1995	June 1996	1994	1995	June 1996
	Percentage share in increase of M3E			Percentage change from previous year		
Official reserves	–4.1	39.3	11.8	–2.4	31.1	11.0
Government credit (net)	34.5	–11.8	–4.0	23.2	–8.6	–4.5
Private credit	102.6	126.1	101.5	11.2	16.6	15.9
Private credit (adjusted)				11.8	11.2	13.3
Other assets	16.5	2.3	–1.6	24.0	3.6	–3.2
Net foreign assets	–12.5	–51.1	–12.2	12.4	60.7	13.3
Capital and reserves	–16.0	–2.4	–7.2	9.6	1.8	7.1
Other liabilities	–21.0	–2.3	11.7	33.1	3.6	–20.8
M3E	100	100	100	10.2	12.4	15.7
Memorandum:						
M3E (adjusted)				10.3	11.6	13.2
Currency	5.3	5.6	4.0	7.4	9.8	9.5
M1	21.1	18.6	18.1	13.2	13.8	18.0
M3	78.7	81.7	85.4	11.2	14.1	18.8
Building Society deposits	18.1	10.1	7.9	8.0	5.6	5.5

Source: Central Bank of Ireland

Table 12. **Sectoral distribution of the growth in private sector credit**

	November 1994	November 1995	May 1996	November 1994	November 1995	May 1996
	Percentage share in credit increase			Percentage change from previous year		
Non-financial sector [1]	32.1	26.3	20.2	3.6	7.3	7.5
Leasing and other financial	2.5	0.0	14.0	0.7	0.0	14.8
Personal credit	4.7	15.4	14.9	1.5	12.3	15.7
Real estate and construction	60.7	58.3	50.9	5.5	13.1	14.9
Total (excluding IFSC) [2]	100.0	100.0	100.0	3.7	9.4	12.5
IFSC non-bank credit	87.6	71.9	42.3	117.3	115.6	56.0
Total (including IFSC)	187.6	171.9	142.3	6.8	15.2	16.2

1. The sum of credits to agriculture, manufacturing, energy, transport, communications distribution and other services.
2. Credit by banks in the International Financial Service Centre to foreign non-banks.

Source: Central Bank of Ireland.

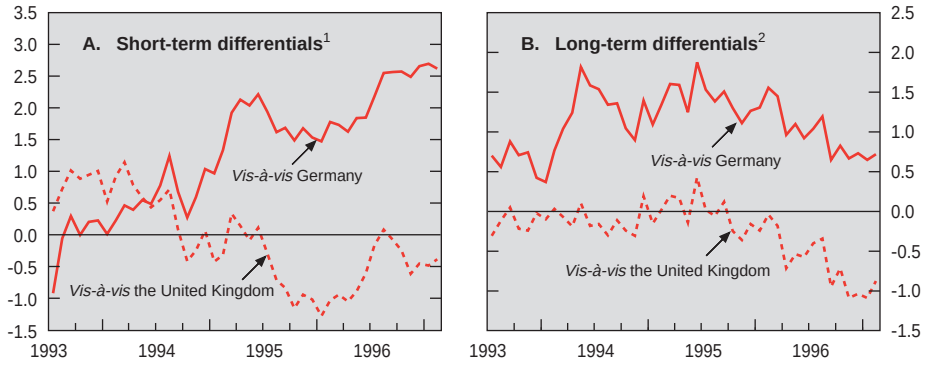
housing mortgages. In addition, there has been strong growth in consumer credit linked to durable purchases such as cars. Overall, about two-thirds of the increase in private credit has been absorbed by the property or consumer credit sectors of the economy. Lending to enterprises in other sectors reflected the generally buoyant economic activity.

... provoked a slight rise in interest rates...

Over the past two years, interest rate policy has continued to focus on keeping inflation low. In March 1995, the authorities raised the official interest rate by 1 percentage point to 7¼ per cent in response to concern about possible inflationary pressures associated with a weakening of the exchange rate (see below). The subsequent decline in rates in other member countries of the ERM gave the Central Bank the opportunity to reverse this increase in its official rates and to let market rates fall even further to 5 per cent. However, the fall in the Irish official rate was kept smaller than that in Germany to avoid generating external inflationary pressures. With the rapid money and credit growth, though,

Figure 10. **INTEREST RATE DIFFERENTIALS**

Percentage points



1. Three-month interbank rate for Ireland, three-month FIBOR for Germany and three-month interbank loan for the United Kingdom.
2. Ten-year central government bonds for Ireland and the United Kingdom, ten-year benchmark government bonds for Germany.

Source: OECD Secretariat.

there was concern that such a policy would be insufficient to ensure the stability of prices. Consequently, the Central Bank allowed market rates to edge upwards from the summer of 1996 onwards, bringing short rates to close to 6 per cent.

... which was reflected in an appreciation of the exchange rate

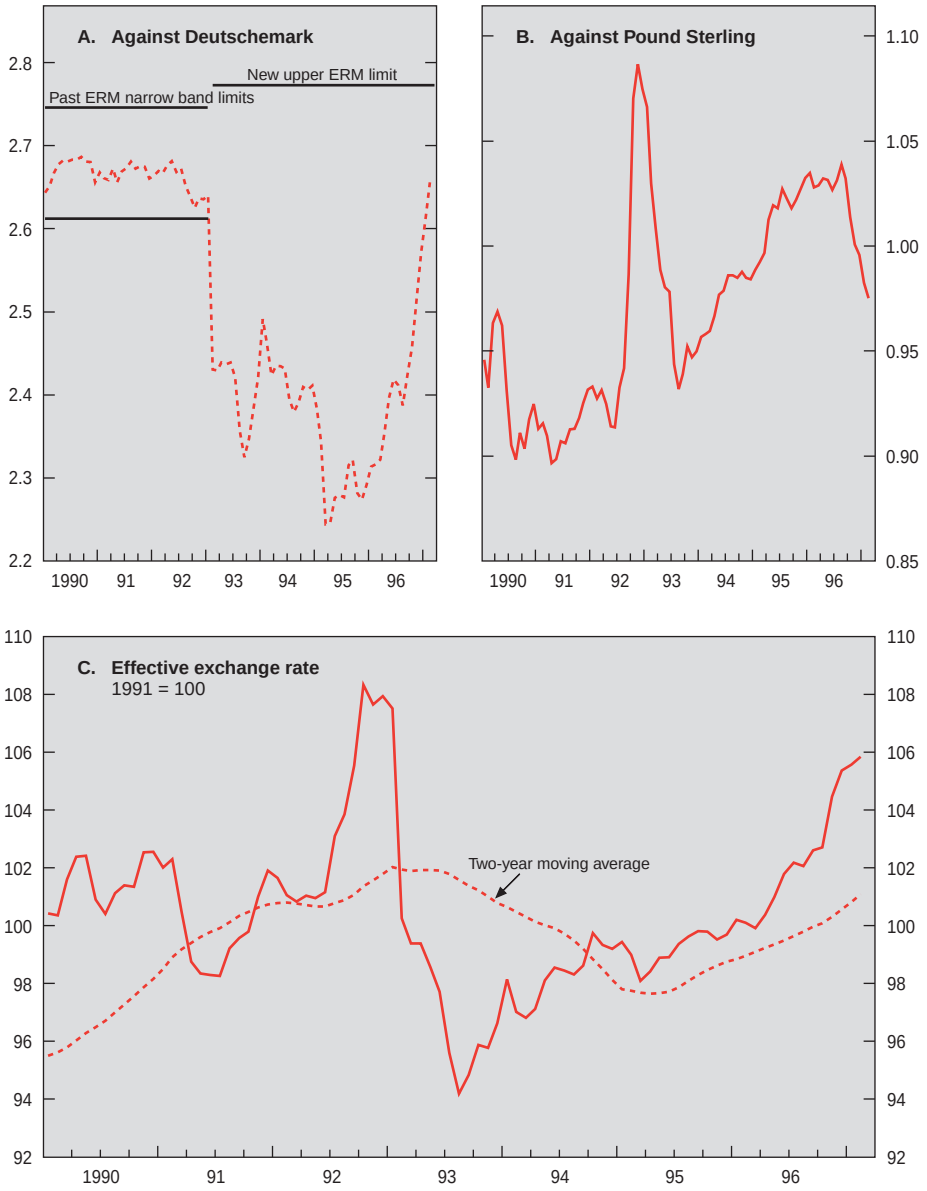
With interest rates higher in Ireland than in most European countries and the economy growing strongly, the Irish pound has shown considerable strength during most of 1995 and 1996. Capital inflows have been large, reflecting the interest rate differential with Germany in both short and long rates (Figure 10). By the end of 1996, the short-term gap had risen to nearly 300 basis points. Faced with these inflows, the authorities intervened to limit the upward pressure on the exchange rate. Consequently, foreign exchange reserves increased 36 per cent between March and December 1995 to a level equivalent to 4 per cent of GNP. During the following year, however, with concern over the domestic liquidity consequences of such intervention at a time of rapid money and credit growth, further pressures for appreciation of the currency were not resisted.

During 1996, the pound largely followed the upward movement of sterling, falling less than 4½ per cent against that currency by the end of the year (Figure 11). In contrast, it climbed by over 12 per cent against the DM in the same period. At the beginning of 1997, the pound had reached its highest level against the German currency prior to its January 1993 devaluation within the ERM. Overall, the effective exchange rate index at end-February 1997 was 3.9 per cent above its 1996 average, according to the Secretariat's index.

Incomes policy

Since 1987, national pay agreements between the unions, employers and government have played a central role in helping monetary policy keep inflation under control by promoting wage moderation. The third of these three-year agreements, the *Programme for Competitiveness and Work* (PCW), ended in December 1996. The estimated 7.7 per cent increase in hourly average industrial earnings between 1993 and 1996 was slightly below the 8.2 per cent allowed by the accord. However, in some areas, such as construction, average earnings have risen faster due to special local bargaining agreements for this sector (see Chapter I).

Figure 11. **EXCHANGE RATE DEVELOPMENTS**



Source: OECD Secretariat.

A new agreement – the *Partnership 2000 for Inclusion, Employment and Competitiveness* (P2000) – was concluded in December 1996 for the period 1997 to 1999. It covers pay policy, government spending, taxation and social measures to reduce poverty and exclusion. In order to increase the extent of social consensus generated by such accords, the number of participants in the negotiations was increased, to include associations representing the Catholic and Protestant churches, the unemployed, women, young people and community groups.

With respect to pay, the new agreement allows for an annual increase of 2.9 per cent in the period 1997 to 1999 in all sectors of the economy (Table 13). Although this is higher than the 2.3 per cent average rise under the PCW, it is similar to the hike permitted in 1996. The 9.3 per cent cumulative rise over three years includes a centrally-set increase of 7¼ per cent and a ''local bargaining'' clause which allows unions to negotiate additional increases up to a ceiling of 2 per cent of the pay bill. Public-sector employees will also benefit from the conclusion in 1997 of many locally-negotiated increases which have their origin in the 1993 central pay agreement. As a result, public-sector pay, including the cost of the recent settlement with the nurses, may rise by about 6½ per cent in 1997 – more than double the increase expected in the private sector. Over the

Table 13. **Pay increases under the Partnership 2000 agreement**[1]

Per cent

		Private sector	Public sector
1997	July	2½	2½ on first Ir£ 200 per week
1998	January	2¼	
	April		2½ on balance
	July		2¼
1999	January	1½	
	July		1½
	September	1	
2000	April		1
Sum of above (simple)		7¼	7¼
Local bargaining		2	2
Total increase (simple)		9¼	9¼
Total increase (cumulative)		9.6	9.6
Annual average		2.9	2.9

1. The Partnership agreement lasts 3¼ years.
Source: Partnership 2000.

three years 1997 to 1999, public-sector pay may rise by 14 per cent and perhaps by up to 16 per cent when allowance is made for wage drift (Table 14).

Reflecting the wider composition of the participants in the process for establishing a national pay agreement, the new accord contains a number of pledges on social policy. The partnership programme places great emphasis on the reduction of social exclusion which it sees as one of the major challenges facing Ireland. In this regard, the government is committed to adopt a national anti-poverty strategy in 1997 focusing on the areas of long-term unemployment, educational disadvantage and low incomes (see Chapter III of the Survey). The programme also calls for a new focus on gender equality and a government strategy for the development and delivery of child care. Measures for social inclusion and equality, including indexation to inflation, will be financed through additional expenditure of Ir£ 525 million (1.3 per cent of GNP) over the three-year period of the agreement.

In addition, the accord contains a number of commitments by the government on tax policy. Income taxation is to be cut by a cumulative Ir£ 900 million between 1997 and 1999 (representing 2.5 per cent of GNP). The government also promised to introduce further business tax reductions of Ir£ 100 million. At the same time, spending on social exclusion is to be raised (see above). However, these pledges will be met while at the same time ensuring that the debt to GDP ratio falls to 70 per cent by 1999 and keeping the general government deficit below 1½ per cent of GDP (equivalent to 63 per cent and 1.1 per cent of GDP,

Table 14. **Estimated impact on pay of Partnership 2000 and earlier agreements**

Annual average, per cent change

	Private sector		Public sector	
	Basic	Including PCW carryover and local deals	Basic	Including PCW carryover and local deals
1997	2.5	3.0	0.5	6.5
1998	2.3	3.3	2.8	3.2
1999	1.8	2.8	2.3	3.7
2000	0.8	0.8	1.5	2.6

Note: Local bargaining increases assumed in July 1998 for the private sector and July 1999 for the public service. Local increases carried over from the PCW are included for the public service.
Source: OECD Secretariat.

respectively, using the current method of computing these fiscal criteria). In addition, the government has prolonged its commitment to limit the growth of public spending to 2 percentage points above the rate of inflation in 1997 and 1998. The implications of these planned tax and spending changes for fiscal policy are discussed below.

The fiscal stance

The 1995 and 1996 budget outcomes

The 1995 budget and spending decisions made during the year resulted in a slight widening in the general government deficit despite exceptionally strong output growth. Indeed, the structural (*i.e.* cyclically-adjusted) deficit widened by 1¾ per cent of GDP to 2.2 per cent for two reasons (Table 15). First, the 7.5 per cent rise in total current government expenditure was Ir£ 177 million (0.5 per cent of GDP) higher than planned (Table 16). Secondly, income tax revenues rose by only ¼ per cent despite a rise in nominal GDP of nearly 11 per cent.

Table 15. **Indicators of fiscal stance**

	1992	1993	1994	1995	1996	1997
	Percentage of GDP					
General government						
Financial balance	–2.5	–2.5	–2.0	–2.3	–0.9	–1.5
Primary balance	2.6	2.2	2.3	1.7	2.8	2.1
Structural balances[1]						
Financial balance	–1.9	–0.6	–0.4	–2.2	–1.0	–1.6
Primary balance	3.2	4.1	3.9	1.8	2.7	2.0
Gross general government debt[2]	92.0	94.5	87.9	81.6	72.8	69.0
Memorandum items						
Transfers from EU	5.4	5.4	3.8	3.8	3.4	3.2
Agricultural funds	4.2	3.8	3.7	3.3	3.0	2.8
Structural funds	2.4	2.8	1.5	1.9	2.0	2.0
Payments	–1.2	–1.4	–1.4	–1.4	–1.3	–1.2

1. The structural balance was improved in 1994 by 0.8 per cent of GDP by a tax amnesty and worsened in 1995 by 0.7 per cent of GDP due to a one-off settlement of a legal award.
2. GDP is measured by the ESA75 concept rather than the national estimate of GDP.

Source: OECD, Department of Finance.

Table 16. **Budgetary developments**

	1994 Out-turn	1995		1996		1997 Budget	1998 Budget	1999 Budget
		Budget	Out-turn	Budget	Out-turn			
Income and expenditure								
Net expenditure	39.0	38.7	38.6	38.0	38.2	37.3	36.4	35.1
Revenue	37.0	36.3	36.8	36.0	37.0	35.7	34.8	34.3
Exchequer Borrowing Requirement	2.1	2.4	1.9	2.0	1.2	1.6	1.6	0.8
General government deficit[1]	1.7	2.5	2.0	2.6	1.0	1.5	1.2	0.5
Gross general government debt[1] (per cent, GDP)	87.9	87.5	81.6	80.6	72.8	69.0	67.0	63.0
Different measures of spending								
Gross expenditure excluding interest	40.8	40.1	39.9	39.5	39.5	38.7	37.9	36.8
Gross expenditure	47.2	46.5	46.3	45.1	45.3	44.4	43.5	42.1
Per cent increase from previous out-turn								
Gross expenditure excluding interest	8.0	5.6	5.8	6.0	7.0	5.5	5.0	3.8
Gross expenditure	5.6	5.8	6.0	4.4	5.7	5.9	4.7	3.7

1. Excludes contingency reserve and does not take into account the new statistical methodology for measuring the deficit to be introduced by EU countries in 1999.

Source: Department of Finance.

The overrun in *expenditure* was due to a court ruling, based on an EU directive, which found that the government had illegally discriminated against women in the payment of certain social security benefits. This decision generated a liability of Ir£ 246 million (0.7 per cent of GDP) of which three-quarters was actually spent in 1995. Excluding this and one other unexpected outlay,[3] total current expenditure was slightly below target. The Exchequer Borrowing Requirement (EBR) was not affected by the spending flowing from this court decision as the government simultaneously sold certain financial assets, so avoiding the need to increase its borrowing (see Box 2). However, the full amount of the court decision was added to the 1995 general government deficit, even though it was not known at the time when final payments would be made. In the event, all reimbursements were made by the end of 1996.

The slow growth in *tax revenue* was in large part due to the end of the tax amnesty, which had boosted tax revenues by Ir£ 220 million in 1994, the

Box 2. **Measuring the Irish budget deficit**

The budget deficit in Ireland has always been measured by the Exchequer Borrowing Requirement (EBR). This measure indicates the flow of cash in and out of the government account at the central Bank. However, consistent with the requirements of the Maastricht Treaty, EU countries decided to use net lending by general government (NLG) as the reference budget deficit. This concept is defined according to the conventions of the national accounts rather than those of the budget.

The three main conceptual differences between NLG and the EBR are the following:

i) The number of public-sector institutions covered by NLG is wider than that covered by the EBR. The latter has a narrow definition of the public sector focusing on just one part of the central government: the general account. The wider concept, NLG, covers the social fund, the National Treasury Management Agency (NTMA, which manages the national debt), postal savings, health boards, local authorities and non-commercial state bodies. While being fully reflected in the EBR, payments from the general account to these other bodies only raise NLG to the extent that the other bodies spend the transfers from the general account outside the general government sector.

ii) NLG does not take into account financial transactions. Therefore, while widening the EBR, loans or equity investments made by the government to state industries do not increase NLG and loan repayments or privatisation proceeds do not reduce it. In the national accounts, such transactions are included in a financial account.

iii) The NLG concept is, in principle, calculated on an accruals basis whereas the EBR is on a cash basis. In the national accounts, the difference between the accruals and cash basis estimates of revenue and expenditure is reflected in the movement of current liabilities and assets in the capital finance account. In practice, however, not all items in the NLG definition of the deficit are on an accruals basis. The most notable exception is tax receipts which are calculated on a cash basis in order to avoid optimistic collection estimates. Another exception is the imputed interest that is due, at redemption, on zero-coupon bonds and on certain small savings instruments issued by the Post Office on behalf of the government. Interest on these liabilities is capitalised and is not captured by the national accounts until it is eventually paid. As a result, the difference between accruals and actual payments does not impact on the current NLG definition of the deficit in Ireland.

Not all of these conceptual aspects are of equal importance in explaining the difference between EBR and NLG. As Ireland is a relatively centralised state, the inclusion of lower levels of the government sector does not greatly increase the size of the deficit. Local authorities, health and education boards only borrow to a limited extent. Moreover, the small deficit of the Social Fund is automatically covered by the general

(continued on next page)

(continued)

account and so, in any case, is included in the EBR. However, the inclusion of the NTMA does make a significant difference to the deficit.

The movement to an accruals base also has a substantial impact on the deficit, as the government has, on several occasions, used the cash definition of the EBR to smooth deficits over a period of several years. Moreover, when zero coupon bonds and postal savings are treated on an accruals basis, under the new ESA95 national accounts standard, the NLG definition of the deficit is widened by 0.4 per cent of GDP. The use of an accruals basis will also mean that Departmental cash balances left over from the previous year will not reduce NLG whereas they are used to reduce EBR.

The application of these principles meant that there have been differences of up to 0.5 per cent of GDP between the EBR and NLG. However in 1996, the difference was just 0.1 per cent as the positive and negative adjustments broadly cancelled out. The reconciliation between the EBR and NLG for that year is as follows:

	Ir£ million	Per cent of GDP
NTMA savings carried over from 1995 spent in 1996	105	0.2
Payment to small savings reserve fund	–162	–0.4
Hepatitis fund payments, disbursed in 1996	24	0.1
Equal treatment cash adjustment	–52	0.1
Sale of State assets	46	0.1
Capital repayments to the Local Loan Fund	79	0.2
Loan and equity investments	–45	–0.1
Payments to reduce pension funds liabilities	–111	–0.3
Deficits of other levels of non-Exchequer agencies, funds and local authorities	56	0.1
Other adjustments (net)	17	–0.1
Total difference (current NLG definition)	–43	–0.1
Postal savings interest	200	0.5
Total difference (forthcoming NLG definition)	157	0.3

Thus, the EBR definition of the budget deficit turned out to be 1 per cent of GNP whereas the NLG definition was calculated at 0.9 per cent of GNP. If the NLG deficit were measured under the forthcoming EU national accounts standards the NLG deficit would have been 1.3 per cent of GDP.

elimination of the temporary income tax levy and other concessions on income and corporate taxes. Another factor was the low growth in corporation tax receipts due to losses on bonds by financial institutions in the previous years.

For the second year running, *the 1996 budget* envisaged a slight increase in the general government budget deficit, to 2.6 per cent of GDP from 2.3 per cent in 1995. It contained measures which were officially projected to raise the deficit by 0.6 per cent of GDP. Their impact was expected to rise to 1.1 per cent of GDP for a full year as the tax measures did not come into effect until April 1996, while the spending increases started only in July (Table 17). However, these official estimates assume that all nominal tax thresholds, social benefits and excise taxes remain unchanged. If the baseline for evaluating tax changes is taken as a constant average tax rate and a constant replacement rate for spending on social benefits, then the discretionary changes in taxation and expenditure would amount to 0.5 per cent of GNP in a full year.

In the event, and despite these measures, the general government deficit in 1996 actually fell to only 0.9 per cent of GDP. The structural deficit is estimated to be about 1.0 per cent of GDP, markedly down from the estimated

Table 17. **Summary of the 1996 and 1997 Budget changes**

Per cent of GNP

	1996	Full year	1997	Full year
Budget changes				
Spending increases	0.3	0.4	0.3	0.6
Income tax and social security contributions reductions	0.4	0.8	0.7	1.1
Indirect tax reductions	–0.1	–0.1	0.0	–0.1
Total changes	0.6	1.1	1.0	1.7
Estimated extent of changes consistent with indexation				
Spending	0.3	0.5	0.2	0.3
Income tax and social security contributions	0.2	0.3	0.2	0.3
Excise tax increases	–0.1	–0.1	–0.1	–0.1
Total changes	0.3	0.7	0.3	0.6
Changes above indexation				
Spending	0.0	–0.1	0.2	0.3
Income tax and social security contributions	0.3	0.6	0.5	0.7
Excise tax increases	0.0	0.0	0.1	0.0
Total changes	0.3	0.5	0.7	1.1

1. The estimated cost of indexation has been calculated by pro-rating the cost of the actual increases in allowance, thresholds and benefits by the estimated increase in wages (3.7 per cent in 1996 and 3 per cent in 1997) or, for excises taxes, the expected increase in the CPI. The abolition of the PSRI income tax allowance has been considered a discretionary change in the 1996 budget, although the enabling legislation has to be renewed each year. The reduction in PSRI contributions is treated as a reduction in revenue rather than an increase in net expenditure, as in the budget.

Source: Budget 1997, Department of Finance.

2.2 per cent in 1995, although still above the 1994 level. A large part of the improvement was due to the absence of one-off special payments such as occurred in 1995. The difference between the actual and the planned EBR measure of the deficit was smaller, reflecting a number of once-off spending decisions, amounting to 0.4 per cent of GNP, which reduced certain government financial liabilities.[4] Consequently, the EBR fell to only 1.2 per cent of GNP, compared with a budget estimate of 2 per cent. Given the small deficit and the reduction in the foreign currency component of government borrowing due to the appreciation of the pound, the debt to GDP ratio fell to 73 per cent, down from nearly 88 per cent at the end of 1994.

The lower-than-expected deficit in 1996 was the result of buoyant tax revenues, with recurring expenditure being held in line with plans. Receipts were 3¾ per cent above budget estimates. Both income and excise tax yields were some 4 per cent above target, while the corporation tax yield was 10 per cent up due to the sharp rise in profits in the modern sector of the economy. Stamp and capital taxes were also buoyant, perhaps reflecting the strength of the housing market. On the other hand, VAT receipts were below their budgeted level due to a new law that restores the standard two-monthly payment regime for this tax.[5] Apart from the once-off spending decisions described above, there was only a small overrun of expenditure, amounting to ½ per cent of initial plans. The growth in gross expenditure was held back by the absence of special factors which had boosted the base for spending in 1995. However, taking 1995 and 1996 together, total current expenditure rose by 12 per cent, with an increase in gross non-interest spending of 13 per cent. Such increases were above the government's target for expenditure growth.

In addition to its macroeconomic objectives, the 1996 budget was aimed at improving the microeconomic performance of the economy. The budget included measures to increase the reward for work, to aid the unemployed to participate in the workforce through the creation and expansion of employment schemes and to improve work incentives by raising unemployment benefits less than the rise in average earnings. With respect to work incentives, further progress was made in reducing the average tax burden and the tax wedge. The government increased the lower limit at which social security contributions and health and training levies are imposed. It also widened the lower rate tax band and raised personal allowances by more than the increase in average earnings. As a result, the

increase in disposable income was greatest for those earning under Ir£ 12 000, while those close to the relatively low threshold for the higher marginal rate of taxation (48 per cent) also gained significantly. Employers' social security contribution rates on low-paid employees were lowered from 9 to 8.5 per cent. In addition, the standard rate of employers' contribution was lowered by 0.2 percentage points to 12 per cent and the threshold for this rate was increased by 8 per cent to Ir£ 250 per week (a level close to the average industrial wage). Finally, for companies not liable to the 10 per cent rate, the corporation tax rate on the first Ir£ 50 000 of company income was lowered to 28 per cent from the standard rate of 36 per cent.

The incentives for the unemployed to take jobs were strengthened by a number of measures on the benefit side. The child dependant allowances paid to the unemployed were frozen for the second consecutive year while the universal child benefit was raised by 7 per cent. This measure is intended to reduce the high replacement ratios of low-income workers with children (see Chapter III and the 1995 Survey). Moreover, the child dependant allowances of the long-term unemployed can now be retained for three months following employment, by which time the Family Income Supplement (FIS) should be payable. Finally, the income limits for the FIS were raised and anyone who has been unemployed for at least one year continues to be eligible for free medical care for three years after entering employment.

The 1997 budget

The 1997 budget plans for a slight widening of the general government deficit from 1 to 1.5 per cent of GDP. Given the falling debt-ratio and an official projection of growth in line with Secretariat estimates of potential, the increase in the structural primary deficit, which excludes interest payments, may be somewhat larger at around ¾ per cent of GDP. Such a rise in the deficit reflects a decision to implement close to half of the tax cuts promised under Partnership 2000 (see above) during the first year of the programme. In total, the budget cut taxes by a total of Ir£ 490 million and raised spending on programmes to reduce social exclusion by Ir£ 215 million in a full year (1.1 and 0.6 per cent of GNP, respectively). After adjusting for fiscal drag, the size of these budget changes falls to 1.1 per cent of GDP in a full year, over twice the scale of those announced in the 1996 budget. Given these changes in the 1997 budget and since

the pledges in the Partnership 2000, after allowing for fiscal drag, are equivalent to a reduction in taxes of about 1½ per cent of GNP, it follows that if these were just limited to those agreed in the accord, then the scope for further reduction in average tax rates would be limited in the second and third years of the agreement. Gross non-interest expenditure is projected to grow by about 7 per cent in nominal terms and 4.7 per cent in real terms in 1997, [6] slightly above the increase seen the previous year, despite government pledges to limit it to 2 per cent above the rise in consumer prices.

The nature of tax cuts and spending increases announced by the government differed somewhat from those implemented in 1995 and 1996. The previous budgets had concentrated tax cuts at the lower end of the income scale and restrained the growth of social benefits in order to improve work incentives at lower income levels. In the 1997 budget, the tax cuts were more broadly distributed, with both the standard rate of income tax and employees' social contributions being lowered by 1 percentage point to 26 and 4.5 per cent, respectively. Social benefits were raised by twice the rate of inflation. As a result, replacement rates for couples with no children and single people, with an earnings potential of half average earnings, are projected to rise by 3 percentage points to 89 and 64 per cent, respectively. On the other hand, the replacement rate for families with children falls somewhat due to the marked improvement in the Family Income Supplement resulting from increased income thresholds and changes in the calculation basis from gross income to income net of social security contributions.

Medium-term fiscal implications of the new pay agreement

In 1997, for the first time, the government introduced a multi-year budget, which covers the period to 1999. Such an approach fits in well with the medium-term fiscal objectives of the new pay agreement. The government plans to keep the general government budget deficit at a maximum of 1.5 per cent of GDP over the next two years. However, this plan includes a contingency reserve that amounts to almost ¾ per cent of GDP by 1999. Moreover, it also makes provision for the introduction of a new statistical convention[7] for measuring the general government deficit, which, by including the capitalised interest paid on postal savings accounts, will raise the deficit by 0.5 per cent of GDP. Thus, the

budgetary plan entails a reduction in the underlying deficit from 1.5 per cent of GDP in 1997 to 0.5 per cent by 1999. Given such a reduction, the public debt should rise by only 4.5 per cent in the three years of the plan, so reducing the ratio to GDP to 63 per cent.

The government expects to reach the deficit target by severely restraining the rise in public spending. The growth of gross non-interest expenditure is projected to slow from the 7 per cent planned in the 1997 budget to less than 4 per cent in 1999 in line with the objective of limiting inflation-adjusted increases to 2 per cent. However, this expenditure target, originally set in 1995, has not stopped actual gross non-interest spending rising by 4¾ per cent per year in real terms in the period 1994 to 1997. On the other hand, it may have been responsible for some slackening in the growth of such expenditure from an annual real rate of 5½ per cent between 1991 and 1994. Achieving such a spending target will be all the more difficult given the government's commitment to increase public-sector pay by 10 per cent between 1997 and 1999.

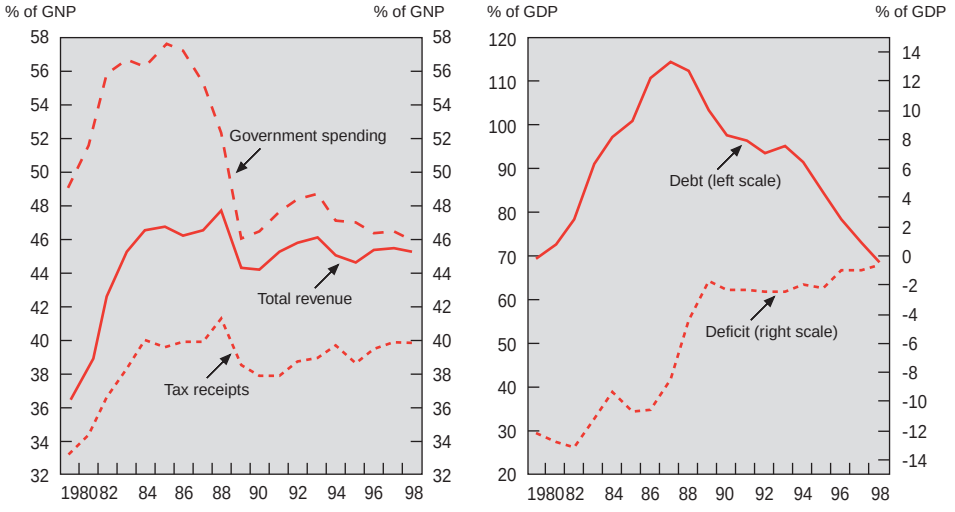
Provided the growth of expenditure is curtailed as planned, the official three-year budgetary projections suggest that there will be room to lower the average tax rate. Government tax revenue is projected to drop to only 32.3 per cent of GNP in 1999, down from 33.7 per cent after the cuts introduced in the 1997 budget. In such a context, it might be desirable to cut marginal rates on low-income workers to improve labour market incentives (see Chapter III).

The government's three-year budget plan is based on GNP growth slowing to an average rate of 4½ per cent in 1998 and 1999, in line with its estimate of potential growth. If output were to increase more quickly, as suggested by Secretariat estimates of potential, there would be scope for the deficit to fall further. However, the economy is unlikely to expand smoothly over the next three years and such fluctuations would affect the deficit. In Ireland, the automatic stabilisers are estimated by the Secretariat to raise borrowing by 0.5 per cent of GDP for each percentage point shortfall in growth. If growth were to drop 1 per cent below the governments projection for two consecutive years (an outcome that would not be out of line with past experience), the deficit would rise by up to 1 per cent of GNP. Such an increase could be accommodated only if the government did not use the contingency reserve for additional spending.

Overall appraisal

Over the past two years, monetary policy has achieved its goal of a low inflation rate. Measured by the CPI, inflation at the end of 1996 was under 2 per cent. So far, the rapid increase in money and credit have not been associated with inflationary pressures in the goods and labour markets. However, asset prices have risen very strongly, surpassing previous peaks in relation to nominal incomes and the growth of housing credit has been accelerating. Such developments point to possible strains in the market for non-tradable goods and services, thus calling for a cautious monetary policy in the period ahead. The new incomes policy may aid monetary policy in its goal of restraining inflation by limiting pay increases to only 3 per cent, although this is somewhat more than allowed under the previous agreement due to the possibility of small increases that can be negotiated at the local level. The taxation commitments in the new central pay agreement are intended to continue the process of reducing average tax rates.

Figure 12. **PROGRESS IN FISCAL CONSOLIDATION**



Source: OECD.

After a decade of successful fiscal consolidation, the 1997 budget continues the policy of keeping a low government deficit and reducing the level of debt relative to GDP which is now down to the EU average, having fallen to 73 per cent by the end of 1996 down from a peak of 117 per cent (Figure 12). Nevertheless, by including about half of the tax cuts and spending increases promised under the three-year Partnership 2000 programme, the deficit will widen somewhat to 1.5 per cent of GDP while the cyclically-adjusted primary surplus (*i.e.* excluding debt interest payments) will fall by ¾ per cent. Simultaneously, however, the government announced a three-year fiscal plan which should result in a steady reduction in the deficit to 0.5 per cent of GDP,[8] provided that the contingency reserve is not drawn down. As long as economic growth remains around 5 per cent and expenditure is restrained to a 2 per cent annual increase in real terms, a lower rate than any seen in the 1990s, this deficit target should be consistent with a fall in the average tax rate.

The movement towards EMU in 1999 strengthens the case for keeping the budget deficit as close as possible to the 1996 outcome of 1 per cent of GDP in the future. This would be essential if the United Kingdom, still Ireland's biggest trading partner, did not join the EMU, as dependence on UK markets, though declining, might expose the Irish economy to shocks that could no longer be countered by monetary policy. Such a perspective also heightens the need for increased labour market flexibility in Ireland, as discussed in the next chapter, to strengthen the self-equilibrating properties of the economy.

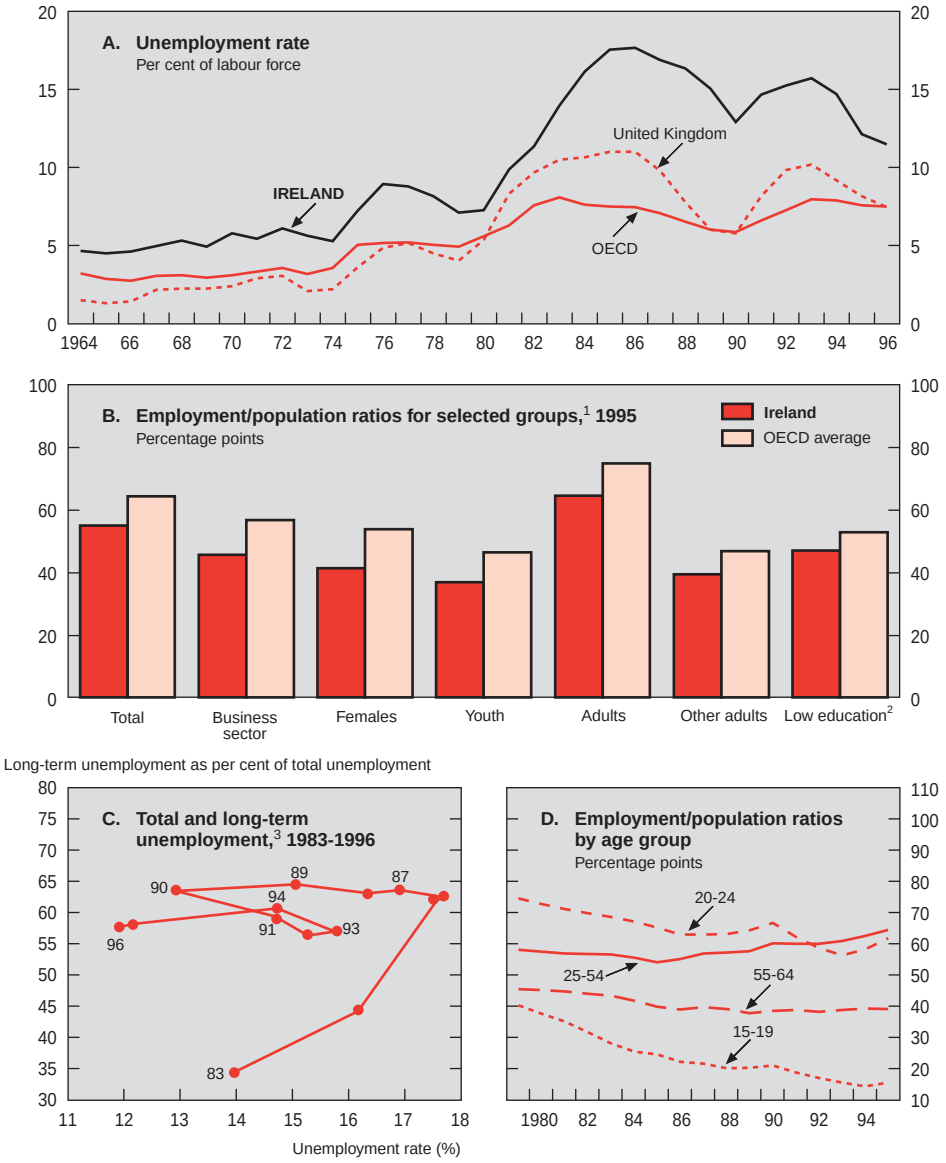
III. Implementing the OECD Jobs Strategy

In the midst of a period of exceptional output growth with low inflation, high unemployment and low employment rates stand out as the major economic problems in Ireland at present. Over the past thirty years, the Irish economy has suffered from an insufficient number of new jobs to employ a rapidly growing population, which has resulted in high levels of emigration and a low rate of participation in the labour force. Despite these two safety valves, the unemployment rate has remained well above the OECD average (Figure 13). The gap was particularly wide in the first half of the 1980s as macroeconomic difficulties raised the unemployment rate above 17 per cent.

The strong expansion during the past three years has generated more new jobs than were created during the previous three decades. Although the unemployment rate fell to an estimated 11¼ per cent in 1996, it still significantly exceeds the OECD average. The problem is exacerbated by the extent of long-term unemployment, which, as a proportion of total unemployment, is among the highest in the OECD area. In addition, a large share of young people are jobless, thus increasing the risk of poverty and social exclusion. This situation represents a serious waste of productive output and increases budgetary pressures, both through higher welfare spending and reduced tax revenue. Increasing labour market flexibility is a high priority, particularly in the context of Ireland joining the EMU, which will limit the use of monetary and fiscal policies to adjust to economic shocks (see Chapter II). The government has already implemented many active labour market programmes and changes in the tax and benefit systems in recent years to address the unemployment problem.

The immense challenge represented by high and persistent unemployment in OECD countries such as Ireland was recognised in the *OECD Jobs Study*, which set out a broad programme of action to tackle it. This chapter, one in a series published in *OECD Economic Surveys*, follows up that programme with specific

Figure 13. **KEY DEVELOPMENTS IN THE LABOUR MARKET**



1. Defined as the percentage of each working-age population group.
2. Completed less than upper secondary education, 1992.
3. Long-term unemployment is defined as individuals looking for work for one year or more.

Source: OECD Secretariat.

recommendations for Ireland. The first section provides an overview of Irish labour market performance, focusing on the extent of long-term unemployment. In the following section, the specific policy recommendations for Ireland that emerge from the *Jobs Study* are discussed. The final sections address recent policy initiatives and the scope for further action.

Labour market and employment performance

Despite rapid employment growth...

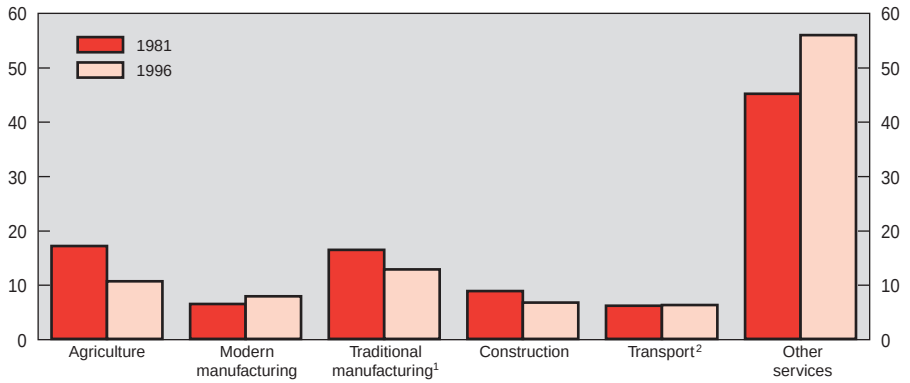
After falling significantly in the first half of the 1980s, employment began to increase again in 1986 (Figure 5). In contrast to most other OECD countries, employment growth has been exceptionally rapid since 1993, expanding at an annual rate of almost 4 per cent despite the continued secular decline in the number of jobs in agriculture (see Chapter I). Moreover, the structure of employment in Ireland has changed significantly during the past fifteen years. The share of the labour force in agriculture fell from 17 per cent in 1981 to 11 per cent in 1996 (Figure 14), still about double the EU average. Given the relatively advanced age on average of the farm workforce, its share of the labour force will continue to decrease. This decline has been offset by increased employment in the service sector, a trend which typically accompanies the transition to higher income levels. It also reflects the expansion of the financial sector following the establishment of the International Financial Services Centre in 1990.

Meanwhile, manufacturing's share of the labour force has fallen over the past fifteen years as a result of the absolute decline in employment in the traditional sector, which includes the textile, food and drink industries. This has been partially offset by job gains in the modern sector, which is dominated by multinational companies that have invested in Ireland. The modern sector – defined as the pharmaceutical, computer, electrical engineering, instrument engineering and other food (primarily soft drink concentrates) industries – now accounts for about three-fifths of manufacturing output but only a third of employment. Average labour productivity is thus about 2.5 times greater in the modern sector than in traditional firms.

There have also been significant changes in the occupational structure of the labour force in line with the growth of the service sector and the increased level

Figure 14. **EMPLOYMENT STRUCTURE**

Per cent of total employment



1. Includes employment in mining and utilities.
2. Includes communications.

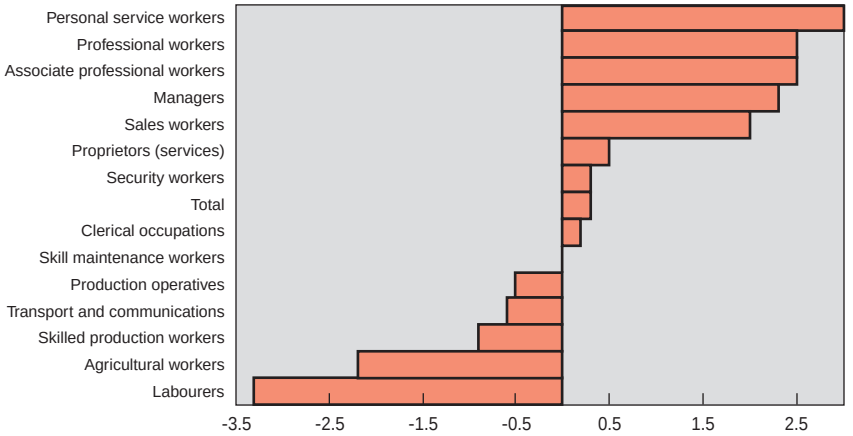
Source: Central Statistics Office, *Labour Force Survey.*

of educational attainment in Ireland (Figure 15). In particular, the number of managerial and professional jobs requiring advanced skills has grown rapidly. At the same time, there has been a corresponding decline in the number of agricultural and unskilled workers.

A significant portion of the new jobs have been on a part-time basis, boosting the share of such employees from 5 per cent of total employment in 1983 to 10 per cent in 1995 (Table 18). A public employment programme accounted for most of the increase. Nevertheless, the share of part-time employment remains substantially below the OECD average of 18 per cent. Most of these employees prefer to work on a part-time basis: in 1995, less than one-fifth were seeking to work longer hours. As in most countries, part-time employment is concentrated among women, who account for 70 per cent of the Irish part-time workforce, and is most prevalent in service industries such as distribution and finance. Meanwhile, temporary workers' share of employment increased from less than 5 per cent in 1983 to 7 per cent in 1995. There is, however, a substantial degree of overlap between the part-time and temporary work forces.[9] Finally, the

Figure 15. **EMPLOYMENT BY OCCUPATION**

1981-94 average annual percentage change



Source: Canny, Hughes and Sexton, 1995.

Table 18. **Labour force by status**

Per cent of labour force at work in selected years

	Part-time employees [1]	Temporary employees [1]	Self-employed		
			Agriculture	Non-agriculture	Total
1983	5.0	4.6	12.8 [2]	7.9 [2]	20.7 [2]
1986	4.9	5.7 [3]	12.0	9.5	21.5
1991	6.7	6.6 [4]	10.3	11.0	21.4
1992	7.3	n.a.	10.6	11.7	22.3
1993	8.6	6.1	9.7	12.2	21.9
1994	9.1	6.3	9.3	12.1	21.3
1995	10.1	7.0	8.9	12.1	20.9

1. The part-time and temporary categories are overlapping to some extent since many temporary employees work part time.
2. 1981.
3. 1985.
4. 1989.

Source: Central Statistics Office, *Census of population* and *Labour Force Survey*; FAS, *Labour Market Review*.

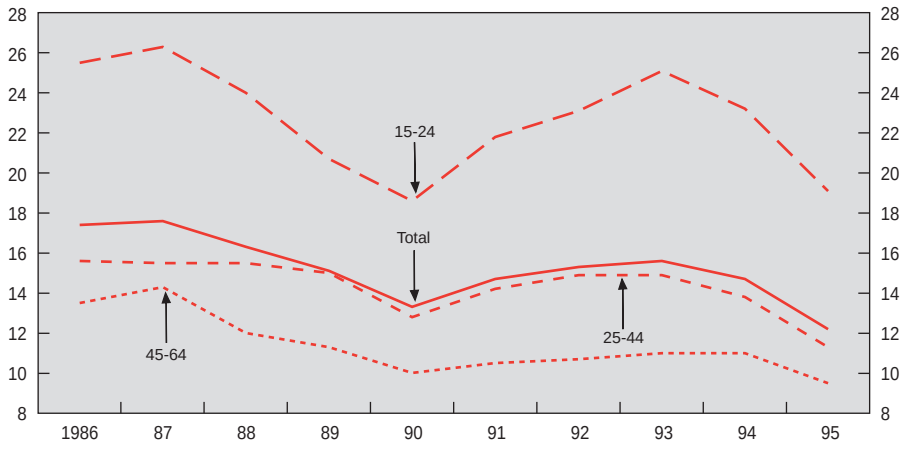
proportion of self-employed persons in the work force has remained relatively stable at about a fifth since 1981 despite a significant decline in the number of such workers in the agricultural sector. In sum, self-employed persons and part-time and/or temporary employees represent about a third of the labour force.

... the unemployment rate remains high

Despite rapid employment gains in recent years, the 11¼ per cent unemployment rate in Ireland in 1996 was still substantially higher than the OECD average of 7.8 per cent. Moreover, the rate for persons in the 15 to 24 age group remained close to 20 per cent (Figure 16), compared with an (unweighted) average of 15.6 per cent in the OECD area. In Ireland, about 30 per cent of the unemployed fall in this age group (Table 19). As a result, single persons account for a disproportionate share of unemployment.

Unemployment is concentrated among persons with only a primary or lower secondary school education. This group, which accounts for about half of the labour force, had an unemployment rate of 18 per cent in 1995, compared to only

Figure 16. **UNEMPLOYMENT BY AGE**
Percentage of labour force in a given age group



Source: Central Statistics Office, *Labour Force Survey.*

Table 19. **Characteristics of the unemployed**

Per cent of total unemployed, 1995

	Share of labour force	Looking for first job	Quit or lost previous job	Total
A. By age and gender				
15-24 age group	18.9	13.2	16.4	29.7
Male	10.5	8.0	9.9	17.9
Female	8.4	5.2	6.5	11.7
24-45 age group	52.4	2.6	42.9	45.5
Male	32.6	1.7	33.2	34.9
Female	19.8	0.8	9.7	10.6
44-65 age group	26.5	0.1	24.3	24.3
Male	19.8	0.1	21.4	21.5
Female	6.7	0.0	2.8	2.8
Over age 65	2.2	0.0	0.5	0.5
Male	1.8	0.0	0.4	0.4
Female	0.4	0.0	0.1	0.1
All ages	100.0	15.9	84.1	100.0
Male	64.7	9.8	65.0	74.8
Female	35.3	6.1	19.2	25.2
B. By marital status				
Single	40.8	15.4	39.3	54.7
Male	24.1	9.4	26.4	35.9
Female	16.7	6.0	12.9	18.9
Married	55.1	0.5	39.4	39.8
Male	38.5	0.4	34.5	34.9
Female	16.6	0.1	4.9	5.0
Separated	2.6	0.0	4.3	4.3
Male	1.4	0.0	3.1	3.1
Female	1.2	0.0	1.2	1.2
Widowed	1.4	0.0	1.2	1.2
Male	0.7	0.0	0.9	0.9
Female	0.7	0.0	0.2	0.2

Source: Central Statistics Office, *Labour Force Survey.*

4 per cent for those with a university education (Table 20). Consequently, those with only a lower secondary education or less accounted for two-thirds of the unemployed in 1995. Despite the improvement in the education system in recent years, many of the poorly educated are young. In 1995, a third of the unemployed were persons under the age of 35 who had failed to complete upper secondary school. Older persons with low levels of education accounted for a smaller proportion of the jobless as many have dropped out of the labour force.

The improvement in the education level of the working-age population appears to have reduced the level of unemployment during the past few years.

Table 20. **Unemployment and education**

Per cent in 1995

	15 to 24 years	25 to 34 years	35 to 44 years	45 to 54 years	55 to 65 years	Over 65 years	Total
A. Unemployment rate by level of education							
Primary [1]	45.5	31.6	24.5	15.5	9.1	1.9	19.0
Lower secondary	27.6	19.1	12.6	10.0	9.7	6.4	17.3
Upper secondary	14.6	7.8	6.1	5.9	6.8	2.7	9.5
Third level, non-university	9.6	5.2	4.0	5.6	3.2	0.0	5.9
University	8.5	4.4	3.1	2.8	2.5	0.0	4.0
Total	19.1	11.3	11.0	10.2	7.7	2.4	12.1
B. Share of the unemployed							
Primary [1]	4.9	5.1	7.8	7.9	2.9	0.2	28.8
Lower secondary	12.1	11.1	7.8	3.9	1.4	0.2	36.4
Upper secondary	10.6	6.8	4.7	1.9	0.7	0.1	24.8
Third level, non-university	2.2	1.9	1.1	0.7	0.2	0.0	6.1
University	0.9	1.5	0.8	0.5	0.2	0.0	3.8
Total	30.6	26.4	22.3	14.9	5.4	0.5	100.0

1. Includes persons that do not state their level of education.
Source: Central Statistics Office.

The proportion of the 25 to 64 age group with only a primary school education has fallen markedly, reflecting the retirement of many of those who completed their education before the introduction of free secondary schooling in 1968 (Table 21). Meanwhile, the share with a third-level education has risen significantly. The rising qualifications would in themselves have reduced the overall unemployment rate from 12.9 per cent in 1989 to 11.7 per cent in 1995, assuming that the unemployment rate for each education level was constant at its 1995 rate during that period. This suggests that there has been a significant decline in structural unemployment in Ireland during the 1990s. Indeed, the OECD Secretariat has estimated that the equilibrium unemployment rate consistent with stable wage increases has fallen in the 1990s. Other factors that may have reduced structural unemployment include the central wage agreements, changes in the tax and benefit systems which increased work incentives, and the falling rate of unionisation.

The rate of short-term (less than one year) unemployment in Ireland was less than the OECD average of 5.2 per cent of the labour force in 1995 (Table 22). The major problem in Ireland is rather concentrated in long-term unemployment:

Table 21. **Unemployment and the rise in educational qualifications**

	Unemployment rate	Highest level of educational qualification				
	1995	1989	1991	1992	1994	1995
	Per cent of labour force aged 25-64	Percent of population aged 25-64				
Primary	18.6	36.9	33.2	31.7	28.1	26.9
Lower secondary	14.7	25.2	26.6	26.1	26.7	25.9
Upper secondary	7.6	23.2	24.3	25.3	26.7	27.3
Third level, non-university	4.9	7.4	7.9	8.6	9.7	9.8
University	3.5	7.3	8.0	8.3	8.8	10.1
Weighted unemployment rate [1]		12.9	12.6	12.3	11.9	11.7

1. The weighted unemployment rate weights the 1995 unemployment rate for each education level by the share of the population aged 25-64 with that educational level.

Source: 1989 to 1994, OECD Education Database; 1995, Central Statistics Office.

7.5 per cent of the labour force was unemployed for more than one year, a rate more than double the OECD average. Long-term unemployment, which accumulated during the problem years of the 1980s, has risen further during recessions, while stabilising during upturns. In 1995, almost 60 per cent of the unemployed had been out of work for more than one year. The duration of unemployment has a strong effect on the probability of finding employment. While two-thirds of those unemployed for less than twelve months find a job within one year, a person out of work for more than two years has only a one in four chance of securing employment.[10]

The unemployment rates for men and women are similar in Ireland, in contrast to some continental European countries where female rates are higher. However, long-term unemployment is most prevalent among men; in 1995, two-thirds of unemployed males had been out of work for more than one year compared to about half of jobless women (Table 23). Information from registered unemployment (see Box 3) indicates that the problem is most severe among men in the 45 to 54 age group.[11] Among men in this age group, those unemployed for more than two years have only a 10 per cent chance of finding a job in the following year compared to 40 per cent for those under age 25. The higher rate of long-term unemployment for older men is partly a result of downsizing since the early 1980s in traditional sectors, such as clothing, textiles, food and drink. In

Table 22. **Long-term unemployment**

1995

	Workers unemployed more than 1 year as per cent of total unemployment			Unemployed as per cent of the labour force		
	All workers	Male workers	Female workers	Short-term	Long-term	Total
Ireland	**58.2**	**63.7**	**49.4**	**4.7**	**7.5**	**12.2**
Other EU countries						
Austria	17.4	17.4	17.4	4.9	1.0	5.9
Belgium	62.4	61.4	63.2	4.9	8.1	13.0
Denmark	27.9	31.9	24.8	7.4	2.9	10.3
Finland	32.3	35.4	28.7	11.6	5.6	17.2
France	45.6	44.5	46.6	6.4	5.3	11.7
Germany	48.3	45.0	50.9	4.9	4.5	9.4
Greece	50.9	42.0	57.4	4.9	5.1	10.0
Italy	62.9	61.9	63.9	4.5	7.5	12.0
Netherlands	43.2	48.6	37.9	4.0	3.1	7.1
Portugal	48.7	46.2	51.2	3.7	3.5	7.2
Spain	56.5	50.7	62.2	10.1	13.2	23.3
Sweden	15.7	17.2	13.6	6.5	1.2	7.7
United Kingdom	43.5	49.5	32.2	4.6	3.6	8.2
Luxembourg	22.4	24.5	20.6	2.3	0.7	3.0
Other European countries						
Czech Republic	30.6	30.2	30.9	2.1	0.9	3.0
Iceland	12.2	10.8	14.0	4.4	0.6	5.0
Norway	26.5	28.6	17.3	3.6	1.3	4.9
Switzerland	32.3	32.3	35.4	2.8	1.4	4.2
Turkey	39.3	34.0	55.1	4.6	2.9	7.5
North America						
Canada	13.8	15.5	11.5	8.2	1.3	9.5
Mexico	1.5	1.3	1.7	6.2	0.1	6.3
United States	9.7	11.0	8.1	5.1	0.5	5.6
Pacific Area						
Australia	30.8	34.2	25.6	6.0	2.6	8.6
Japan	18.1	23.9	9.9	2.6	0.6	3.2
New Zealand	22.9	26.8	18.0	4.9	1.4	6.3
Unweighted average	**30.4**	**30.6**	**29.9**	**5.2**	**3.3**	**8.6**

Source: Central Statistics Office, *Labour Force Survey*, OECD, *Employment Outlook.*

filling new jobs, which tend to be higher-skilled positions in the service sector or modern manufacturing, employers generally prefer to choose among the large cohort of younger workers with higher educational qualifications. Consequently, older workers, who on average have less education, get pushed down the unemployment queue or leave the labour force. Moreover, the work incentives for

Table 23. **Unemployment by duration**

Per cent of total unemployment, 1995

	Number of unemployed (000s)	Less than 6 months	6 months to 1 year	More than 1 year	1 year to 2 years	More than 2 years
A. By age and gender						
Male	105	19.4	13.9	66.9	16.7	50.2
15-24 age group	30	27.4	21.7	50.9	20.6	30.3
Over age 25	75	16.2	10.9	72.9	15.2	57.7
Female	63	26.8	20.9	52.3	20.3	32.0
15-24 age group	20	31.2	24.7	44.1	21.5	22.6
Over age 25	43	24.7	19.1	56.2	19.8	36.4
Total	168	22.0	16.5	61.5	18.1	43.4
15-24 age group	50	29.0	22.9	48.1	20.9	27.2
Over age 25	118	19.3	13.9	66.8	16.8	50.0
B. By situation						
Lost or left previous job	66	24.0	14.8	61.2	16.1	45.1
Looking for first job	29	13.3	20.8	65.9	19.3	46.6
Previously out of labour force	73	23.5	16.5	60.0	19.4	40.6

Source: Eurostat.

these workers tend to be weaker as a result of the tax and benefits system in place. A 1993 survey found that the long-term unemployed accounted for only 6 per cent of the 8 000 new recruits in private-sector firms. In addition, there is a substantial degree of long-term unemployment among persons looking for their first job; in 1995, two-thirds had been searching for at least one year. Many in this group left school without any qualifications. Overall, persons with only a primary or lower secondary school education account for about three-fourths of the long-term unemployed. In addition, this problem tends to be geographically concentrated in certain neighbourhoods in Dublin, Cork and Limerick, which are also affected by high rates of crime and drug abuse. However, unemployment rates for Ireland's eight regions were grouped within 3 percentage points of the national average in 1995.

Moreover, participation rates are low...

In addition to high unemployment, Ireland's labour market is characterised by a relatively low participation rate. In 1995, 63 per cent of the 15 to 64 age group was economically active, significantly below the OECD average of 71 per

Box 3. **The measurement of unemployment in Ireland**

The standard measure of Irish unemployment is the annual Labour Force Survey, which has a sample of 46 000 households, approximately 4 per cent of the population. This survey determines respondents' employment status according to the criteria established by the International Labour Organisation (ILO) in order to calculate a standardised unemployment rate. In addition to this survey-based measure of unemployment, there is also a monthly count of the number of persons registered as unemployed in order to qualify for social welfare benefits or to protect their pension entitlements.* Prior to 1996, trends in registered unemployment were used to calculate the unemployment rate in the months following the survey, which is taken in April of each year.

Until the mid-1980s, the gap between registered unemployment and that reported by the Labour Force Survey was quite small. However, the two series have diverged significantly since the 1985 Social Welfare Act, which required government programmes to treat men and women equally. This encouraged many women, including some who classify themselves in the survey as engaged in ''home duties'', to register as unemployed. Consequently, women accounted for over 85 per cent of the increase in registered unemployment over the past ten years. By April 1996, this measure was almost 50 per cent higher than the number of unemployed reported by the survey, a difference of more than 90 000 persons. This prompted the Central Statistics Office, which produces both series, to investigate the causes of the large gap using a sample of persons included in both measures.

The results of this study showed that half of those registered for unemployment benefits would not be classified as unemployed by international definitions. First, 25 per cent were no longer economically active according to their responses to the Labour Force Survey. Approximately three-fourths of this group reported that they were neither looking for nor wanted jobs, while the remainder were primarily students or retired persons. Second, about 10 per cent were engaged in part-time work, which is permitted for persons who are registered as unemployed. Finally, 11 per cent were working full time, with about three-quarters of this group indicating that their jobs were permanent. This result may indicate fraud, as well as administrative delays in removing those who find jobs from the list of registered unemployed. In addition, 28 per cent of the registered unemployed did not live at the address reported to the Department of Social Welfare and thus were not included in the study. While the results of the survey may have to be explored further given several potential sampling problems,** the Central Statistics Office has decided to suspend the publication of the monthly estimate of the unemployment rate using the number of registered unemployed. Instead, it will conduct the Labour Force Survey on a quarterly basis beginning in mid-1997 in order to provide more accurate information on

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developments in the labour market. In addition, the government has introduced measures to ensure that all claimants meet the conditionality requirements for receiving benefits.

The significant number of unemployment benefit recipients who are not searching for work is reflected in the high rate of economic inactivity in some age groups (Table 24). Among men between the ages of 45 and 64, for example, almost a fifth were inactive in 1995. On the other hand, there are many unemployed according to the ILO definition who are not included among the registered unemployed. This is particularly the case for young people who do not have a sufficient employment record to qualify for benefits and who live with their parents and thus fail to satisfy the means test for unemployment assistance. Consequently, they have little incentive to register.

Table 24. **Population by principal economic status**[1]

Per cent of population aged 15 to 64 in 1995

	Male	Female	Male	Female	Male	Female	Male	Female	Total
	15 to 24		25 to 44		45 to 64		15 to 64		
At work	36.9	32.3	81.4	52.4	69.5	26.5	65.7	39.3	52.6
Unemployed	10.9	7.4	13.6	4.0	11.8	1.6	12.3	4.2	8.3
Labour force (1 + 2)	47.8	39.7	95.0	56.4	81.3	28.1	78.0	43.5	60.8
Students	50.5	54.9	1.1	0.8	0.1	0.1	14.2	14.8	14.5
Labour force and students (3 + 4)	98.3	94.6	96.1	57.2	81.4	28.2	92.2	58.4	75.3
Sick or disabled	0.4	0.3	2.1	1.0	7.3	2.1	3.2	1.2	2.2
Inactive[2]	1.3	5.0	1.6	41.8	2.0	66.8	1.6	39.6	20.6
Retired	0.0	0.0	0.2	0.0	9.4	2.9	2.9	0.9	1.9
Total population	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

1. ''Principal Economic Status'' is a method of classification based on the individual's own assessment of their status. This approach is used in both the census of population and the Labour Force Survey, where it accompanies the ILO-based questionnaire.
2. Includes persons on ''home duties''.

Source: Central Statistics Office, *Labour Force Survey*.

Ireland is not the only country where the number of persons receiving unemployment benefits exceeds the level of unemployment reported by labour force surveys. In 1995, this was also the case for Belgium, New Zealand, the Netherlands, Sweden and Finland (Table 25). However, the gap was significantly higher in Ireland than in other OECD countries.

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Table 25. **Unemployment benefit recipients**[1]

As per cent of unemployed[2]

	1990[3]	1995[4]
Ireland	**124**	**149**
Belgium	126	138
New Zealand	112	127
Netherlands	124	125
Sweden	102	109
Finland	111	108
Australia	82	101
Denmark	100	100
Norway	81	94
United Kingdom	73	94
Austria	89	90
Germany	67	87
France	87	76
Switzerland	99	71
Spain	54	40
Japan	36	39
United States	33	36
Greece	30	36
Unweighted average	85	89

1. Includes recipients of unemployment insurance and assistance as well as guaranteed income.
2. The number of unemployed according to labour force surveys.
3. 1991 for Australia, Austria, Belgium, Denmark, Finland, Greece, Norway and Switzerland.
4. 1994 for Greece and the United Kingdom.

Source: OECD.

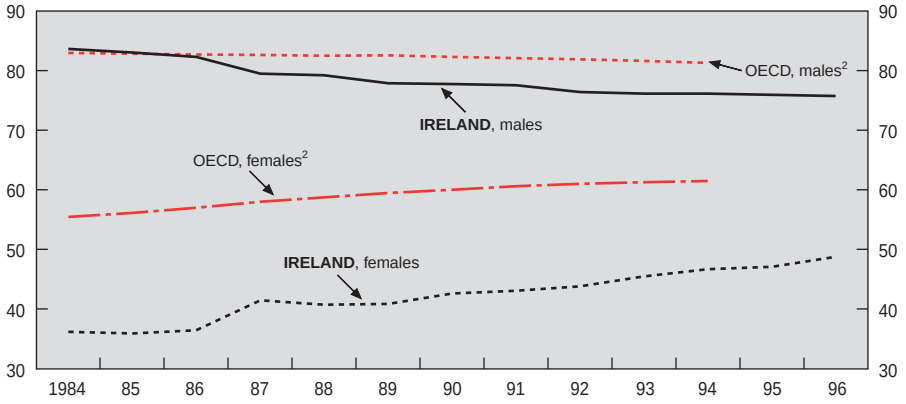
* About 95 per cent of those registered receive unemployment benefits or assistance.

** These problems include incorrect responses by those interviewed and interviewer bias.

cent. The low participation rate in Ireland is primarily a result of the relatively small proportion of women who work. Despite increasing from 34 per cent in 1973 to 48 per cent in 1995, the female participation rate remains far below the OECD average (Figure 17). The increase in female labour force activity in Ireland during the past twenty years has been concentrated among the 25 to 55 age group. In contrast, the participation rate of older women has fallen due to earlier retirement, while that of the 15 to 24 age group has declined sharply as a result of the increased number of young women in higher education.[12] By 1994,

Figure 17. **LABOUR FORCE PARTICIPATION RATES**[1]

Percentage of labour force



1. Age 15 to 64.
2. Nineteen countries for OECD.

Source: Central Statistics Office, *Labour Force Survey,* OECD Secretariat.

over 30 per cent of all 20 year olds were enrolled in third-level education, with women accounting for more than half. This is substantially above the average of 23 per cent in the other OECD countries for which similar data are available.

The higher level of educational attainment has encouraged a larger proportion of women above the age of 25 to work. In addition, the decline in the fertility rate and the change in the structure of employment, notably the increased number of service sector and part-time jobs, has boosted the female participation rate. Another factor has been shifting social attitudes and legal changes, such as the removal of the ban on married women working in the public service in 1973, the Anti-Discrimination Acts of 1974 and 1975, the Employment Equality Act of 1977, and the introduction of a statutory right to maternity leave in 1981. Moreover, the adoption of equal treatment of women under the social welfare code has encouraged them to join the labour force (see Box 3). Given the rise in female participation, their share of employment has increased from 26 per cent in 1971 to 38 per cent in 1996.

Until 1990, the rising participation rate of Irish women was more than offset by a falling rate for men. Consequently, the overall participation rate, which had been as high as 70 per cent in the mid-1960s, fell to a low of 60 per cent in 1989. Since then, however, the overall rate has risen despite the continued decline in the proportion of men who are economically active. By 1995, the male participation rate had fallen to under 78 per cent, somewhat below the OECD average. As in the case of women, though, much of the gap reflects the high and rising enrolment in education by young men. Consequently, the labour force participation rate for males aged 15 to 24 is 48 per cent compared to an OECD average of 57 per cent. With half of this age group in school, only 2 per cent are inactive (Table 24). A second reason for falling male participation in Ireland is the fact that the rate of older men, while still above the OECD average, has declined significantly. This is primarily a result of declining demand for older, less-skilled workers in certain manufacturing industries, the increased availability of private pensions and social welfare programmes offering benefits to workers approaching retirement age. The secular decline in the farm labour force has also reduced the participation rate of older men.

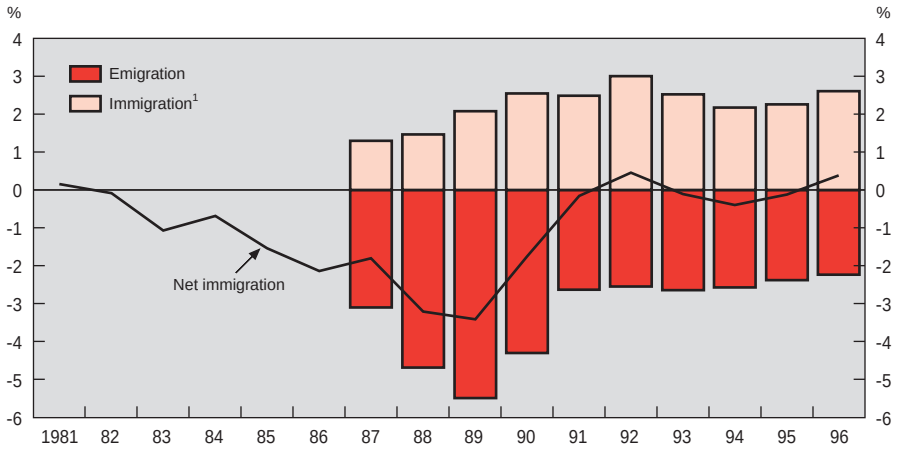
As a result of the relatively low participation rate and high unemployment rate, the employment-to-population ratio is below the OECD average for every category shown in Figure 13. For men in the 55 to 64 age group, the employment-to-population ratio fell from 82 to 59 per cent between 1977 and 1994. Consequently, there were fewer men at work in 1994 than in 1977.

... and net emigration has been substantial

In addition to low participation rates, Ireland has experienced a high level of emigration over the past two centuries. During the 1980s, net emigration totalled almost 180 000 – equivalent to 14 per cent of the labour force – while domestic employment declined by 2 per cent over the decade. Without the ''safety valve'' of overseas migration, unemployment in Ireland might have been substantially higher.[13] Since peaking at 5.5 per cent of the labour force in 1989, gross emigration has stabilised at about 2½ per cent (Figure 18), reflecting improved employment prospects in Ireland relative to the United Kingdom and the United States, the major destinations for Irish emigrants. At the same time, this has boosted the return of Irish workers from abroad, resulting in a net inflow by 1996. The so-called ''Irish Diaspora'' – the estimated 20 million persons of Irish descent living

Figure 18. **INTERNATIONAL MIGRATION**

Per cent of labour force



1. Primarily Irish citizens returning to Ireland.
Source: Central Statistics Office.

abroad – is a factor facilitating emigration, in addition to representing a large potential source of ‘‘immigrants’’. Approximately 3 million of these persons are Irish citizens, of which 1.2 million were born in Ireland.[14] An estimated 2 million reside in the United Kingdom and ½ million in the United States, with the remainder located mainly in other English-speaking countries. A recent survey reported that 10 to 15 per cent of the Irish working-age population has lived abroad for at least one year. Among those with university degrees, the proportion rises to 30 per cent.

The United Kingdom remains the most common destination of emigrants, reflecting its proximity, the large number of Irish persons already living there and a common language. Flows between the two countries have also been facilitated by the freedom from controls on the movement of workers across the Irish Sea. However, the UK share of Irish emigrants declined from 69 per cent in 1989 to 43 per cent during the past three years, while the percentage moving to the United States increased (Table 26). Most emigrants are young persons who leave Ireland for economic reasons. During the past five years, almost two-thirds of

Table 26. **Estimated emigration by destination and age**

Thousands of persons

A. By destination

Year (ending April)	United Kingdom	Other EU countries	United States	Other countries	Total
1987	21.8	3.1	9.9	5.4	40.2
1988	40.2	2.8	7.9	10.2	61.1
1989	48.4	3.9	8.2	10.0	70.6
1990	35.8	5.1	7.7	7.6	56.3
1991	23.0	3.1	4.8	4.4	35.3
1992	17.3	7.7	3.8	5.9	34.6
1993	17.1	7.3	5.9	6.0	36.3
1994	15.4	5.7	9.7	5.1	35.9
1995	13.5	4.9	8.9	6.5	33.9
1996	14.5	4.9	6.0	7.4	32.8

B. By age group

Year (ending April)	0-14	15-24	25-44	45-64	65 and over	Total
1987	2.8	24.0	11.8	1.7	0.0	40.2
1988	8.3	31.2	18.3	3.2	0.0	61.1
1989	7.8	37.0	21.9	3.8	0.0	70.6
1990	6.7	30.8	16.9	1.9	0.0	56.3
1991	4.6	19.9	10.5	0.4	0.0	35.3
1992	0.2	22.1	11.2	1.0	0.1	34.6
1993	0.2	23.3	11.5	1.3	0.0	36.3
1994	0.3	23.8	11.2	0.7	0.0	35.9
1995	0.0	21.8	11.7	0.4	0.0	33.9
1996	0.4	20.3	10.9	1.1	0.0	32.8

Source: Central Statistics Office.

those moving overseas were between the ages of 15 and 25, while most of the remainder were in the 25 to 44 age group. Although emigrants in the past were typically unemployed, this is no longer the case. According to a 1992 survey of emigrants, while 44 per cent went abroad to find work, another 21 per cent left jobs in Ireland to take up employment overseas. Another 10 per cent left to complete their education.[15] Of the unemployed who do emigrate, most have been out of work for less than a year.

Given the importance of international migration, the unemployment rate does not accurately reflect labour market conditions in Ireland. Econometric studies confirm that the level of emigration in any given year depends on labour market conditions in Ireland relative to those in other countries, particularly the

United Kingdom (O'Connell and Sexton, 1994). For example, net emigration was more than 40 000 in 1988, a year when the Irish unemployment rate was more than 16 per cent, about double that in the United Kingdom. With the decline in Irish unemployment coinciding with a rising UK rate during the following four years, there was net immigration into Ireland by 1992. The mobility of Irish workers has resulted in similar trends in unemployment rates in Ireland and the United Kingdom (Figure 13). The gap, which remained constant at about 3 percentage points between 1963 and 1983, has widened considerably since then. Econometric studies also suggest that differences in real incomes have had an impact on net emigration that is broadly equal to that of relative unemployment rates, a result consistent with the surveys showing that many emigrants are seeking better employment abroad. Given the close link between labour market conditions abroad and migration, the concept of an equilibrium unemployment rate consistent with stable wage increases is less relevant for Ireland than for larger countries with labour markets that are less open to international migratory flows.

The fact that those who have completed higher education are more likely to go abroad has generated concern that emigration has a negative ''brain-drain'' effect on the Irish economy. The percentage of third-level graduates who emigrated to find employment increased from 8 per cent in 1980 to over 11 per cent in 1994. However, the rapid convergence in living standards between Ireland and other OECD countries, particularly the United Kingdom, has encouraged emigrants to return after having acquired useful work experience and improved skills. According to the 1992 survey cited above, less than half of those who emigrated in 1986 were still living abroad six years later. Migration is encouraged by the tendency of some employers to prefer hiring returning migrants for technical positions, given their broader experience, rather than recent graduates in Ireland. This pattern of temporary migration thus constitutes an additional method of acquiring human capital.

Policy requirements

The main challenge facing Ireland is to sustain strong employment growth in the coming years in the context of a rapidly growing population and thereby to reduce the unemployment rate. As a result of the high birth rate in the past, the

number of working-age persons is increasing at a 2¼ per cent annual rate.[16] In addition, the upward trend in the female participation rate is boosting the growth of the labour force. In this context, reducing unemployment will require focusing on the problems of the long-term unemployed. This would further necessitate changes in the tax and welfare systems to increase the financial reward for working, especially for those with low levels of skills and education. Regarding the tax system, it is essential to reduce further the high marginal tax rates imposed on low-income workers. In addition, some features of the benefit system discourage the transition from unemployment to work. In particular, the indefinite payment of unemployment assistance at a flat rate and the additional child benefits, without requiring the unemployed to actively look for work and to accept any job offers, weakens work incentives. While the government has implemented a very wide range of Active Labour Market Policies (ALMPs), there is some uncertainty about their effectiveness. It would be beneficial to rationalise the efforts in this area, while concentrating resources on programmes that appear to have the greatest impact on the job prospects of the unemployed. Well-targeted employment subsidies, such as the Back to Work Allowance Scheme, and skill-specific training appear to be most effective in this regard. In the long run, reducing the number of students who leave the secondary school system without qualifications appears to be a key to lowering unemployment. The paragraphs below provide some background analysis to these recommendations for Ireland, which are summarised in Box 4.

Enhancing the ability to adjust and adapt

Taxes and transfers

The interaction of the tax and social welfare systems has weakened the work incentives of many persons, particularly those with low educational qualifications.[17] The government has increased social welfare benefits in real terms since 1980, while taxing relatively low incomes at high marginal rates. The lack of co-ordination between tax and welfare policies reduced the gap between the level of income support available to the unemployed and net income from work, thus making the transition from unemployment to work less attractive and lessening the incentives for low-paid workers to remain in employment. In recent years, priority has been given to closer co-ordination of changes in the tax and benefit systems to reduce these disincentives.

Box 4. **The OECD Jobs Strategy: synopsis of recommendations for Ireland**

The *OECD Jobs Study* sets out a strategy based on nine recommendations for improving the ability of OECD economies to cope with structural change, through enhancing the ability to adjust and adapt, and increasing the capacity to innovate and be creative. The nine distinct policy areas covered included the macroeconomic policy framework and the creation and diffusion of technological know-how. With respect to labour and product market flexibility, it identified the need for initiatives in the following areas: working-time flexibility, the entrepreneurial climate, wage and labour cost flexibility, employment security provisions, active labour market policies, labour force skills and competencies, and unemployment and related benefit systems.

Ireland is in the process of implementing reforms in some of these areas. The detailed review of Ireland's labour market and employment performance and the assessment of policy requirements below suggests the following priority areas for policy action:

The tax system

– *Reduce the high marginal tax rate on low-paid workers.* This could be accomplished by eventually equalising personal allowances and the exemption limit in the income tax system, thus eliminating the 40 per cent ''marginal relief'' rate, which is paid by one-eighth of taxpayers in 1997.

The benefit system

– *Change the rules of the unemployment benefit system to strengthen work incentives* by making the payment of benefits conditional on recipients' willingness to look for jobs, improve their skills and to accept any job offers. In addition, the government should consider introducing a system of decreasing unemployment benefits over time.
– *Reform the child benefits system, which favours the unemployed.* Replacing the three benefits currently provided with one unified payment for all children is one approach that would improve incentives for families with children.
– *Improve the system of housing-related benefits* by eliminating features that contribute to poverty and unemployment traps. Allowing unemployed persons to retain rent and mortgage supplements for a limited time period after beginning work would be one approach. In addition, the link between incomes and rents for local authority housing should be weakened.

(continued on next page)

(continued)

Active labour market policies

- *Rationalise the schemes currently in operation.* The large number of schemes, currently numbering over 30, in a relatively small economy suggests a high degree of overlap and administrative cost. Reducing the complexity of the system should increase its effectiveness and allow cost savings.
- *Integrate the activities of the government agencies assisting the unemployed.* At present, the payment of social welfare benefits by the Department of Social Welfare is separated from the National Training Agency's responsibilities of providing job placement, counselling and placement in programmes for unemployed persons. Integrating the activities of these agencies into ''one-stop shops'' would help ensure that unemployed persons receiving benefits meet the eligibility requirements suggested above and enhance the effectiveness of active labour market programmes.
- *Focus on targeted employment subsidy programmes.* Schemes of this type appear to have increased the number of long-term unemployed who have accepted jobs or become self-employed.
- *Emphasise training programmes that help the unemployed acquire specific skills that are demanded by employers.* A skill-specific approach appears to have had a more positive impact than general training.

Increase labour force skills and competencies

- *Intensify efforts to reduce the proportion of students who leave school without qualifications.* Given the high rate of unemployment among early school-leavers, it is important to increase the share of youth who finish upper secondary school and obtain a recognised qualification.
- *Further increase the resources devoted to schools in disadvantaged areas.* The school dropout rates of children from lower socio-economic backgrounds are relatively high. Using some of the additional resources made available by the decline in the birth rate could improve the prospects of this group.

Enhance the diffusion of technology

- *Encourage the diffusion of technology* from the modern sector to domestic firms.

Enhance product market competition

- *Reducing the role of state-owned firms* through privatisation and by ending their monopoly position would increase competition.

Ireland's income tax system, which is based on the joint taxation of married couples, has a top rate of 48 per cent and a standard rate of 27 per cent. Each taxpaying unit is also granted a personal allowance from income tax based on whether the spouse works and other factors such as the amount of pension contributions and interest paid on housing loans. To avoid imposing a high average tax rate on low incomes, an exemption limit was established in 1977, below which income is not taxed. The exemption limit increases with the number of children in a family.[18] However, such a system would result in a sharp increase in tax liability as one passed the limit and all income above the personal allowance was taxed at the standard rate. To avoid this disincentive to work, a ''marginal relief'' tax rate of 40 per cent was introduced on income above the exemption limit.[19] One result of this system, however, is that low-income workers face a marginal rate of 40 per cent rather than 27 per cent, which weakens incentives. The marginal relief rate was paid by about a sixth of taxpayers in 1996, primarily married persons with children.

In addition to income tax, employers and employees are subject to the Pay Related Social Insurance (PRSI) contribution. The revenue raised from this contribution, which accounts for 12 per cent of gross government receipts, flows into the Social Insurance Fund, which finances unemployment benefits and pensions. In the 1996 budget, an employee's first Ir£ 80 of earnings a week (30 per cent of the average manufacturing wage) were exempted from the PRSI,[20] a change intended to encourage low-skilled persons to work (Table 27). Earnings above that threshold are taxed at a 5.5 per cent rate. A lower rate also applies to employer contributions on wages up to a level close to the average manufacturing wage. Above that threshold, employers pay a rate of 12 per cent. In addition, employees pay an additional 2.25 per cent tax on earnings above Ir£ 188 a week, which includes the Health Contribution (1.25 per cent) and the Employment and Training Levy (1 per cent). Overall, employers contributed 70 per cent of the Social Insurance Fund revenues in 1996, while employees accounted for just over 20 per cent, with the self-employed paying most of the remainder. The combined employer and employee social insurance contribution rate of 17.5 per cent, plus the 2.25 per cent health and employment levy, is slightly below that in the United Kingdom and only about half that in continental EU countries that impose such taxes.[21] The relatively low rate in Ireland reflects the young age structure of

Table 27. **Social insurance contributions**

Per cent of weekly earnings in 1996

	Per cent of average wage in manufacturing	PRSI[1]		Levies[4]
		Employers[2]	Employees[3]	Employees
Below Ir£ 30	11	0.5	0	0
Ir£ 30 to Ir£ 80	11 to 30	8.5	0	0
Ir£ 80 to Ir£ 188[5]	30 to 71	8.5	5.5	0
Ir£ 188 to Ir£ 250[6]	31 to 95	8.5	5.5	2.25
Over Ir£ 250[6]	95	12.0	5.5	2.25

1. Pay Related Social Insurances contributions provide most of the revenue for the Social Insurance Fund, which finances pensions and unemployment benefits.
2. These rates apply to total pay.
3. These rates apply to earnings in excess of Ir£ 80 a week.
4. Levies include the Health Contribution (1.25 per cent) and the Employment and Training Levy (1 per cent).
5. This upper limit was raised to Ir£ 197 in the 1997 Budget.
6. This limit was raised to Ir£ 260 in the 1997 Budget.

Source: Department of Social Welfare (1996).

the population and a flat-rate pension that amounts to only 28 per cent of average earnings.

The average tax rate, including income and employee PRSI contributions, has declined from 35 per cent of average earnings for a single person in 1985 to 28.6 per cent 1996 (Table 28), reflecting the declines in tax rates. An international comparison of the total tax wedge, which includes employer social security contributions, shows that taxes in Ireland are slightly below the OECD average and substantially less than the levels in some continental European countries (Figure 19). Ireland, though, is marked by very high marginal rates at low income levels. In 1996, a single person entered the top tax bracket at an annual income that was almost 10 per cent *below* the level of average earnings. Consequently, such workers face the maximum marginal tax rate, including payroll taxes, of 55.8 per cent. In contrast, in the United Kingdom, which has an average tax wedge that is very close to that in Ireland, a worker begins to pay the top 40 per cent tax rate at an earnings level that is almost double the average production worker's wage. Moreover, low-income workers in Ireland face a high marginal tax rate of 47.8 per cent due to the exemption system described above. A married couple with two children face this high rate until they reach an income only 4 per cent below average earnings, at which point the marginal rate falls to its lowest level of 34.8 per cent.

Table 28. **Income tax and social levies**

Per cent of average gross earnings of a single person

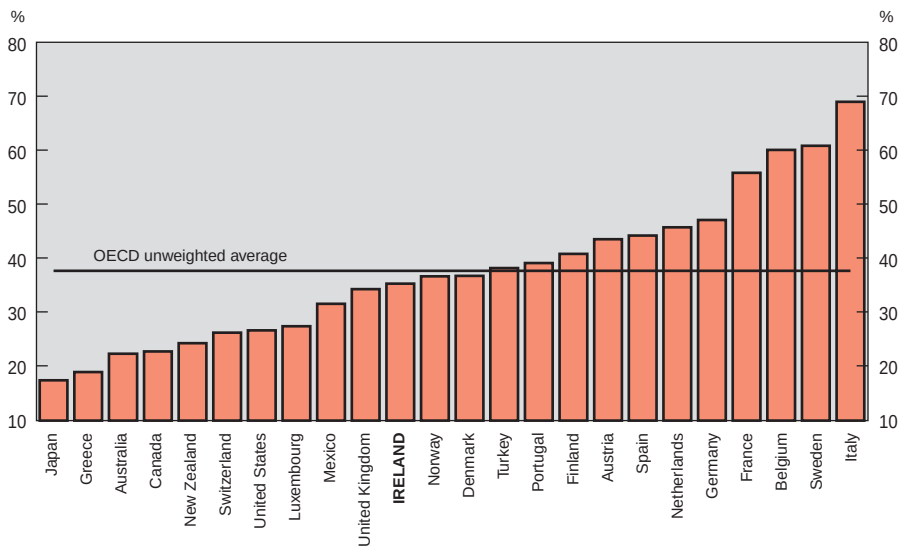
	Average earnings[1] Ir£	Tax rates[2]		Proportion taxed[3] at:		
		Average	Marginal	Zero	Standard Rate	Higher
1985	8 939	35.0	56.5	31.2	50.3	18.5
1986	9 606	34.2	55.5	31.1	48.9	20.0
1987	10 096	35.5	55.8	29.6	46.6	23.9
1988	10 575	34.5	55.8	29.7	53.9	16.4
1989	11 000	33.2	55.8	28.5	55.5	16.0
1990	11 424	32.3	55.8	27.5	56.9	15.7
1991	11 925	32.3	55.8	26.7	56.2	17.1
1992	12 405	30.8	55.8	25.7	60.3	14.1
1993	13 073	32.4	55.8	24.9	58.7	16.4
1994	13 454	30.7	55.8	25.5	61.0	13.5
1995	13 752	29.1	55.8	25.0	64.7	10.3
1996		28.6	55.8			

1. Industrial workers in the manufacturing sector.
2. Includes social security taxation and temporary income tax levies.
3. Income taxation only.

Source: Department of Finance.

Figure 19. **TOTAL TAX WEDGES**[1]

Per cent of an average production worker's income in 1994



1. The total tax wedge includes income taxes and employer and employee social security contributions.

Source: OECD, *Tax/Benefit Position of Production Workers.*

These disincentive effects are compounded by aspects of the social welfare system and its interactions with the tax system. Insurance-based *unemployment benefits* are limited to fifteen months. Those who are unable to find a job within this time period are entitled to receive *unemployment assistance*, a means-tested payment which has no maximum duration. Persons with an insufficient employment record to qualify for insurance-based benefits are eligible for payments under the assistance programme.[22] In theory, benefits under both schemes are conditional on job search by the recipient and willingness to accept any job offers. In practice, however, there is evidence that these conditions have not been rigorously enforced. Indeed, as noted in Box 3, a recent study found that a fourth of those registered as unemployed had withdrawn from the labour force while another 11 per cent were working full time. In addition to the basic flat-rate benefit, both schemes include additional payments for adult dependents, such as non-working spouses, and for children. The government has equalised payments under the two systems by sharply increasing unemployment assistance and by eliminating the pay-related component of unemployment benefits in July 1994. Between 1977 and 1994, unemployment assistance for a married couple without children increased by 58 per cent in real terms,[23] far faster than the 26 per cent rise in net earnings for a worker earning the average manufacturing wage. The rise in payments for single persons and families with children also far exceeded real wage increases. The number of persons getting *unemployment assistance* increased four-fold from 50 000 to 202 000 between 1977 and 1994, while those receiving *unemployment benefits* rose slightly from 55 000 to 66 000.

Compared with other OECD countries, payments to jobless persons are not high in Ireland. A summary measure of the *replacement rate* – defined as the unemployment benefit as a share of previous earnings – was 26 per cent before taxes, somewhat below the average of 30 per cent in the OECD countries for which data are available (Table 29). On an after-tax basis, the replacement rate was 37 per cent compared to an OECD average of 50 per cent. The replacement rate in Ireland, though, varies sharply by family size, given the child dependent allowances of Ir£ 13 per week (about a fifth of the standard benefit) for unemployed persons. At average earnings in 1994, for example, the net replacement rate rises from 49 per cent for a childless couple to 64 per cent for a family with two children. In addition, rates are higher at low income levels. For a childless couple, the replacement rate rises to 67 per cent as earnings fall to two-thirds of

Table 29. **Gross and net unemployment benefit replacement rates**

Benefit entitlement as a percentage of previous earnings

	Summary measures[1]		Net replacement rates for a couple[2] in 1994			
			Average earnings[3]		2/3 of average earnings[3]	
	Gross replacement rate, 1995	Net replacement rate, 1994/95	No children	2 children	No children	2 children
Japan	10	45	43	42	49	48
United States	12	16	60	68	66	60
United Kingdom	18	51	35	51	52	67
Italy	20	19	37	47	36	45
Greece	22	n.a.	n.a.	n.a.	n.a.	n.a.
New Zealand	22	34	48	64	70	80
Ireland[4]	**26**	**37**	**49**	**64**	**67**	**70**
Austria	26	n.a.	n.a.	n.a.	n.a.	n.a.
Germany	26	54	60	71	60	70
Australia	27	31	49	64	66	76
Sweden	27	67	81	84	82	85
Canada	28	47	63	67	64	67
Switzerland	30	62	77	89	75	87
Spain	32	49	75	75	74	78
Portugal	35	n.a.	n.a.	n.a.	n.a.	n.a.
France	38	55	69	71	79	81
Norway	39	62	67	73	66	75
Belgium	42	59	64	66	75	76
Finland	43	59	63	75	67	83
Netherlands	46	69	77	77	79	78
Denmark	71	81	69	73	92	93
Unweighted average	30	50	60	68	68	73

1. The summary measure is an unweighted average of 18 different replacement rates covering two income levels, three duration categories and three types of family circumstances. The gross measure is for 1994 for the United States and 1995 for all other countries. The net measure is for 1995 for Australia, Ireland, Sweden and the United Kingdom and 1994 for all other countries. See Martin (1996) for further information and comparison with other indicators of generosity.
2. Single-earner household.
3. The average earnings of a production worker.
4. The figures for Ireland do not include housing benefits since few unemployed persons receive them. If housing benefits had been included, the summary measure of the net replacement rate for Ireland would have been 58 per cent instead of 37 per cent.

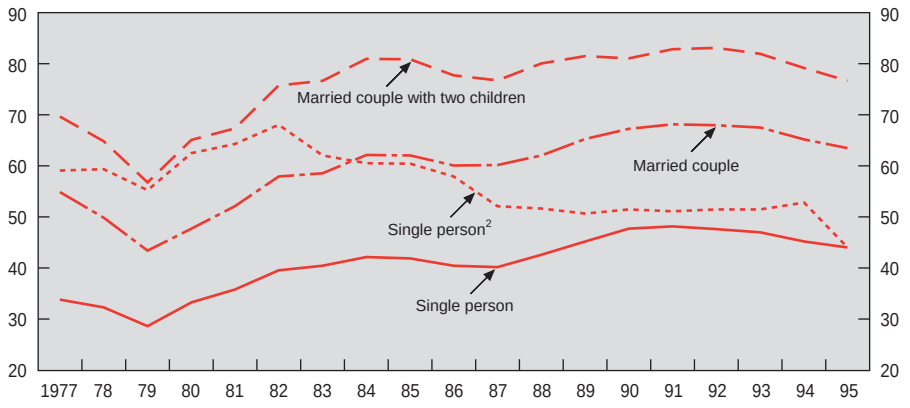
Source: Martin (1996) and OECD, *Employment Outlook* (1996).

the average. Moreover, entitlements are noticeable for their indefinite duration at a fixed level. In contrast, seven of the countries shown in Table 29 pay no unemployment benefits after three years,[24] while another six provide payments at a reduced rate. Given the positive cross-country link reported in the *OECD Jobs Study* between the duration of unemployment benefits and the average length of

unemployment, the indefinite duration of assistance in Ireland is likely to have increased long-term unemployment.

While their level is not especially high by international standards, the sharp increase in unemployment benefits relative to wages in Ireland has affected the functioning of the labour market. Net replacement rates increased sharply between 1979 and 1984, particularly for those earning wages that were less than average (Figure 20). Following a further rise in the late 1980s, rates peaked in 1991. A recent econometric study of seventeen OECD countries concluded that the generosity of unemployment benefits is positively related to the level of unemployment (Scarpetta, 1996). In the case of Ireland, it found that the rise in the net replacement rate was responsible for about half of the estimated 9.7 percentage point increase in the Irish structural unemployment rate between 1971 and 1993. The effect of the rising replacement rate in Ireland was the second strongest among the countries studied and was about seven times higher than the OECD average. The modest decline in Ireland since 1991 has likely reduced the level of structural unemployment.

Figure 20. **NET REPLACEMENT RATES**[1]
At two-thirds of the average manufacturing wage



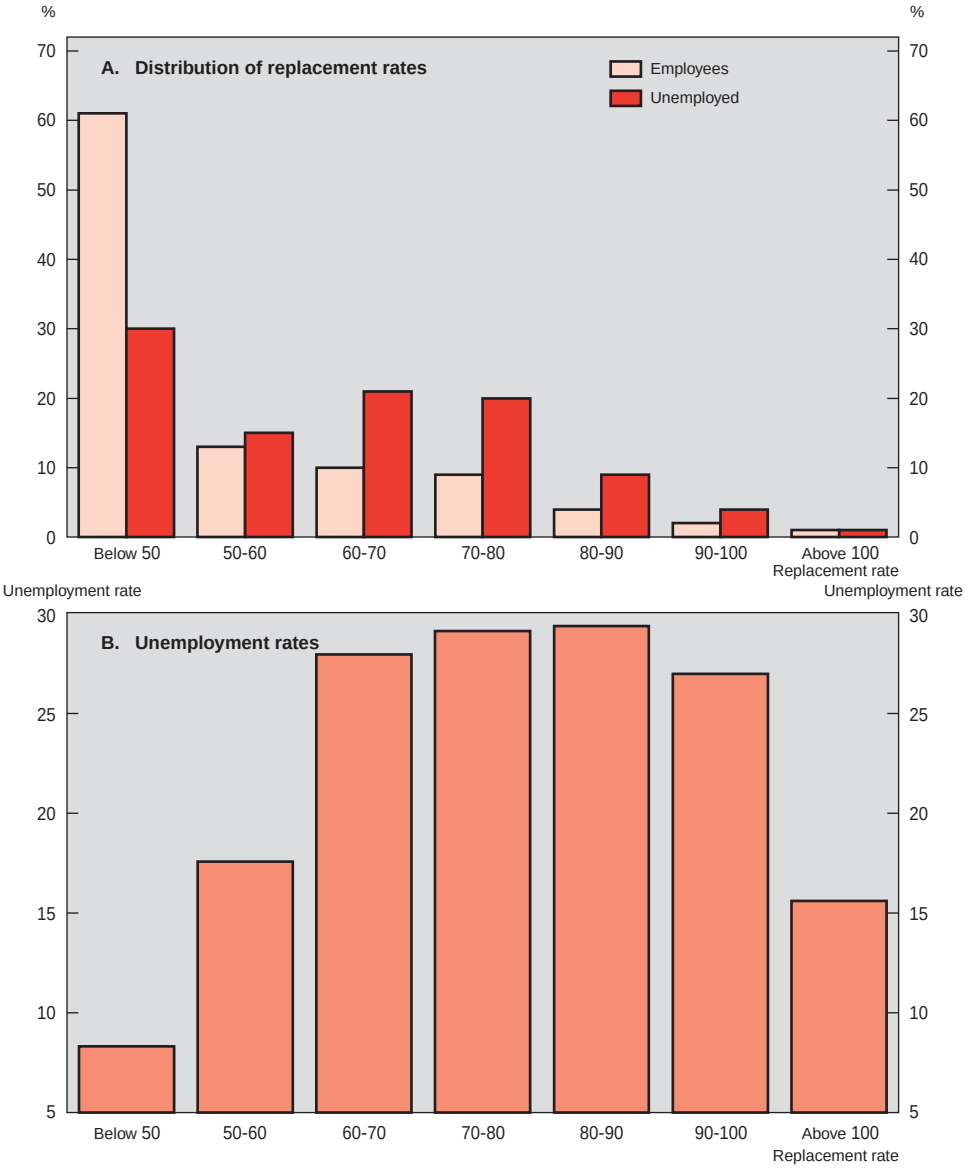
1. Unemployment Assistance as a per cent of earnings.
2. A single person receiving Unemployment Benefits, which included a pay-related component until 1994.
Source: Export Working Group, 1996; 1995, Secretariat estimate.

Although some econometric studies of Ireland have not found a significant impact from replacement rates on unemployment, other evidence, such as the divergent trends in Ireland and the United Kingdom, also suggest a link. In contrast to Ireland, the UK net replacement rate declined steadily from 34 per cent in 1977 to 19 per cent in 1993 (Scarpetta 1996). The Irish unemployment benefit of Ir£ 64.5 per week is now about a third higher than that in the United Kingdom. The relative increase in Irish benefits and replacement rates appears likely to have encouraged some Irish residents of the United Kingdom to return home. This would account for some of the wider gap between the unemployment rates in the two countries since 1983.

Moreover, estimates of net replacement rates for different segments of the Irish population are positively correlated with the level of unemployment.[25] It has been estimated that a little more than half of the labour force face replacement rates of less than 50 per cent (Figure 21, Panel A).[26] Among this group, the unemployment rate is about 8 per cent (Panel B). For the almost 30 per cent of the labour force with replacement rates above 60 per cent, however, the unemployment rate is more than 25 per cent. In addition, more than four-fifths of persons on ''home duties'', primarily women married to men working full time, also face replacement rates above 60 per cent. This reflects, in part, their low earnings potential because of time spent out of the labour force while raising children. In addition, the income-splitting provisions of the tax code mean that a wife faces the same tax rate on initial earnings as that imposed on the last pound of her husband's earnings. The high replacement rate faced by many on home duties contributes to the low participation rate of women.

In addition to its effect on unemployment and participation rates, the interaction of the tax and social welfare systems also reduces the incentives of low-paid employees to seek better-paid jobs or to work longer hours. In some cases, increases in gross income result in declines in net earnings. This so-called ''poverty trap'' occurs at an annual income of between Ir£ 7 000 and 11 000 (50 to 80 per cent of average earnings) for a married couple with two children (Figure 22). This problem is partly due to the Family Income Supplement (FIS), which was introduced in 1984 to provide an additional child allowance to low-paid workers.[27] This programme has reduced replacement rates by ensuring that no family is worse off as a result of leaving unemployment and giving up the Child Dependent Allowance which accompanies unemployment benefits and

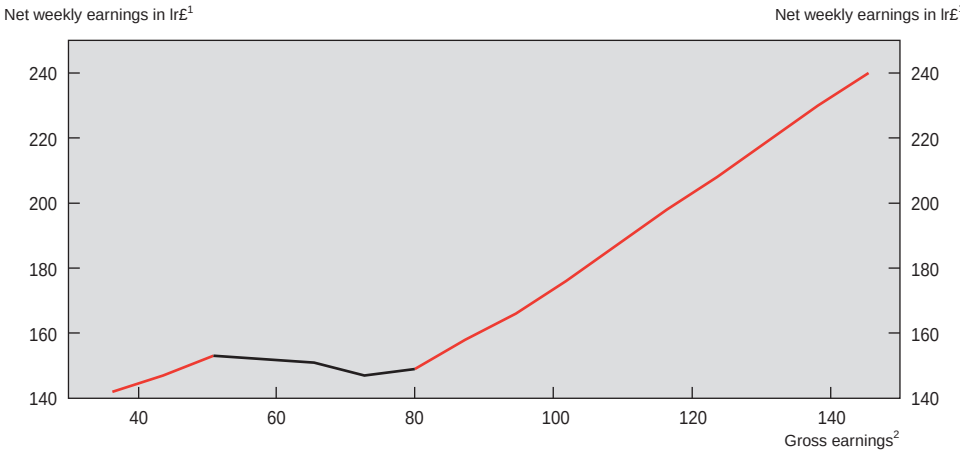
Figure 21. **UNEMPLOYMENT RATE BY LEVEL OF REPLACEMENT RATE**[1]
1994



1. This measure does not include housing benefits and non-cash benefits such as the medical card.
Source: Callan and Nolan, 1996.

Figure 22. **POVERTY TRAP**

Married couple with two children, 1995/96 tax year



1. Includes value of non-cash benefits and assumes that the Family Income Supplement is fully taken up.
2. Gross earnings as a per cent of average wages in the manufacturing sector.

Source: Expert Working Group, 1996.

assistance. At least one parent must work a minimum of 19 hours a week to qualify for the FIS, which varies depending on the level of income and the number of children. As income increases, the benefit is withdrawn at a 60 per cent rate (*i.e.* an Ir£ 1 000 rise in earnings reduces the FIS by Ir£ 600). The phasing out of the benefit, combined with the 40 per cent tax rate, results in an overall marginal effective tax rate of more than 100 per cent over the range identified above for a family with two children.

The extent of the poverty trap, however, is minimised by the fact that only 25 to 40 per cent of the roughly 30 000 eligible families receive FIS benefits. Consequently, only 32 000 children are covered by this scheme (Table 30). The low take-up is thought to reflect uncertainty over the size of the payment and the stigma attached to the certification of income that applicants must obtain from their employer. Moreover, families subject to income tax would face a marginal effective tax rate of more than 100 per cent if they sign up for FIS benefits. Those recipients who pay income tax – an estimated 1 500 to 2 000 families – are subject to the poverty trap. In addition, the FIS recipients who are not subject to

Table 30. **Child income support in 1995**

	Number of children covered	Average annual payment per child (Ir£)	Total budget expenditure (millions of Ir£)
Child Benefit [1]	1 066 000	286	305.3
Child Dependent Allowances [2]	515 000	622	320.5
Family Income Supplement	32 078	717	23.0
Total			648.8

1. Paid for all children, regardless of the level of family income, employment or marital status.
2. Supplementary payments accompanying unemployment benefits, unemployment assistance and disability benefits.
Source: Expert Working Group, 1996.

income taxes are presumably content to earn less than the exemption limit since they would face a marginal effective tax rate of more than 100 per cent on income above that level. However, the time-lag in re-assessing a family's eligibility for the FIS reduces the extent of the poverty trap. Once a family qualifies, it retains the payment at the same rate for a year subject to remaining employed. Moreover, this problem is rare among single persons or married people without children who have no entitlement to FIS.

However, the poverty trap is exacerbated by the withdrawal of other benefits, such as free medical cards and housing-related subsidies.[28] The medical card, which entitles low-income persons to free health care, is estimated to have an annual value of about Ir£ 100 per person (Expert Working Group, 1996). The card is withdrawn when one's income surpasses a given level, which in 1995 was set at about 90 per cent of the tax exemption limit (about half of average earnings).

In addition, housing-related subsidies, which are not included in the replacement rates discussed above, reinforce the disincentive effects of the withdrawal of FIS benefits and the medical card and the marginal relief tax rate. In particular, the ''differential rent'' system relates the cost of local authority housing to the income of tenants. Each local authority has the right to determine its own system subject to the requirement that the proportion of income that a household pays in rent should increase with the level of income. In some areas, though, sharp increases in rent can act as a disincentive to work. In Cork, for example, higher rent would absorb more than 30 per cent of a pay increase for a family earning about half of the average wage. In 1993, about 94 000 tenants (almost one-tenth

of all households) lived in local authority housing. In addition, the Supplementary Welfare Allowance paid Ir£ 54 million (0.2 per cent of GDP) in rent and mortgage supplements in 1994. Payments are almost half as large as the basic unemployment benefit for a single person. Expenditures on this scheme, which increased five-fold between 1989 and 1994, provided rent subsidies to 67 600 persons and mortgage subsidies to 16 500. This programme is limited to the unemployed and those working less than thirty hours a week, thus discouraging participants from accepting low-paid, full-time work. In addition, the supplements are reduced pound for pound by increases in net earnings. This high withdrawal rate discourages recipients from earning more.

Flexibility of employment and working time

The rising participation rate and the significant migratory flows of workers in and out of Ireland suggest a high degree of labour market flexibility. One reason may be that employment security legislation, which can discourage new hiring by firms, is less strict in Ireland than in most European countries. A comparison of sixteen European countries in the *OECD Jobs Study* found that Irish laws in this area were the third least restrictive after Switzerland and the United Kingdom (Table 31). This finding was corroborated by a recent study of legislation covering working time, irregular hours, temporary employment, dismissal protection and minimum wages in EU countries which found Ireland to be the second least regulated country after the United Kingdom (Koedijk and Kremers, 1996). Irish firms are required to give a minimum period of notice of between one and eight weeks, depending on the length of service, before terminating an employee's contract. Moreover, the 1977 Unfair Dismissals Act compels employers to show ''substantial grounds'' before firing employees who work at least eight hours per week and have been with the firm for more than one year. Grounds for dismissing a worker include incompetency, misconduct or redundancy. In 1993, the definition of unfair dismissals was widened and the penalties to firms were increased. Workers can appeal to the Employment Appeals Tribunal, which ruled on almost 5 000 cases in 1995 regarding dismissals, minimum notice and terms of employment. Finally, workers have the right to three weeks of vacation each year in addition to the nine public holidays.

In line with EU directives, Ireland introduced legislation in 1977 that requires firms contemplating making collective redundancies to initiate discus-

Table 31. **Employment protection legislation**[1]

	Regular procedural inconveniences	Notice and severance pay for no-fault dismissals	Difficulty of dismissal	Overall ranking for strictness of protection against dismissals
Ireland	**6**	**3**	**6**	**3**
Other EU countries				
Belgium	4	13	3	5
Denmark	1	11	5	4
France	9	7	6	6
Germany	13	2	12	9
Greece	8	12	10	12
Italy	3	16	15	14
Netherlands	16	1	8	7
Portugal	12	15	16	16
Spain	15	14	13	15
United Kingdom	4	5	1	2
EFTA				
Austria	10	10	11	13
Finland	14	9	4	9
Norway	6	6	14	8
Sweden	11	8	9	11
Switzerland	2	4	2	1

1. The country rankings are based on the situation in the late 1980s. Rankings increase with the strictness of employment protection, *i.e.* 1 is the least strict.

Source: OECD, *Jobs Study*, 1994.

sion with representatives of employees at least 30 days in advance and to notify the Minister for Enterprise and Employment. The purpose of these consultations is to avoid redundancies or minimise the number. Workers who are thus dismissed are to be compensated by the firm or by the government if the company is unable or unwilling to pay. Compensation is set at half a week's pay for each year of continuous employment between the ages of 16 and 41 and one week's pay for each year above that age. While job security legislation in Ireland is judged to be relatively weak, it still appears to have some negative effect on hiring. In a 1994 poll, insufficient flexibility in shedding staff was cited by 62 per cent of firms as an important reason for not employing more workers compared with 50 per cent in 1989 (European Commission, 1995). Among firms with more than 500 workers, however, the share fell from 62 to 46 per cent over the same period.

The rising importance of part-time and temporary work is another element of labour market flexibility in Ireland. Regular part-time workers have been covered by the range of protective legislation since 1991. In addition, some aspects of the tax and benefit systems bias workers and firms towards part-time employment. As noted above, the high 55.8 per cent marginal effective tax rate on relatively low incomes discourages full-time work. Working part time is one way to reduce the marginal rate to 34.8 per cent. Moreover, earnings of less than Ir£ 80 per week were exempted from employee's payroll contributions (PRSI) in 1996, while employers pay a lower rate below Ir£ 250 (Table 27). The *Part-Time Job Incentive Scheme*, run by the Department of Social Welfare, provides a weekly supplement of Ir£ 40 per week (15 per cent of average weekly wages in manufacturing) to previously unemployed persons who accept part-time employment. In addition, all part-time workers are eligible for unemployment payments, subject to their level of income, employment status and history. For example, a part-time employee working three days a week and earning two-thirds of the average hourly wage would receive Ir£ 940 in annual benefits (7 per cent of average annual manufacturing earnings). However, an employee earning the average wage would surpass the maximum earnings level for receiving unemployment assistance after two days of work.[29] Almost 30 000 part-time workers, approximately a fifth of all such employees, receive benefits, which can be regarded as a form of employment subsidy. Another factor encouraging part-time employment is the means-testing for the Adult Dependent Allowance that is paid with unemployment benefits. The spouse of an unemployed person is allowed to earn up to Ir£ 60 per week without losing any of this benefit. The complete withdrawal of benefits above that threshold discourages spouses from working longer hours. Overall, the tax and benefit systems appear to encourage part-time work. A part-time employee with a working spouse would be able to retain about 70 per cent of the corresponding earnings, assuming a 20-hour work week at the average wage rate (Doudeyns 1995). If the spouse is unemployed, the ratio rises to 77 per cent.

The wage formation system

The institutional framework for wage formation has an important effect on an economy's ability to adapt real and relative wages to changes in labour demand and supply conditions. Since 1988, wages in Ireland have been governed by a series of three-year national pay agreements negotiated by the government,

employers and the trade unions. As a result, the standard collective bargaining process has been effectively constrained by these agreements in which workers accepted moderate wage hikes, essentially in line with expected price rises, in exchange for a reduced tax burden and increased government spending on welfare, health and housing. In particular, the 1994 to 1996 Programme for Competitiveness and Work (PCW) committed the government to reduce the burden of taxation on workers with low incomes and to raise the threshold at which the higher (48 per cent) tax rate came into effect. It also established a target of reducing government debt from 97 per cent of GDP in 1993 to 91 per cent in 1997 (see Chapter II). This incomes policy covered both public and private-sector employees. Workers are represented by the Irish Congress of Trade Unions, whose members account for about half of all employees in Ireland. Between 90 and 95 per cent of companies have observed the wage guidelines. The 1991 to 1993 accord contained a ''Local Bargaining Clause'', which allowed firms to grant higher pay in exchange for productivity improvements. In fact, more than two-thirds of the agreements incorporated such a clause. This provision was absent from the PCW[30] which did, however, include an ''affordability'' clause that takes into account firms' ability to pay. Consequently, companies in difficulty have been able to negotiate locally to limit pay increases below the norm.

The central wage agreements have helped to reduce industrial unrest; the number of days lost to labour disputes in 1994 was the lowest since 1923.[31] Moreover, they have had a positive effect in moderating pay settlements during a period of rapid growth, thus preserving Ireland's international competitiveness and helping to keep inflation low (see Chapter I). Despite the central wage agreements in force since 1987, wage dispersion has widened.[32] The real hourly earnings of the lowest decile have stagnated, or even fallen slightly, while those of the top decile have grown by over 4 per cent annually. As a result, the ratio of earnings of the top decile to the bottom for men working full time increased from 3.5 to 5 between 1987 and 1994 (Table 32). The change in the distribution of earnings, though, has been less marked among women. Given the stagnation of earnings at the lower end of the pay scale, the share of the labour force earning less than two-thirds of median earnings is now 23 per cent, one of the highest rates in the OECD area (Figure 23). Low-paid workers are concentrated among

Table 32. **The distribution of earnings in Ireland**

Hourly earnings of employees as a per cent of median earnings

	All employees		Male employees	
	1987	1994	1987	1994
Bottom decile	47.1	46.5	53.0	44.9
Bottom quartile	72.7	67.1	75.7	69.8
Top decile	196.1	225.9	186.5	222.6
Top quartile	136.6	149.6	135.1	147.6
Ratio of top to bottom decile	4.2	4.9	3.5	5.0

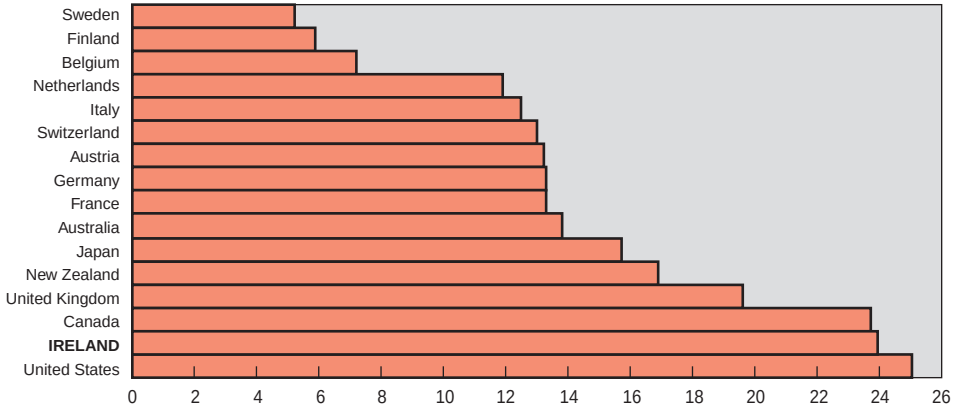
Source: Nolan and Hughes (1997).

the young and among women; more than half are under the age of 25, while under 15 per cent are men in the 25 to 64 age group (Table 33).

Low pay is rarely the cause of poverty in Ireland as most such employees are members of a household with other sources of income. In fact, only 13 per cent of the low-paid are in households that have a net income of less than 60 per

Figure 23. **THE PROPORTION OF THE LABOUR FORCE EARNING LESS THAN TWO-THIRDS OF MEDIAN PAY**

1994 or most recent year



Source: OECD and Nolan and Hughes, 1997.

Table 33. **The distribution of low pay by age and sex**

1994

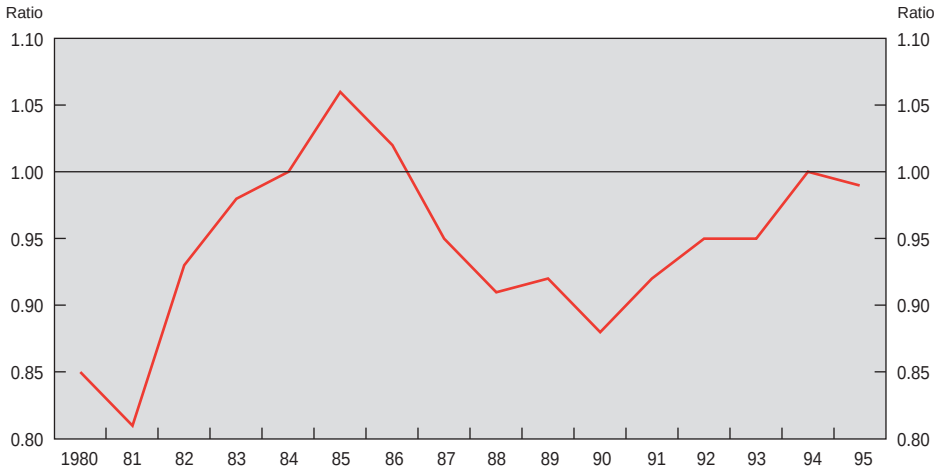
	Low-paid workers as percentage of age group			The age and sex distribution of low-paid workers		
	Male	Female	All	Male	Female	All
Under 25	59.4	62.4	60.8	28.6	26.8	55.4
25-34	11.5	23.7	16.7	8.2	12.9	21.0
35-44	4.4	26.2	11.5	3.1	8.7	11.8
45-54	3.9	30.5	11.8	1.9	6.4	8.3
55 and over	7.3	24.5	12.3	1.4	2.0	3.4
All ages	16.7	35.6	23.9	43.3	56.7	100.0

1. Low pay is defined as two-thirds of median full-time weekly earnings.
Source: Nolan and Hughes (1997).

cent of the national mean (Nolan and Hughes, 1997). Moreover, only 5 per cent of households with income below that level include a low-paid worker, a rate similar to that in the United States and the United Kingdom. Between 1987 and 1994, the poverty rate among households headed by someone in the labour force (*i.e.* employee, self-employed or unemployed) declined. In contrast, the rate has risen among those not in the labour force, who depend primarily on welfare benefits. However, the extent to which the income of these households falls below the poverty line has declined.

Moderate wage growth may also be the result of market forces. Econometric studies suggest that the two dominant factors in determining nominal wage growth are the expected growth of prices and the movement of real after-tax wages in the United Kingdom (Curtis and Fitz Gerald, 1996). The former is closely linked to exchange rate policy, given the open nature of the economy, while the dependence on UK wages reflects the extent of flows of workers between the two economies. Indeed, the ratio of labour costs in Ireland and the United Kingdom has remained close to unity since 1980 (Figure 24). As a result of this linkage, productivity growth in Ireland has little impact on domestic real wage growth. Instead, faster productivity gains reduce costs and raise employment. This process can continue as long as the supply of labour in Ireland remains highly elastic. Moreover, there is little evidence that either the level or the change in Irish unemployment has much effect on the growth of wages. However, some domestic factors do appear to have an impact on wage develop-

Figure 24. **RELATIVE COST OF LABOUR**
Irish industry compared to United Kingdom



Source: Baker, Fitz Gerald and Honahan, 1996.

ments. In particular, income taxes affect nominal wage growth. Given the trend towards an equalisation of after-tax wages in Ireland and the United Kingdom, a reduction in Irish taxes tends to lower wage growth in Ireland. The fall in the rate of unionisation also appears to have exerted some downward pressure on real wages in recent years.

Union membership in 1994 was close to the 1980 figure of half a million. Consequently, its share of the labour force fell from 57 to 53 per cent during that period. This declining trend reflects the growing importance of foreign-owned firms, most of which are not unionised. During 1994 and 1995, for example, only two of the 32 companies that established operations in Ireland recognised unions (EIRR, 1996). Workers in the public sector are much more likely to be organised; in 1993, an estimated 80 per cent belonged to a labour union or staff association compared to only 36 per cent of employees in the private sector. The tradition of more than one union in most firms has made collective bargaining difficult. Moreover, in craft occupations, such as in the printing industry, union membership is a requirement for employment. However, the 1990 Industrial Relations

Act has limited the scope for conflict by establishing the Labour Relations Commission to provide conciliation and other services. In addition, it requires unions to hold a secret ballot to authorise strikes and restricts secondary picketing.

Ireland does not have a legally-imposed national minimum wage. However, Joint Labour Committees, composed of employers and employees, establish minimum rates of pay in some industries, such as clothing, hairdressing, retail grocery and agriculture, where collective bargaining is not well established. These rates apply to about 100 000 workers (about one-tenth of all employees). The rate for waiters, for example, is set at 48 per cent of average earnings in manufacturing, while that of workers in the retail grocery industry is 64 per cent. The rate varies, though, with respect to the age and length of service of the employee and working conditions. Labour unions have argued for the introduction of a national statutory minimum wage to improve the patchy occupational and regional coverage of the current system. Employers, in contrast, favour the flexibility of the present approach.

Active labour market policies

Expenditures on ALMPs[33] have increased from 1.4 per cent of GDP in 1991 to 1.8 per cent in 1995, financed in part by the EU Social Fund, as part of the government's efforts to increase the employment opportunities for the long-term unemployed. Spending on these programmes, which was almost two-thirds as large as the benefits to unemployed persons, was double the OECD average (Table 34). Moreover, expenditure per jobless person in Ireland is significantly above the OECD average. The high level of spending on ALMPs reflects, in part, the importance of public employment, which supports a wide range of public services through voluntary and community organisations. In other countries, such services are usually provided by local government authorities. However, even excluding public employment, Irish spending on ALMPs as a share of GDP is still the fourth highest in the OECD area, reflecting the importance of programmes focused on worker training and employment subsidies (Table 35).

The largest ALMP, with spending equivalent to 0.7 per cent of GDP, is the *Community Employment* programme, which was launched in 1994. It provided 40 000 part-time jobs in 1995 in more than 3 000 projects sponsored by public sector organisations or community groups. Participants in this scheme accounted

Table 34. **Expenditures on active labour market programmes**

(Latest year available)

	Share of GDP (per cent)	Expenditures relative to unemployment benefits (per cent)[1]	Expenditure intensity[2]
Japan	0.1	31.4	0.04
Czech Republic	0.2	106.7	0.05
United States	0.2	51.7	0.03
Luxembourg	0.3	33.8	0.09
Greece	0.4	70.6	0.04
Austria	0.4	25.7	0.06
Switzerland	0.5	37.6	0.10
United Kingdom	0.5	37.6	0.06
Canada	0.6	42.4	0.05
New Zealand	0.7	53.1	0.09
Spain	0.7	27.7	0.03
Portugal	0.7	64.0	0.11
Australia	0.7	44.5	0.08
Italy	0.9	90.3	0.11
Netherlands	1.1	34.6	0.15
France	1.2	60.0	0.10
Germany	1.3	62.1	0.14
Norway	1.4	122.7	0.25
Belgium	1.4	47.3	0.11
Finland	1.6	38.3	0.09
Ireland	**1.8**	**63.4**	**0.15**
Denmark	2.3	50.9	0.23
Sweden	3.0	118.1	0.39
Average (unweighted)	0.9	57.4	0.11

1. Expenditures on early retirement for labour market reasons are included with unemployment benefits.
2. Expenditure as a percentage of GDP divided by the number of unemployed persons.

Source: OECD, *Employment Outlook*, 1996.

for about 3 per cent of total employment. One-fourth of the jobs are reserved for persons older than 35 who have been unemployed for more than three years. These participants have an annual contract that is renewable for up to three years. The remainder of the jobs are for persons over the age of 21 who have been unemployed for more than one year. These posts are for one year, although they can be extended for a second year in some cases. A participant must wait six months from the end of one job to become eligible for another post.[34] All participants are paid an allowance set by the government in place of unemployment benefits. In 1995, the allowance was about 25 per cent higher than benefits for the unemployed. Given the additional expenses for supervision and materials, the cost of the programme per worker is about a third higher than providing them

Table 35. **Active labour market programmes[1] in 1995**

	Expenditure (Ir£ million)	Participants (thousands)[2]
A. Labour market training	**82.8**	
Training for unemployed adults and those at risk	61.7	
Vocational Training Opportunities Programme (VTOS)	24.9	n.a.
FAS[3] Specific Skills Training (SST)	22.3	16.2
FAS Local Training Initiatives	5.0	2.0
DSW[4] Third Level Education Allowance	3.3	1.3
CERT Training for the Unemployed	3.0	1.6
FAS Enterprise Training	1.9	1.2
FAS Return to Work	1.4	1.1
Training for employed adults	21.1	
FAS Training Support Scheme (TSS)	6.7	32.5
FAS Job Training Scheme	6.6	2.9
Development Agency Training Grants	6.1	n.a.
CERT Continuing Training	1.3	5.8
Adult Agricultural Training	0.4	1.5
B. Youth measures	**97.52**	
Measures for unemployed and disadvantaged youth	42.2	
Basic Training for Early School Leavers[5]	13.4	2.1
FAS Training Workshops[5]	13.1	4.0
FAS Community Youth Training Programme (CYTP)	12.6	3.2
FAS Skills Foundation Programme	1.3	0.8
FAS Linked Work Experience	1.0	0.5
FAS Teamwork	0.8	0.3
Support for apprenticeships and general youth training	55.3	
Statutory Apprenticeships	28.5	n.a.
CERT Initial Training	14.3	4.0
Initial Agricultural Training	12.5	1.5
C. Subsidised employment	**374.2**	
Subsidies to regular employment in the private sector	97.6	
DSW Smallholders Unemployment Assistance	35.2	10.1
DSW Back to Work Allowance (BTWAS)	25.2	10.1
Development Agency Employment Grants	23.9	n.a.
DSW Students Summer Job Scheme	10.1	15.8
DSW SI Exemption Scheme	2.0	n.a.
DSW Part-Time Job Incentive Scheme	1.2	0.5
Support for unemployed persons starting enterprises	6.0	
DSW Area Based Enterprise Allowance	4.0	0.8
FAS Community Enterprise Programme	2.0	n.a.
Direct employment creation (public or non-profit)	270.6	
FAS Community Employment	270.6	54.6
D. Measures for the disabled	**35.7**	
National Rehabilitation Board Measures	35.7	n.a.
Total	**698.2**	

1. There is an additional Ir£ 108 million of expenditure on administration and the Public Employment Service.
2. Number of participants entering each programme.
3. National Training Agency.
4. Department of Social Welfare.
5. Part of the Youthreach programme.

Source: OECD.

with unemployment compensation alone. The implementation of this programme, as well as other ALMPs, is facilitated by the 38 Area Based Partnerships established by the government to promote local development.[35] The *Community. Employment* programme differs from previous public employment schemes in three respects. First, it is much larger; the previous programme, the *Social Employment Scheme,* only employed about 15 000 persons a year in the early 1990s. Second, it provides twenty days of training in addition to work experience for those in projects with more than eleven participants. Third, participants are allowed to keep secondary welfare benefits, such as medical cards and rent and mortgage supplements.

The *Community Employment* programme is intended to improve the job prospects of the long-term unemployed by helping them to develop or maintain good work habits. However, previous public employment programmes have not been very successful in persuading firms to hire persons who have been out of work for long periods. A survey of participants in the old *Social Employment Scheme* found that only 18 per cent were employed eighteen months after completing the programme (Table 36). In fact, compared to other ALMPs, those enrolled in public employment schemes had the lowest probability of finding jobs after finishing their programme.[36] While another 27 per cent were enrolled in further education and training or in employment schemes, more than half were unemployed or had withdrawn from the labour force. The initial results of the Community Employment programme appear to be more positive, perhaps reflecting the training aspects and other changes, as well as improved labour market conditions. In 1996, 36 per cent of participants went into employment, while another 30 per cent received further training.

Public training programmes are targeted at early school-leavers and the long-term unemployed. The largest programme is the *Vocational Training Opportunities Scheme* (VTOS), which is intended to give the long-term unemployed a second chance to obtain secondary school leaving certificates. This scheme, which is administered by the Department of Education, enrols about 5 000 persons a year, primarily over the age of 25. In addition, the *Specific Skills Training* (SST) programme, which is run by the National Training Agency (FAS), helps to meet particular skill demands in local labour markets by training the unemployed. In 1995, more than 16 000 persons, who had to meet certain minimum standards of competency to enter the programme, were trained. Almost

Table 36. **Employment experience of participants in ALMPs**

Status of participants 18 months after completing the programme, per cent of total[1]

	At work	Enrolled in education and training/employment scheme	Unemployed	Not in labour force
A. Public employment				
Social Employment Scheme	17.5	27.0	47.8	7.7
FAS Teamwork	44.4	20.4	31.5	3.7
B. Labour market training for unemployed adults				
Vocational Training Opportunities Programme (VTOS)	18.8	28.1	47.7	5.5
FAS Specific Skills Training (SST)	60.3	6.9	24.5	8.3
FAS Local Training Initiatives	40.7	15.5	19.6	24.2
FAS Return to Work	37.4	6.5	23.9	32.2
C. Measures for unemployed and disadvantaged youth				
Youthreach	30.4	6.2	56.5	6.8
FAS Community Youth Training Programme (CYTP)	40.9	20.7	35.0	3.4
FAS Skills Foundation Programme	45.1	16.1	36.3	2.6
FAS Linked Work Experience	47.2	4.2	45.8	2.8

1. Based on a survey from May to August 1994 of 3 200 persons who had completed ALMPs between April and July 1992.
Source: O'Connell and Sexton, 1995.

half of the participants in SST had finished secondary school, compared to less than a third for all ALMPs. For a course to continue, more than two-thirds of the trainees have to find relevant jobs following the training. Participants in FAS schemes receive allowances, which, in total, account for over 40 per cent of the FAS training budget. VTOS and SST are supplemented by the following programmes: the *Local Training Initiative,* which funds activities organised by communities; the *Third Level Education Allowance,* which pays the unemployment benefits of persons over the age of 23 who are admitted to third-level classes after being unemployed for more than six months; and the *Return to Work* scheme, which provides training primarily for women over the age of 25 who wish to re-enter the labour force. In sum, spending on these programmes for unemployed adults and those at risk amounted to almost 0.2 per cent of GDP in 1995.

The government spends a slightly larger amount, about 0.25 per cent of GDP, on training programmes aimed at young people. The *Youthreach* pro-

gramme, which is run jointly by FAS and the Department of Education, is aimed at providing early school-leavers with skills needed for employment and further training. At present, there are 3 500 places in this programme. Those who have finished the initial stage can advance to the *Linked Work Experience* option, which allows participants to work for a private-sector firm while receiving an allowance.

The post-training experience of workers suggests that the return from state-run programmes for the unemployed in Ireland, as elsewhere, is not very high. A year and a half after completing training, less than 20 per cent of the participants in the VTOS were employed, while 28 per cent were enrolled in further training and 48 per cent were unemployed (Table 36). It is striking that the two main programmes to help the long-term unemployed – VTOS and public employment (under the old *Social Employment Scheme*) – yielded almost identical results, suggesting that the quality of the programmes may not be responsible for the disappointing outcomes. Moreover, these disappointing outcomes raise questions about the feasibility of integrating the long-term unemployed in the labour market through training and public employment. The other training programmes, which generally offer a second chance at education, also had low rates of employment after eighteen months. In contrast, 60 per cent of the participants in the SST programme to train the unemployed in specific skills were at work. Even after controlling for the higher education levels of the trainees, it was found that the SST had a substantially greater impact on post-programme employment probabilities than other schemes (O'Connell and Sexton, 1995).

In addition to government-sponsored programmes, there is a considerable amount of training of employees by their firms. A FAS survey[37] found that 43 per cent of private-sector employees (excluding the agricultural sector) went on formal training courses in 1994. The average duration of courses was 3.9 days, implying an average of 1.7 days of training per employee. In addition, another 37 per cent of employees received structured training in the work place, which lasted an average of three days. In total, employees received almost five days of training on average in 1995. Training within enterprises is especially important in the high-tech sector, where employers tend to hire young people with a secondary education or a Regional Technical College degree and than provide firm-specific training. As a result, the traditional statutory apprenticeship is being increasingly by-passed and the number of apprentices has fallen in recent years. Firm-based

training is promoted by seven government programmes for employed adults. The most important is the *Training Support Scheme,* which helped finance programmes for more than 32 000 workers in 1995 in small and medium-sized enterprises (SMEs), internationally-traded services and the distribution sector. The *Job Training Scheme* provides a subsidy to firms to train new employees who were previously unemployed or enrolled in another training programme. Finally, the *Development Agency Training Grants* fund activities associated with new foreign investment. It is likely, however, that a substantial amount of the training in firms that is funded by public money would have taken place even in the absence of government grants. Moreover, such programmes primarily benefit employees who already have a high level of human capital and probably do little to foster the employment or earning prospects of workers with low educational qualifications.

In comparison to public employment and training programmes, Ireland spent a relatively low 0.2 per cent of GDP in 1995 on subsidies for regular employment in the private sector. The largest programme, the *Back to Work Allowance Scheme,* was introduced by the Department of Social Welfare in September 1993 to allow the long-term unemployed to accept jobs while retaining the financial security provided by their social welfare benefits. Participants, who must be at least 23 years old and work more than 20 hours a week, receive 75 per cent of their benefits during the first year, 50 per cent in the second year and 25 per cent in the third year. In addition, they retain their secondary welfare benefits. This scheme reduces the replacement rate of the long-term unemployed, thus encouraging them to take jobs that they would not otherwise accept because of the low wages on offer or, alternatively, to test the option of self-employment. Employers hiring the long-term unemployed under this programme must show that the positions do not displace existing jobs and that they are not temporary. A second DSW programme introduced in 1993, the *Social Insurance Exemption Scheme,* reduces firms' PRSI taxes for two years on the wages of unemployed persons that they hire.

The number of participants in the *Back to Work* scheme approached 17 000 by the end of 1996, with self-employed persons accounting for about three-fifths of the total. The effectiveness of the scheme will depend on what proportion of these workers remain employed after three years when their social welfare benefits are completely phased out, as well as on their future earning

prospects. A survey at the beginning of 1996 (Ronayne and Eason, 1996) found that 81 per cent of the participants were still employed after two years in the programme, at which point they had lost half of their benefits. The survey found other positive results. *First,* the *Back to Work* scheme appears to activate discouraged workers; almost half of the participants had made little or no effort to find a job in the six months before entering the programme. *Second,* 18 per cent of those working under the programme had refused earlier job offers because of the low wages offered (the so-called unemployment trap). *Third,* the programme cut wage costs, thus encouraging employers to hire. About 60 per cent of the firms in the scheme paid lower wages to new employees hired under the scheme. *Fourth,* it promoted the hiring of the long-term unemployed; 72 per cent of the firms said that they would have been unlikely to employ such persons in the absence of the *Back to Work* scheme. *Fifth,* it encouraged persons working in the underground economy to legitimise their activities. The recent study by the Central Statistics Office showing that 11 per cent of the unemployed worked full time (see Box 3) suggests that the underground economy is significant in Ireland.[38] Of course, not all of the additional employment can be attributed to the *Back to Work* scheme. It is estimated that 30 to 40 per cent of the new jobs or self-employment would have been created even without the programme, indicating a significant degree of deadweight loss.

The Employment Services Offices (ESO), administered by FAS, provide a free vacancy notification and recruitment service to workers. In 1994, the ESO was notified of approximately a fourth of private-sector vacancies and about 40 per cent of these were filled by persons referred by the Service. Consequently, only about 10 per cent of all new hires in the private sector found employment through the ESO. However, the number of vacancies referred to these offices has increased sharply during the past two years. In addition to the recruiting service, the ESO provides guidance to unemployed persons as well as information about the education and training courses available. It also administers the FAS work experience programmes, such as *Community Employment.* However, unemployed persons are not obliged to register with the ESO, which has relatively few resources. There are approximately 650 unemployed persons per ESO staff, a figure that is relatively high compared to other European countries.

The large number of schemes in operation may boost administrative costs, which accounted for 15 per cent of ALMP expenditures in 1995. While the

number of programmes reflects an effort to meet the needs of the unemployed, it may lead in some cases to a lack of policy coherence and conflicts, particularly when the government agencies involved do not effectively co-ordinate their activities. This is particularly the case in the area of private-sector employment subsidy schemes administered by the DSW and the FAS. The introduction of the *Back to Work* scheme by the DSW in 1993, for example, led the FAS to discontinue several less-generous employment subsidy programmes the following year. In addition, the payment of unemployment compensation, which is handled by the DSW through its 120 local offices, is separated from the FAS network of 50 offices, which oversee the placement of workers in employment and training. This is one reason that the availability for work and job-search conditions for receiving unemployment compensation have not been effectively implemented.

Increasing the knowledge base, efficiency and innovative activity

Product market competition

Ireland, a small open economy, appears to have a high degree of competition in its product markets. The common external tariff of the EU declined as a result of the WTO agreement, while the level of Irish imports has increased from 55 per cent of GDP in 1990 to 66 per cent in 1995. In addition, the relatively low level of government regulation encourages competition. Indeed, Ireland was found to have the lowest degree of product market regulation in the EU, according to a recent study that examined regulations on business establishment and shop hours, the degree of public ownership and industry-specific support, competition policy and the implementation of the EU single market programme (Koedijk and Kremers, 1995).

There is still room for improvement, though, in the area of regulation. In particular, public utilities remain subject to heavy controls which will have to be re-examined in light of the EU Single Market initiative. In addition, the government has asked the European Commission to allow Telecom Eireann, which is 80 per cent government-owned, to retain control of a major potential competitor, a cable television company. Entry barriers also remain a problem in some areas. In 1995, regulations were introduced that limit the entry of new pharmacies while similar restrictions apply to taxis and public houses. The number of licenses for public houses has not increased in 80 years.

The establishment of the Competition Authority in 1991 has focused attention on preventing anti-competitive practices. Enforcement of the law, which depends primarily on actions by injured parties, appears to have been strengthened with the rising number of private actions. In addition, a main activity of the Authority has been to evaluate requests for exemption from the law. The large number of notifications – 1 347 by the end of 1996 – suggests that a wide range of restrictive practices existed in the past. The Authority had dealt with 1 074 of these requests by the end of last year; most practices have been certified as not violating the 1991 Competition Act, although some were modified to satisfy various concerns. A 1996 amendment to the Competition Act empowers the Authority to begin investigations on its own initiative. Its ability to act, though, is constrained by its relatively small size of four members aided by a staff of thirteen, although there are plans to increase this to 20. Moreover, it does not have its own budget. Instead, its costs are borne by the Department of Enterprise and Employment. In addition, under the Merger Act, the Authority can consider mergers only if requested to do so by the Minister of the Department of Enterprise and Employment. The Authority has made three such reports since 1992. However, an independent commission is reviewing this system and is expected to make a report by the end of 1997.

The privatisation of state-owned firms may also facilitate the emergence of competition. In 1995, the government sold its remaining shares in the insurance company, Irish Life Plc. and, in 1996, disposed of its holdings in Irish Steel Ltd. In addition, a strategic alliance for Telecom Eireann was finalised at the end of 1996, which included the sale of 20 per cent of the government's share to a consortium consisting of the Dutch and Swedish telephone companies. The terms of the sale, which were cleared by the EU Commission, allow the consortium to purchase another 15 per cent of the equity of Telecom Eireann after three years, or alternatively, for the government to sell that amount to the public. There has also been restructuring of a number of loss-making state-owned enterprises. This process has included a focus on the core business through the sale of non-core assets, strategic alliances and a reduction in operating costs through redundancy programmes and changes in production processes. In some cases, the government has injected new equity to improve a company's balance sheet. Restructured companies include the Bord na Mona (the Irish Peat Board) and Aer Lingus.

Technology and innovation

R&D spending has increased rapidly in recent years, almost doubling from 0.8 per cent of GDP in 1989 to 1.4 per cent in 1995. The business sector funded 67 per cent of these expenditures and the government 23 per cent, while institutions of higher education and foreign sources accounted for the remainder. Foreign-owned firms performed about two-thirds of business R&D in 1995. Given the fact that these firms accounted for a similar share of output and are concentrated in such high-tech industries as electronics, computers and pharmaceuticals, their large share of expenditures is not surprising. Overall, foreign firms financed 44 per cent of total R&D spending.

The important role of high-tech foreign firms in Ireland raises the possibility of spillovers that would benefit indigenous firms. However, a 1992 study found that the links between the modern and traditional sectors were weak. The office machinery industry, for example, sourced only 2 per cent of its industrial input from domestic firms. The scope for closer relationships between domestic and foreign companies is limited by the fact that most traditional Irish firms are located in industries such as clothing, textiles, food and beverages. Moreover, the ability of traditional Irish firms to benefit from such technology spillovers depends on their own innovative capacity, access to skilled labour and capital, and knowledge of new production and organisational practices. The government has attempted to encourage ties between Irish suppliers and foreign firms through the *National Linkage Programme.* While there is little direct evidence of closer links, the emergence of a vibrant base of indigenous firms in high-tech industries such as software and biotechnology may suggest a positive impact from the foreign companies in Ireland. Moreover, surveys at the firm level suggest that domestic companies in industries such as food processing are adopting more flexible production and organisation methods (OECD, 1996*c*).

In 1994, the government established the Science, Technology and Innovation Council to improve co-ordination and policy coherence between the government and the private sector. Its 1995 report recommended strengthening R&D in traditional areas, such as food, marine products and timber. Efforts in R&D have been boosted by support from EU structural funds and participation in the EU framework programmes.

Fostering entrepreneurship

Regulations on starting new businesses appear to be relatively less onerous in Ireland than in other EU countries. The European Commission reported in 1995 that administrative formalities took a maximum of 24 days in Ireland compared with up to 42 days in the United Kingdom and 158 days in Germany. The reductions in tax rates in recent years have improved the entrepreneurial climate. There are also government programmes to help unemployed persons to start new businesses. The *Enterprise Training* programme, administered by the FAS, provides an initial period of instruction followed by practical field work that allows potential entrepreneurs to test the feasibility of their business ideas. About 1 200 persons participated in this scheme in 1995. The DSW also provides an *Area Based Enterprise Allowance,* which allows persons starting businesses in designated disadvantaged areas to continue receiving unemployment compensation for one year. After that, entrepreneurs may transfer to the *Back to Work* scheme, which phases out benefits gradually. Finally, the *Community Enterprise* programme provides start-up grants and continuing support for up to a year for community-based groups and worker co-operatives that establish new enterprises.

The recommendations of the Task Force on Small Business have helped improve the environment for SMEs. The role of the County Enterprise Boards has been expanded to assist small firms, while their access to bank finance has been improved. In addition, the government has introduced the *Small Business Operational Programme* to assist small firms. This plan, which covers the period 1995 to 1999, envisages the expenditure of Ir£ 53 million (0.03 per cent of GDP), of which Ir£ 20 million is to be provided by the private sector. The programme is intended to improve further the access of small firms to financial resources and to government procurement plans. In addition, it will provide technical assistance and promote worker training and the adoption of best practices among small businesses. As noted above, the *Training and Support Scheme* helps finance the training of workers in SMEs.

Upgrading skills and competencies

Over the past three decades, Ireland's education system has improved significantly (see the detailed analysis of this topic in the 1995 Survey). In 1994, the participation rate in secondary education was relatively high; 83 per cent of

17 year olds in Ireland were enrolled in formal education compared to the OECD mean of 78 per cent. In addition, with the expansion of higher education, the proportion of young workers with a scientific or engineering degree in Ireland was the highest in the OECD. Despite a much higher number of students relative to the adult population, public expenditures on education were not much above the OECD average, indicating a relatively efficient use of resources in this area. However, the level of educational attainment of the population as a whole is still low compared with other OECD countries, reflecting the relatively late introduction of free secondary education in Ireland in 1968. Consequently, secondary education was not available to a significant proportion of today's adult population. In 1994, 55 per cent of the population between the ages of 25 and 64 had completed only lower secondary or primary school compared to an OECD mean of 41 per cent (Table 37). Moreover, upper secondary was the highest level of education completed by 27 per cent of Irish adults, less than the OECD mean of 39 per cent.

A major weakness of the education system is a higher variation in educational attainment across primary schools than is found in most other countries (see 1995 Survey). This is one reason why a significant proportion of children enter secondary school without the necessary skills. The problem is concentrated in disadvantaged areas; young people from lower socio-economic backgrounds are almost five times more likely to leave school with low qualifications and have a 50 per cent probability of being unemployed one year after leaving school. Overall, 18 per cent of students fail to finish secondary school As noted above, these early school-leavers account for 65 per cent of unemployed persons.

The traditional apprenticeship programme, which was based on the length of time served, was replaced in 1995 by a system aimed at the achievement of specific standards. The required skills, which were developed after an extensive period of consultation with the appropriate industries, are acquired through a seven-phase training programme, four of which are on-the-job. Trainees are able to update their skills through participation in different modules. Training is certified by the FAS and international institutions. The number of registered apprentices declined from 13 000 in 1990 to 11 800 in 1996, reflecting the shorter duration of apprenticeships. About 4 000 persons entered apprenticeships in both years.

Table 37. **Educational attainment**

Percentage of the population 25 to 64 years of age
by the highest completed level of education, 1994

	Lower secondary education[1]	Upper secondary education	Non-university tertiary education	University-level education	Total
North America					
Canada	26	28	29	17	100
United States	15	53	8	24	100
Pacific Area					
Australia	50	27	10	13	100
New Zealand	43	34	14	9	100
European Union					
Austria	32	60	2	6	100
Belgium	51	27	12	10	100
Denmark	40	40	6	14	100
Finland	36	44	9	11	100
France	33	50	8	9	100
Germany	16	62	10	13	100
Greece	55	27	6	12	100
Ireland	**55**	**27**	**10**	**9**	**100**
Italy	67	26	a	8	100
Netherlands	40	38	a	21	100
Portugal	81	8	3	7	100
Spain	74	11	4	11	100
Sweden	28	46	14	12	100
United Kingdom	26	54	9	12	100
Other OECD countries					
Czech Republic	27	63	x	10	100
Norway	19	53	11	16	100
Switzerland	18	61	13	8	100
Turkey	80	13	a	7	100
Country mean	**41**	**39**	**8**	**12**	**100**

1. Includes those who only finished primary school.
Source: OECD, *Education at a Glance* (1996).

Recent policy initiatives

The 1996 budget included significant changes in the tax system, which reduced replacement rates, especially for low-income workers. In addition, it reformed the benefit system so as to improve work incentives. The rise in unemployment benefits was kept below the increase in average earnings. In addition, the universal child benefit was increased by 7 per cent, while the child dependent allowance paid to unemployed parents was left unchanged for the

second straight year, thus making employment relatively more attractive. Moreover, long-term unemployed persons who start work can retain the dependent allowance for three months, at which point they should begin to receive the Family Income Supplement. In addition, they are now allowed to keep their free medical card for three years after becoming employed, irrespective of their level of income, rather than for only two years.

The 1997 budget, however, reversed some of these changes (see Chapter II). Most notably, the rise in social benefits was twice the rate of inflation. As a result, replacement rates for those without children have increased by about 3 percentage points in the case of workers earning half the average wage. However, a marked rise in universal child benefits has reduced the rate somewhat for families with children. In addition, eligibility for the FIS will be based on income net of social security contributions and levies. By 1999, FIS will be calculated net of income tax as well. As recommended by the Expert Working Group on the Integration of the Tax and Social Welfare Systems, personal allowances were increased by more than the exemption limits. In addition, the threshold for the 48 per cent marginal rate was raised by about 6 per cent while the standard rates for income tax and employee's PSRI were both cut by 1 percentage point (to 26 and 4.5 per cent, respectively). The 1997 tax reductions were distributed more broadly than previous budgets, which had focused more on low-paid workers. The top marginal tax rate fell from 55.8 to 54.8 per cent, while the lowest rate was reduced from 34.8 to 32.8 per cent. In addition, the rate paid under the marginal relief rate has declined from 47.8 to 46.8 per cent and the number of workers subject to this rate is expected to fall by a fourth to about 12 per cent of all taxpayers. These reductions, combined with a new lower corporation tax rate of 28 per cent on the first Ir£ 50 000 of company income, should improve the entrepreneurial climate.[39]

The individual income tax cuts were part of the new three-year central wage agreement, the Partnership 2000 for Inclusion, Employment and Competitiveness, which was announced in December 1996 (see Chapter II). The pact allows wages to rise by 7¼ per cent between 1997 and 1999. In addition, there is a ''local bargaining'' clause, which allows unions to negotiate an additional 2 per cent wage hike depending on the circumstances of the particular firm. The addition of this provision, which was absent from the PCW, should allow a somewhat greater degree of relative wage flexibility. Finally, it was agreed that

consideration would be given to alleviating the unemployment trap resulting from the withdrawal of the mortgage and rent supplements.

The 1997 budget also added 5 000 places to the existing 17 000 in the *Back to Work Allowance Scheme*. In addition, new ALMPs have been implemented during the past year. FAS introduced *Jobstart*, another employment subsidy, which pays Ir£ 80 a week to employers who hire persons who have been unemployed more than three years. Although the take-up of this subsidy was initially lower than expected, the target of 2 000 participants by the end of 1997 is likely to be achieved. A second new programme introduced in 1996 by FAS is *Workplace*, which allows the unemployed to work for up to five weeks while keeping their unemployment benefits, in addition to being reimbursed for travel costs. This programme, which has 5 000 positions, is targeted at persons who have been unemployed for more than six months and are at risk of drifting into long-term unemployment. There are few administrative formalities for employers and they are not obliged to guarantee continued work after the five weeks are over.

A third new initiative is the establishment of fourteen *Local Employment Services* offices under the direction of the Department of Enterprise and Employment to assist the long-term unemployed. The offices, which are placed in areas designated as disadvantaged by the government, are intended to work with local community groups and employers to help integrate the long-term unemployed into the labour market. In addition, they are supposed to provide intensive guidance, counselling and job preparation assistance to these persons. In essence, the *Local Employment Service* provides intensive assistance to the long-term unemployed in disadvantaged areas to supplement the ESO service available to all job-seekers. In this regard, their intended function is similar to a new programme in Denmark, which draws up individual action plans for each person after three months of unemployment, and *Restart* in the United Kingdom. Both of these schemes have been credited with increasing the employment prospects of the long-term unemployed. However, in contrast to Denmark and the United Kingdom, the Irish scheme is completely voluntary. The size of the *Local Employment Services* office network is significantly smaller than the original proposal by the National Economic and Social Forum to expand the Employment Service staff by 800 positions at a cost of Ir£ 30 million (0.1 per cent of GDP). Given that there are approximately 100 000 long-term unemployed in Ireland,

this would have provided one counsellor for 125 unemployed persons. The government instead decided to allocate Ir£ 6 million to develop the programme on a pilot basis in the most disadvantaged areas. The service was recently extended to four new areas.

The Department of Education has also introduced programmes, such as *Early Start* and *Breaking the Cycle of Disadvantage,* to help schools respond more effectively to the needs of students from lower socio-economic backgrounds. *Early Start* is a one-year pre-school programme targeted on children deemed to be the most at risk of educational failure. The second programme provides grants for the purchase of teaching materials and equipment by disadvantaged schools as well as a reduction in student-teacher ratios during the first four years of primary school. The Department has also implemented reforms to encourage students to finish secondary school. These changes are intended to help schools respond more adequately to the diversity in the abilities and aspirations of students. *First,* the Leaving Certificate Vocational Programme has been restructured to increase its vocational dimension. *Second,* an ''applied'' option has been added to the leaving certificate. This new programme will qualify a student for vocational training and education, but not directly for third-level schools. *Third,* a transition year, which includes work experience, has been added at the beginning of upper secondary school. Finally, the government has assumed further growth in third-level enrolments in its planning for higher education. Despite the decline beginning in 2000 in the number of young people finishing secondary school, enrolment is projected to increase from 90 000 in 1994 to 121 000 in 2010 as a result of higher participation rates and an increased number of mature students.[40] Fees for third-level education were abolished in 1996.

Finally, in October 1996, the government issued a White Paper on Science, Technology and Innovation which recommended the establishment of new national structures to oversee public support in this area. The report, noting a lack of R&D capability in traditional firms, called for measures to strengthen education and training, improve public awareness of science and technology and improve public research, particularly in natural resource-based industries. The report, though, did not establish any priorities or targets for public expenditures on science and technology. The government allocated Ir£ 4 million in 1996 to begin implementing these proposals.

Assessment and scope for further action

Ireland has experienced the largest fall in unemployment of any OECD country between 1993 and 1996, a period of generally high and stable unemployment rates in the OECD area. The decline is linked to Ireland's strong economic expansion, which has generated an unprecedented increase in employment. The growth rate of employment in Ireland during the past three years has been the second highest in the OECD area. The decline in unemployment has also been aided by structural factors, such as the improvement in the educational level of the working-age population and changes in the tax and benefit system which have reduced replacement rates, thus improving work incentives. Finally, moderate wage growth, supported by central wage agreements, and the falling rate of unionisation may have had a favourable impact.

Despite this progress, Ireland's unemployment rate, estimated at 11¼ per cent in 1996, is considerably above the OECD average. Moreover, a large share of the unemployed have been out of work for more than one year and many have effectively withdrawn from the labour force, although they continue to receive unemployment and social welfare benefits. In addition, the employment-to-population ratio is low, reflecting a female participation rate that is significantly below the OECD average. Finally, there is a serious youth unemployment problem concentrated among early school-leavers.

Government policies have been intended to reduce unemployment through improving incentives to work. To accomplish this objective, benefits that are conditional on employment have been targeted at persons with high replacement rates. However, two decades of piecemeal change have generated a system where the incentives at the lower end of the income scale are weak despite some improvements in recent years. Many of the unemployed have limited prospects of significantly boosting their net income by working. Some of those who do work face very high effective marginal tax rates.

High and persistent unemployment has created pressure for public spending on ALMPs. As a result, there are more than 30 such programmes with total expenditure amounting to 1.8 per cent of GDP. A rationalisation of ALMPs, coupled with further reform of the tax and benefit systems, would appear to be necessary. In short, a fundamental shift in approach – comparable in magnitude to the sweeping changes that were successfully implemented in fiscal policy in

the 1980s – appears necessary to resolve the unemployment problem. The labour market, as well as economic prospects more generally, have improved sufficiently to make such an approach conceivable.

At present, the objective of providing an adequate income for those who are unemployed conflicts with ensuring that individuals will benefit from accepting jobs at relatively low pay. The 1996 report of the Expert Working Group on the Integration of the Tax and Social Welfare Systems established five guidelines in this regard: *i)* there must be a reward for working; *ii)* the transition to work must be facilitated; *iii)* tax on the lower paid should be reduced; *iv)* the tax and social welfare systems should be simplified; and *v)* tax and social welfare reforms should be co-ordinated.

To strengthen the incentives to work, it is essential to reduce the high effective marginal tax rates imposed on low-income workers. One possible approach in this regard would be to continue increasing the personal allowances more rapidly than the exemption limit in the income tax system, thus reducing the share of taxpayers subject to the 40 per cent marginal relief tax rate. According to the Expert Working Group, the complete elimination of the marginal relief system would have reduced the replacement rate in 1995 by as much as 4 percentage points for a married couple with two children earning between Ir£ 11 000 to 14 000 a year (80 to 102 per cent of average earnings in manufacturing). However, the cost of this approach, in terms of lost tax revenue, must be carefully weighed against the benefit of improved incentives.[41]

The issue of unemployment benefits also needs to be addressed. At present, benefits to jobless persons are viewed as long-term income support rather than temporary assistance to workers between jobs. The authorities should consider allowing benefits to decrease over time, a policy which would help reduce the wage expectations of the long-term unemployed. In addition, the unemployed should be required to search for work and to accept any job offers in return for continued receipt of benefits. Several studies in other OECD countries suggest that programmes to strengthen the conditionality of benefits by ensuring that unemployed persons are looking for work and to help them search effectively reduce the extent of unemployment. This is especially the case when job placement services are accompanied by effective monitoring of the job-search activity of the unemployed. Moreover, such studies also suggest that the lifetime employment prospects of people in such programmes are significantly improved.

Imposing such conditions would require closer co-ordination between DSW and FAS, an area where some progress has been achieved. The two agencies currently run separate networks of offices to accomplish their assigned tasks. Studies of other OECD countries suggest that the separation of benefit administration from the public employment service functions of job referrals and placement in ALMPs is inefficient. Moreover, greater integration of these activities in ''one-stop shops'' in Ireland would facilitate the increased monitoring necessary to enforce eligibility requirements. Such offices could also integrate an expanded network of *Local Employment Services* offices, which are intended to provide intensive support services to the long-term unemployed. Finally, with unemployment benefits in Ireland having increased to near the OECD norm, their level should not continue to rise more rapidly than net income in work, as they have during much of the past fifteen years and again in 1997. In addition, unemployment benefits available to part-time workers, the majority of whom are not seeking to work longer hours, should be reconsidered.

Child benefit payments in Ireland also create disincentives to work since they are larger for unemployed families than for those at work. One possible solution would be the payment of an enhanced, universally-available child benefit, financed in part by the elimination of both the FIS and the Child Dependent Allowances that accompany unemployment support. If this universal benefit were taxable,[42] it would cost about Ir£ 200 million (0.5 per cent of GDP) and would need to be offset by reduced spending elsewhere or a smaller reduction in taxes. Alternatively, if adjustments on this scale were not feasible, improvements in the FIS benefit available to low-income earners could be introduced. In that regard, the recent decision to calculate FIS on income net of social security contributions and levies will improve incentives. However, since this programme has had a limited impact to date, with only 40 per cent of those eligible receiving these payments, measures would be needed to improve take-up.

In addition, the withdrawal of housing-related benefits can contribute to poverty and unemployment traps. One possible solution would be to let unemployed persons retain rent and mortgage supplements for a period of time after beginning work, followed by a phased withdrawal. Moreover, the disincentive effects resulting from the link between income and rents for local authority housing should be alleviated.

As noted above, attempts to address the unemployment problem have been pursued sequentially and partially, leading to a proliferation of government schemes in this area. Many of the existing programmes appear to overlap. Consequently, different agencies administer schemes aimed at the same groups of people, making unclear the division of responsibility between the FAS employment service the Department of Enterprise and Employment and the Department of Social Welfare. Moreover, the complexity of the system may lead to administrative failure. For example, by August 1996, only 50 families had taken advantage of the scheme included in the 1996 budget to allow long-term unemployed persons who start work to retain the child dependent allowance for three months. A major effort to rationalise and simplify ALMPs may lead to greater effectiveness, as well as reduced administrative costs. In addition, there is a need for more rigorous analysis to determine which programmes are most effective in order to help the government to concentrate its efforts in areas where they will have the most benefit. Employment subsidies that are carefully targeted are best placed to reduce long-term unemployment. In terms of training, programmes aimed at specific marketable skills, such as the SST, seem to have the most positive effect.

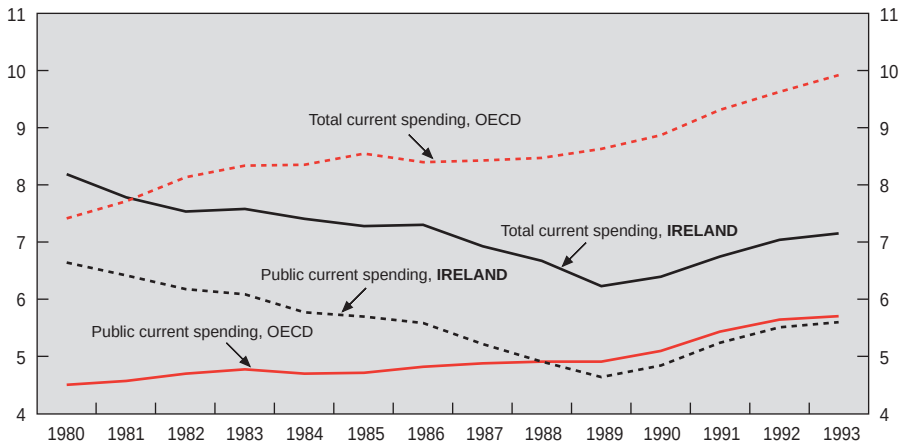
Improving the education and training system is another key to reducing long-term unemployment, which is concentrated among persons who failed to complete secondary school. The innovations introduced during the past few years should help schools to cater to the wide diversity in students' abilities and aspirations. In addition, utilising some of the resources that are being made available by the decline in the birth rate to improve standards in schools with a high concentration of socially-disadvantaged children should provide a high return to society. There is also a need for standardised tests on a school-by-school basis. Such transparency could intensify pressure on schools to improve their performance and help identify areas in need of additional resources. Another way to increase quality would be to boost the time spent on core subjects; Irish secondary schools devote a relatively low proportion of teaching time to mathematics and science compared to other OECD countries (OECD, 1996*a*). Finally, the government should be cautious in expanding the capacity of third-level education, given the projected decline in the number of secondary school leavers and the need to improve schools in disadvantaged areas.

IV. The health care system

The Irish health care system has a unique structure. It is a mixture of a universal health service, free at the point of consumption, and a fee-based private system where individuals have to subscribe to insurance if they wish to be covered for their medical expenses. There appears to be a general social consensus in Ireland that, apart from the most economically disfavoured parts of society, first-level medical care, such as general practitioners should not be a publicly provided service. However, while until 1991 higher income groups were expected to provide for a significant part of the cost of their own hospital care, since then all residents are eligible for free state hospital care subject to a modest daily charge. Despite this widening in provision of public medical care, the costs of the government health system have remained broadly under control, with the private sector expanding and now facing a more competitive environment following the ending of the monopoly of the government-owned health insurance company. Reflecting this, public health spending is in line with the OECD average (Figure 25), and private spending is markedly below average. While health status indicators have improved substantially, they remain relatively low compared to most other European countries, with a lower life expectancy suggesting that health care outcomes could be improved.

In this chapter, the basic structure of the Irish health care system as presently designed is first described. This is followed by a review of the operation of the system, both in a national and international context, and by an examination of the major policy initiatives that have been launched to reform the system during the past decade. The paper concludes with a discussion of key challenges currently being faced by the system, including attempts to improve the organisation and financing of the acute hospital sector, the restructuring of medical training and the introduction of competition within the private health insurance market.

Figure 25. **HEALTH EXPENDITURE**
Percentage of GDP



Source: OECD Health Database.

Main features of the system

Components and operation

Since the foundation of the Irish State, legislation relating to the health services has been based on a recognition of ‘‘the obligation to ensure a high standard of health and medical services for the whole population and equitable access to medical services’’. As the capacity to prevent illness has increased over time, the quality of medical service provision has improved and health service eligibility has been expanded throughout the population. Legislative initiatives after the foundation of the State, and particularly the Health Service Act of 1947, were concerned with controls over food hygiene and infectious diseases. The expansion of the maternity and child medical services was the subject of legislation in the 1950s and the fluoridation of the public water supplies and controls on the distribution of drugs derived from legislation in the 1960s. The health legislation of the 1970s and 1980s was very much concerned with the organisation, management and financing of the health service system, together with issues relating to health education and promotion. In the 1990s, a reorganisation of

some elements of the health care system has been on the agenda. Competition was introduced into the market for private health insurance by the 1994 Health Insurance Act.

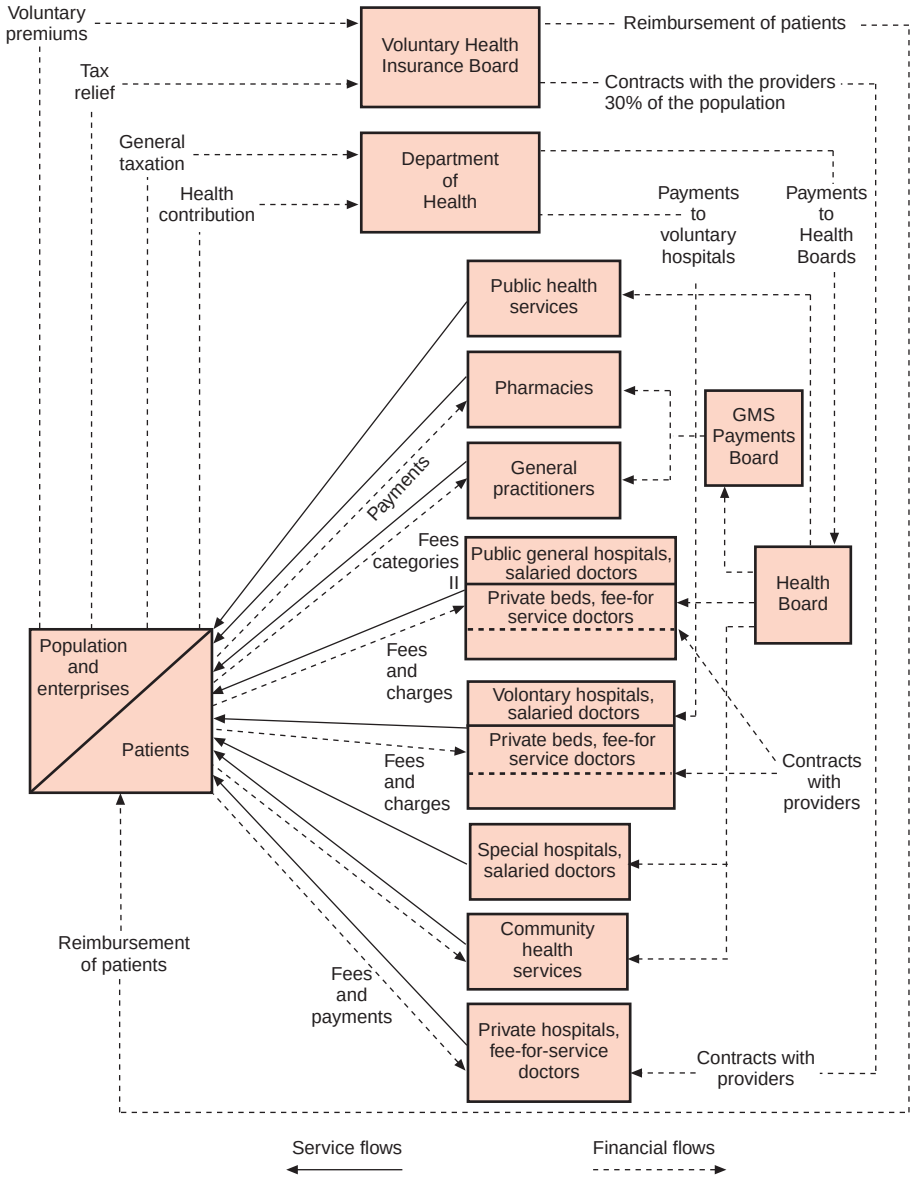
Irrespective of the changes which have been introduced in recent decades, one of the characteristics of the Irish health care system has been that all of those below a relatively low income level have had free access to all health services which are mainly funded out of general taxation. Currently such free access is granted to about one-third of the population, though this number has been falling slightly as more people become employed. The remaining two-thirds of the population must pay privately for general practitioner and pharmaceutical services, but they are entitled to public hospital services on payment of a modest statutory charge. Despite this, about two-thirds of the better-off category of the population continue to purchase private medical insurance. A thriving private health industry therefore exists, covering one-third of all acute hospital beds and two-thirds of general practitioner care. The full linkages and complexity of the Irish health system is best described in a diagram (Figure 26) in which the population is on the left, service providers are on the right and the third party payers, including the Department of Health and the Voluntary Health Insurance Board are at the top and where the broken lines represent financial flows and the solid lines display service flows.

Public health care

Under the Health Act of 1970, a regionalised system of health service provision was introduced with the establishment of eight health boards. The populations served by these boards range from 200 000 to 1.2 million, with annual budgets ranging from Ir£ 100 million to Ir£ 399 million in 1997. Membership of the boards include elected local representatives, ministerial nominees and delegates of consumer and health professional organisations.

The operation of the health board structure is essentially built around three core programmes: general hospitals; special hospitals; and community care programmes (Table 38). Depending on the size of the health board, the *general hospital programme* may incorporate regional, general or community and district hospitals. Regional hospitals are located in large centres of population and provide a comprehensive range of services. General or county hospitals have consultant-staffed units for the provision of services to the immediate catchment

Figure 26. **THE SYSTEM OF HEALTH CARE IN IRELAND**



Amended from OECD, "The Reform of Health Care, A Comparative Analysis of Seven OECD Countries", *Health Policy Studies,* No. 2, Paris, 1992, Chart 6.1, p. 74.

Table 38. **Structure of public health expenditure by programme**

	1980	1984	1990	1994	1994	
	Ir£ million				Per cent of total	Per cent of GNP
General hospitals	393	593	801	1 147	50.1	3.7
Special hospitals	153	262	320	438	19.1	1.4
Handicapped	60	113	152	223	9.7	0.7
Psychiatric	93	139	169	216	9.4	0.7
Community care	150	254	383	613	26.8	2.0
Health	90	147	231	371	16.2	1.2
Welfare	47	50	127	201	8.8	0.6
Protection	13	18	24	40	1.7	0.1
Support	35	58	71	93	4.1	0.3
Total	731	1 167	1 575	2 291	100	7.3

Source: Department of Health.

area. District hospitals contain small units staffed on a part-time basis by general practitioners. The *special hospital programme* is essentially concerned with service provision for the mentally ill and the mentally handicapped. Following a major review[43] of the psychiatric services in 1984, care of the mentally ill has been reoriented away from the large psychiatric hospitals to include more care in acute general hospitals and in the wider community. Health boards enter into agreements with a wide range of voluntary organisations, particularly those caring for the mentally handicapped, for the provision of designated services.

When the health boards were established in 1971, the voluntary public hospitals were allowed to maintain their direct financial links with the Department of Health. Many of these hospitals have traditionally been run by religious orders and some function as teaching hospitals. Alternatively, voluntary public hospitals may be incorporated by charter or statute and work under lay boards of governors. The greatest concentration of these hospitals is found in Dublin and other large centres of population.

The *community care programme* within the health board has three components: community protection; community welfare; and community health. Responsibilities for *community protection* include the prevention of infectious diseases, food hygiene, child health examinations and health education. *Commu-*

nity welfare provision is more assistance than health related and covers a wide range of services including provision for the disabled, home helps, meals-on-wheels, support for homes for the elderly and home nursing services.

The *community health* programme includes general practitioner services provided for low-income families through the General Medical Service (GMS) scheme and home nursing services. The current income threshold for GMS means that 36 per cent of the population in the lowest income group (known as Category I patients) are entitled to free general practitioner services, medicines and home nursing services. The remaining 64 per cent of the population (Category II patients) must procure and pay for general practitioner services on a private basis. Since 1989, the government has paid for general practitioner services provided within the GMS on a capitation basis, with the rate determined by demographic and geographic factors and special fees payable for designated services.

Private health care

As noted, a significant private health sector has been developed alongside the public health sector in Ireland. The private sector comprises the following elements: general practitioner services for the better-off two-thirds of the population; the purchase of pharmaceuticals by the same two-thirds of the population; privately-financed hospital stays; and all optical and dental care. While there are 60 acute hospitals within the Irish public sector, there are an additional 35 private hospitals (including private psychiatric hospitals). However, the private hospitals are small, having an average of less than 30 beds while the public hospitals average almost 200 beds. Moreover, the public hospitals have a different case load than the private hospitals which undertake no accident and emergency treatments. Overall, about 16 per cent of hospital beds are privately-owned. In addition, about 20 per cent of the beds in the public sector are used by the private-sector, representing about half of the total provision of private-sector health care. In total, one-third of all hospital beds are used for privately financed medical care.

Funding arrangements

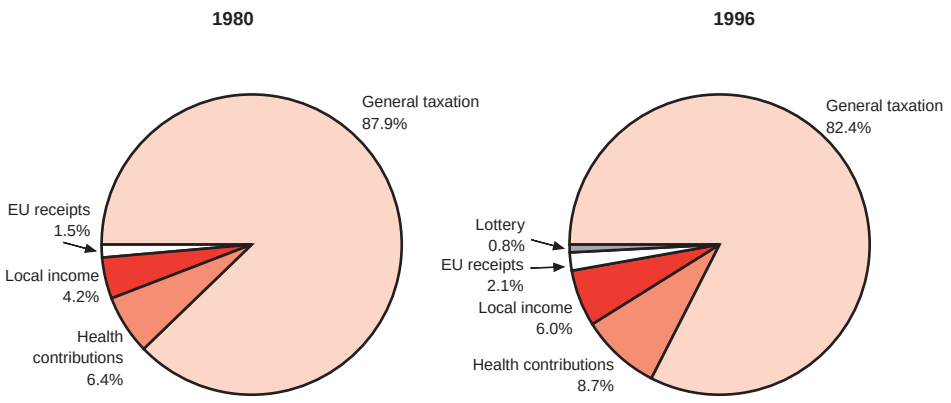
The Irish health service is essentially a mixed system, with approximately 75 per cent of funding drawn from public sources and 25 per cent by health

insurance companies and households. The public sources are mainly constituted by general taxation and a designated health contribution.

Public health funding

Public health care expenditure has become somewhat less reliant on general taxation as a source of funding in the past sixteen years. The tax-funded share has dropped from 87.9 per cent in 1980 to 82.3 per cent in 1996, though part of this fall has been offset by an increase in income-based compulsory health contributions (Figure 27). On the other hand, the share of health service spending funded by the national lottery, EU receipts and local income sources has risen by 3.2 percentage points. Charges for outpatient services and inpatient care in the public wards in public hospitals were introduced in 1987, and private usage of beds in public hospitals has always been charged. All medical card holders are exempt from charges. Since their introduction, charges have been revised a number of times, making a small but increasing contribution to health expendi-

Figure 27. **SOURCES OF FUNDING FOR HEALTH EXPENDITURE**



Source: Department of Health, 1996.

ture. Despite this development, general taxation continues to constitute the main source of public health service funding in Ireland.

Exchequer spending on the health services is determined in negotiations between the Department of Finance and the Department of Health. The latter has responsibility for policy development and overall planning within the health services. Funding for the provision of public health services is provided on the basis of annual budgets to the eight regional health boards. In general, these budgets constitute a global allocation with the health board having responsibility for allocating funding across the different programmes. In so doing, the Health Department has in recent years developed an approach of setting a budget allocation for each programme based on an expected activity level, with monitoring by the Department and the Health Boards during the year to ensure that the activity targets are met and the budget limits are respected. This allocation is determined by changing service needs and the general economic guidelines being applied to the operation of the public service as a whole. These guidelines will have particular reference to public sector pay levels which generally constitute the largest component of health expenditure.

As previously noted, the voluntary public hospitals have been maintained outside of the health board structure and funding for these hospitals is negotiated individually with the Department of Health. In addition to service delivery agencies, other corporate bodies and registration boards are supported by the Department of Health. These include such bodies as the Hospital Council and the Health Research Board.

Private health funding

Until 1996, private health insurance was provided only by the Voluntary Health Insurance (VHI) Board and various occupational health schemes.[44] The VHI is currently the main provider of health insurance services in Ireland. Following the enactment of the Health Insurance Act of 1994, another company has entered the market (see below). Premiums paid by the population for health insurance are currently tax-deductible at the standard tax rate. The VHI reimburses members directly for in-patient services and has some arrangements with providers, including hospitals and doctors, for direct payment of fees. VHI insurance covers mainly hospital expenditure. Visits to general practitioners (GP) and outpatient services are also covered but the threshold for reimbursement, set

on a cumulative annual basis, is relatively high so that nearly all GP visits are paid entirely by the customer. All private patients (whether insured or not) pay the entire cost of prescribed medicines up to a quarterly ceiling beyond which the government pays.

The VHI operates through the provision of a series of plans with different levels of entitlement. The base plan offers a full range of care in semi-individual accommodation in public hospitals, covers the payment of the Ir£ 20 ($32) per night charge for the use of public hospital facilities and offers partial refunds in respect of accommodation charges in private hospitals. The company then progressively adds to its products a range of options covering different accommodation and treatment levels in public and private hospitals and nursing homes. For a family of two adults and two children, annual premia range between Ir£ 445 ($700) and Ir£ 892 ($1 400), representing between 3 and 13 per cent of average industrial earnings, respectively. Since 1996, subscribers have had to pay 3 per cent of charges in a majority of private hospitals. In addition to hospital care, the first five plans cover other eligible expenses in excess of Ir£ 250 for an individual and Ir£ 400 for a family. Eligible expenses include general practitioner, specialist and physiotherapy fees up to a specified level. Public hospital accident and emergency care (but not other out-patient care) as well as home nursing are also covered, as are specific medical and surgical appliances and emergency dental care, subject to a pre-determined limit.

Private insurance provided by VHI covers about 37 per cent of the population. Those with higher incomes (*i.e.* the Category II population group) have the greatest liability for out-of-pocket expenditure and are the most likely to be VHI subscribers. When Nolan (1991) undertook an analysis of the VHI population, there were actually two eligibility categories (II and III) in addition to Category I. At that time Category II incorporated about half the population while Category III accounted for the 15 per cent at the top of the income distribution. Nolan (1991) estimated that 33 per cent of those in Category II and 74 per cent of Category III members were VHI subscribers. The bulk of the clients would appear to be either in the age range 25 to 54 or to be the dependants of such people. Under 5 per cent of the customers were retired people, though this group made up 11 per cent of the population (Table 39).

Table 39. **Age structure of Voluntary Health Insurance customers**

1988

	Male	Female	Total
	Per cent of total		
0 to 14	30	26	28
15 to 24	16	15	16
25 to 34	14	16	15
35 to 44	15	15	15
45 to 54	11	11	11
55 to 64	8	8	8
65 to 74	4	5	5
74 and over		2	2

Source: Department of Health.

International comparison

The Irish health care system can be considered a mixture of different models. There are considerable elements of a social welfare approach. What makes an understanding of the Irish system more complex is that, until 1996, the health insurance-based sector has operated as part of the broader state health system. Indeed, since its establishment, the Voluntary Health Insurance Board has been a statutory non-profit organisation owned by the Department of Health. As a provider of health insurance, the VHI is therefore unique and has been operated in a way that tries to ensure that a significant number of people stay in the private system, so minimising the cost of hospital care to public finances. This has also ensured that working in public hospitals remains attractive to consultants. In particular, the premia paid do not vary with age, so making it more likely that older people, with the highest demand on hospital care, stay in the system.

Similar mixes of public and private care do exist in other countries or regions. A few US states and Canadian provinces, Australia and, to a certain extent the Netherlands, have a similar mix, leaving aside the role of the VHI in Ireland. A fundamental difference with most continental European systems is that, in Ireland, two-thirds of all patients receiving first-level medical care are faced with a price that reflects marginal costs and which they have to pay themselves, so limiting demand. Moreover, the state-funded system for the poorest third does not function on a fee-based system. Rather, the doctor is paid a

fixed fee per patient per year, the latter being free to change doctors annually. In addition, the doctor is given an indicative target for the pharmaceutical costs of his state-financed patients and can keep a proportion of savings relative to this target, so giving an incentive to restrain the growth of costs.

An additional characteristic of the Irish system is that both public and private services may be secured from the same providers. The majority of general practitioners and hospital consultants provide services to both public and private patients; pharmacists serve the public and private sectors; and the major public hospitals have public and private beds. In addition, depending on the nature of their contracts, hospital consultants may practise in both public and private hospitals. This mix of public and private services, which facilitates complementary roles rather than conflict, is widely regarded as one of the strengths of the Irish health system.

The performance and cost of the system

Health outcomes

The overall health of the Irish population has improved considerably over the past forty years. The risk of dying before age 70 is now quite close to that in other countries. As a result, life expectancy, at birth, in Ireland has increased substantially but it is still below that for most other countries in the European Union. By 1993, it was 72.7 for men and 78.2 for women, having increased by eleven years for women and nine years for men since 1950. This placed Ireland eleventh for men and thirteenth for women in the fifteen-country EU area. At older ages, the relative position of Ireland deteriorates: by age 65, life expectancy is indeed the lowest in the EU area. The fact that heart disease and cancer each accounted for approximately one-quarter of all deaths in Ireland in 1992, with incidence rates being substantially above the EU average (Table 40), suggests that there is still considerable room for improvement in health status.

The extent of loss of life amongst the young has been declining. There has been a substantial reduction in infant mortality since 1970, dropping from 19.5 to 5.9 in 1993. In the same period, maternal mortality has declined from 31 per 100 000 live and still births to 2 per 100 000. Overall premature death (defined as the number of years remaining to age 70 for those who die before the age of 70

Table 40. **Causes of death in Ireland**

	1991 percentage of EU death rates				Actual death rate			
	Men		Women		Men		Women	
	All ages	0-64	All ages	0-64	All ages	0-64	All ages	0-64
Diseases of the circulatory system	130	142	120	128	523	132	308	42
Ischaemic heart disease	183	188	174	164	330	101	147	21
Cerebrovascular diseases	86	82	99	116	84	12	79	11
Cancer of the digestive organs and peritoneum	104	102	114	126	90	32	54	20
Malignant neoplasm of female breast	. .	. .	125	127	. .	. .	39	25
Trachea/Broncus/Lung cancer	89	72	179	157	68	24	25	10
Externally caused injury and poisoning	88	. .	81	. .	61	. .	23	. .
Bronchitis/Emphysema/Asthma	81	91	130	119	22	4	12	2
Motor vehicle accidents	80	. .	100	. .	17	. .	7	. .
Cancer of the cervix	. .	. .	103	117	. .	. .	3	3

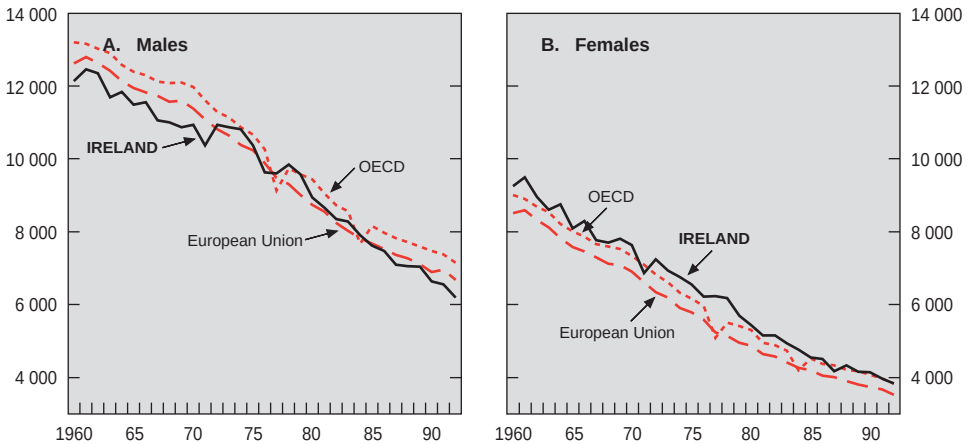
Source: *World health statistics*, HFA, 1994.

from all causes except suicide) has fallen by 2 per cent annually between 1960 and 1992 for men and 2.7 per cent for women (Figure 28), at almost exactly the same pace as in the rest of the OECD and the EU. By 1992, the avoidable death rate for men was some 13 per cent below that in the rest of the OECD area and 7 per cent below that in the EU. For women, though, the death rate was similar to that in the rest of the OECD area and above the EU average. This creditable performance was achieved with a low absolute level of resources spent on health care in Ireland and despite relatively low incomes.

Thus the health disadvantage seen in Ireland, compared with other developed countries, is very much concentrated on the elderly.[45] Younger Irish people (under 45) seem, if anything, to have a health advantage compared with most other countries, while middle-aged people conform relatively closely to international norms. Signs of substantial falls in older-age mortality did start to emerge in the second half of the 1980s, though they were not sufficient to close the gap between Ireland and other EU countries. It could be that, as income levels

Figure 28. **AVOIDABLE LOSS OF LIFE**

Per 100 000, per year



Source: OECD Health Database.

converge (see Chapter I) and as healthier younger generations age, life expectancy in the coming decades will increase independently of the resources devoted to health care.

Health expenditure growth

Ireland is among the few OECD countries that has experienced a reduction in the share of GNP devoted to health – taking public and private expenditure together – in the period 1980 to 1993. The drop in the share has been about 23 per cent and this contrasts with a 24 per cent increase seen in thirteen other OECD countries for which data is available (Table 41). As a result, the share of GNP devoted to health has fallen markedly, from amongst the highest to the lowest, reaching 7.2 per cent by 1995 (Figure 29, Panel C). However, this comparison gives perhaps an exaggeratedly adverse picture of health spending in Ireland. Within Europe, the average share of GNP devoted by Ireland to public health spending is only slightly below average, though the absolute level of per capita spending is markedly lower than in other European countries.

Table 41. **Evolution of health care spending**

1980-1993

	Percentage change in:	
	Total health expenditure as a percentage of GDP	Total health expenditure per capita
Australia	16	123
Belgium	26	173
Canada	38	167
Denmark	–1	118
Finland	35	162
France	29	158
Germany [1]	2	122
Ireland	**–23**	**104**
Italy	23	162
Netherlands	10	118
New Zealand	7	112
Norway	24	185
Spain	28	193
Sweden	–20	46
United Kingdom	27	168
United States	52	209

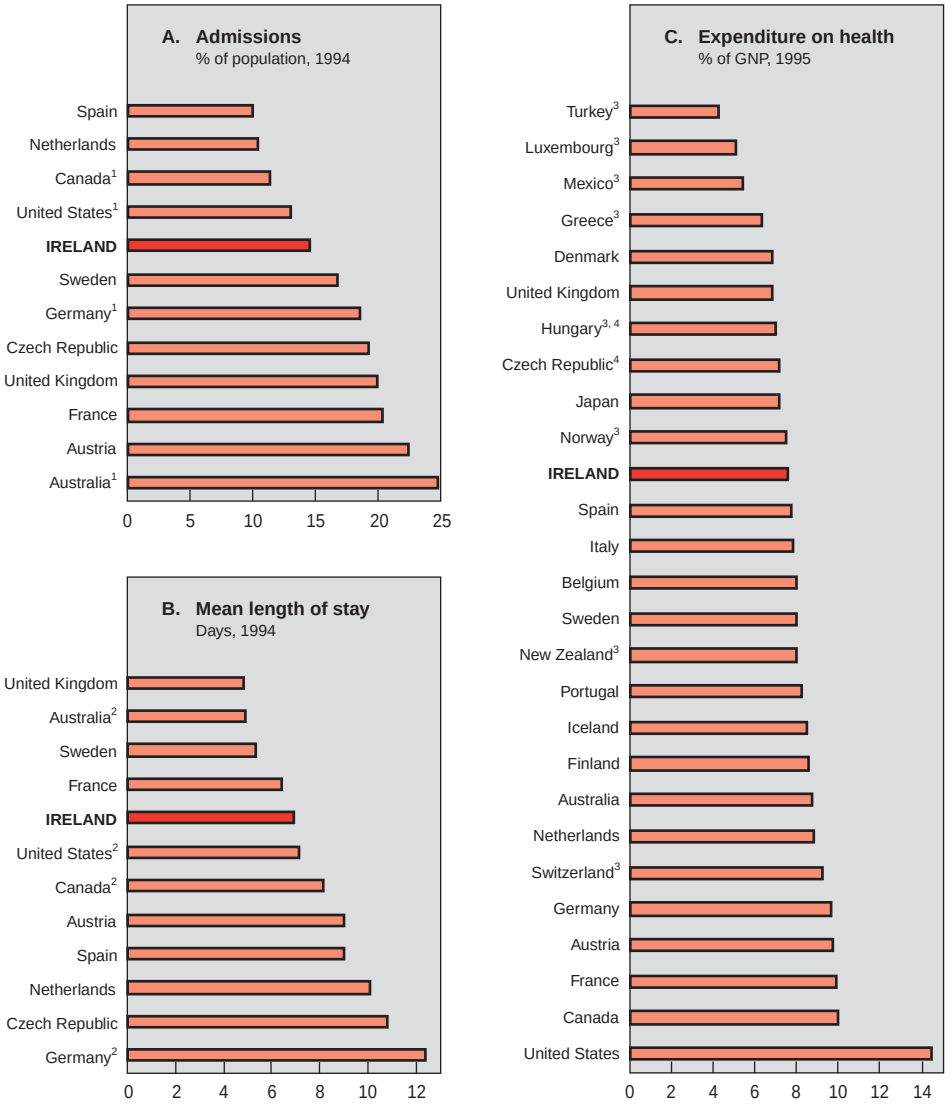
1. Pre-unification boundaries are used for Germany.

Source: OECD Health Data, 1995.

The growth in the volume of resources devoted to health care by the public sector has been well controlled. Over the past sixteen years, the movement of the volume of gross non-capital public health expenditure (*i.e.* before the deduction of income from public bed charges and private treatment in public hospitals) has followed three distinct phases. It declined by 2 per cent between 1980 and 1984, and again by 6 per cent in the four subsequent years. Spending then started to grow in the eight years ending 1996, rising by 16 per cent – an annual volume growth rate of 2 per cent (Figure 30). This was equivalent to the annual increase in health service employment in the same period. As a result, the fall in gross public resources devoted to health in the 1980s has been reversed and gross expenditure is now some 10 per cent higher than in 1980, though due to the increase in charges and private treatment, the level of net public expenditure has not increased so rapidly.

With the volume of public spending remaining practically stable through the 1980s, gross health expenditure as a percentage of GNP has shown a general

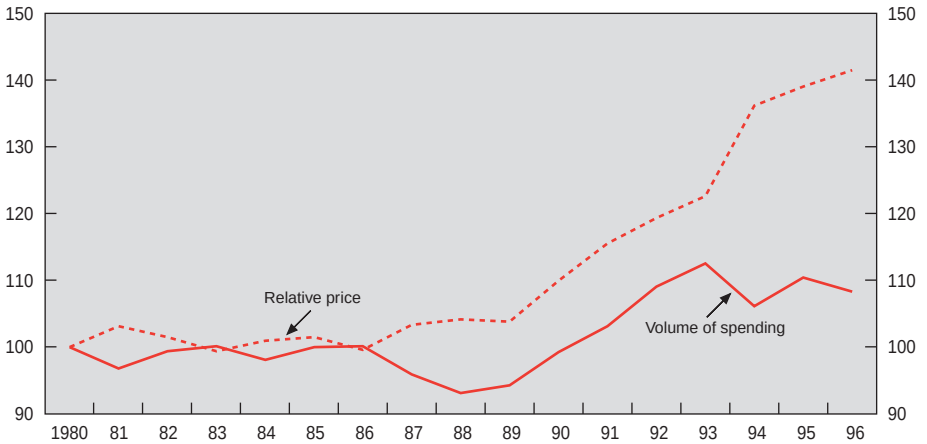
Figure 29. **MAIN HEALTH INDICATORS: AN INTERNATIONAL COMPARISON**



1. 1992 for Australia, Canada and United States, 1990 for Germany.
2. 1991 for Canada, 1992 for Australia and United States, 1993 for Germany.
3. 1994.
4. GDP has been used for the calculation.

Source: OECD.

Figure 30. **VOLUME AND RELATIVE PRICE OF HEALTH CARE SPENDING**
Index 1980 = 100



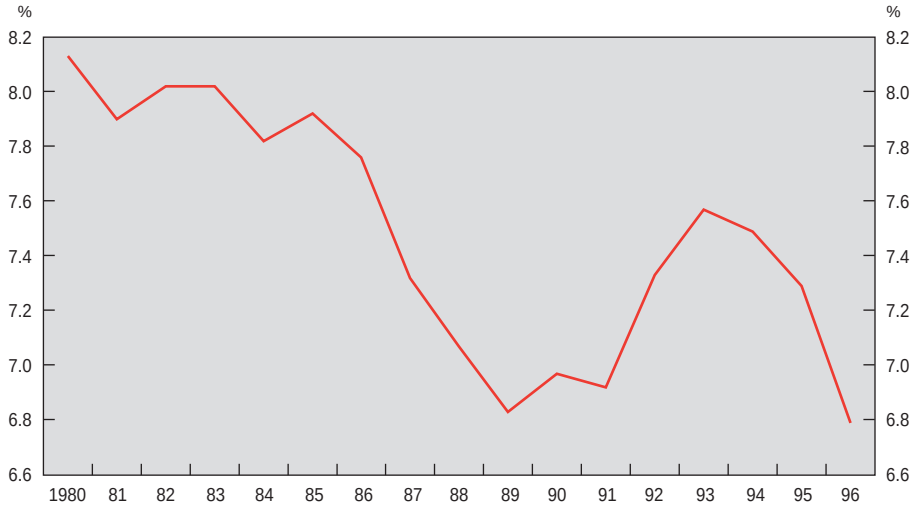
Source: Department of Health, 1996.

downward trend from a high of 8.1 per cent in 1980 to 6.8 per cent in 1989 (Figure 31). Despite a marginal drop in 1991, health expenditure as a proportion of GNP generally increased after 1989 to 7.6 per cent in 1993, the highest level reported for this decade. Since then the economy has grown at an exceptionally fast rate, so that the share of GNP devoted to health expenditure has fallen to 6.8 per cent in 1996 – the lowest level since 1980.

The major increase in health expenditure in the 1990s has not been in the core general hospital programme but in the community care programme (Figure 32). This programme, which covers community health, protection, welfare and services for the handicapped, has shown the largest proportional increases between 1990 and 1994. These have expanded at over 12 per cent annually in contrast to a 9 per cent annual increase in the hospital care programme. These developments follow the trend of the 1980s when, despite constant overall health expenditure, the community care programme grew in size.

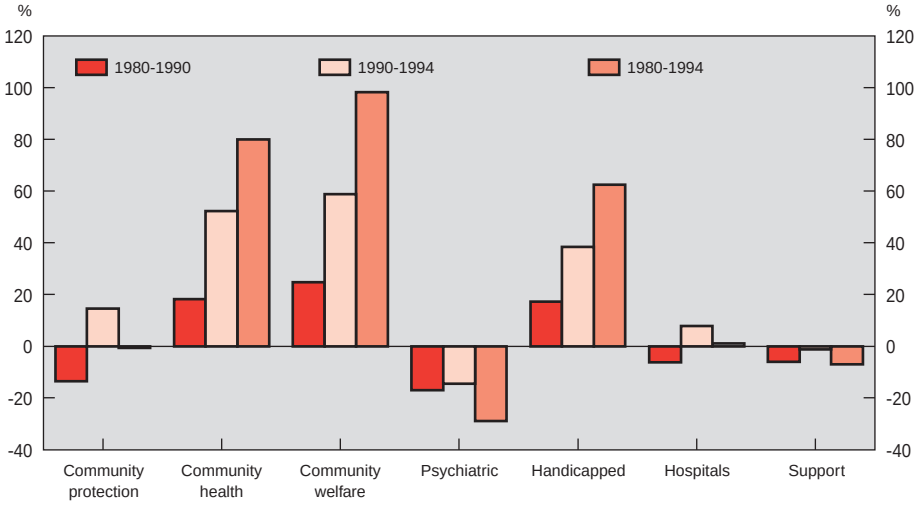
There is some evidence of complementarity in the growth of private and public expenditure. In the 1980s, when spending on public hospitals was falling

Figure 31. **GROSS NON-CAPITAL HEALTH EXPENDITURE RELATIVE TO GNP**



Source: Department of Health.

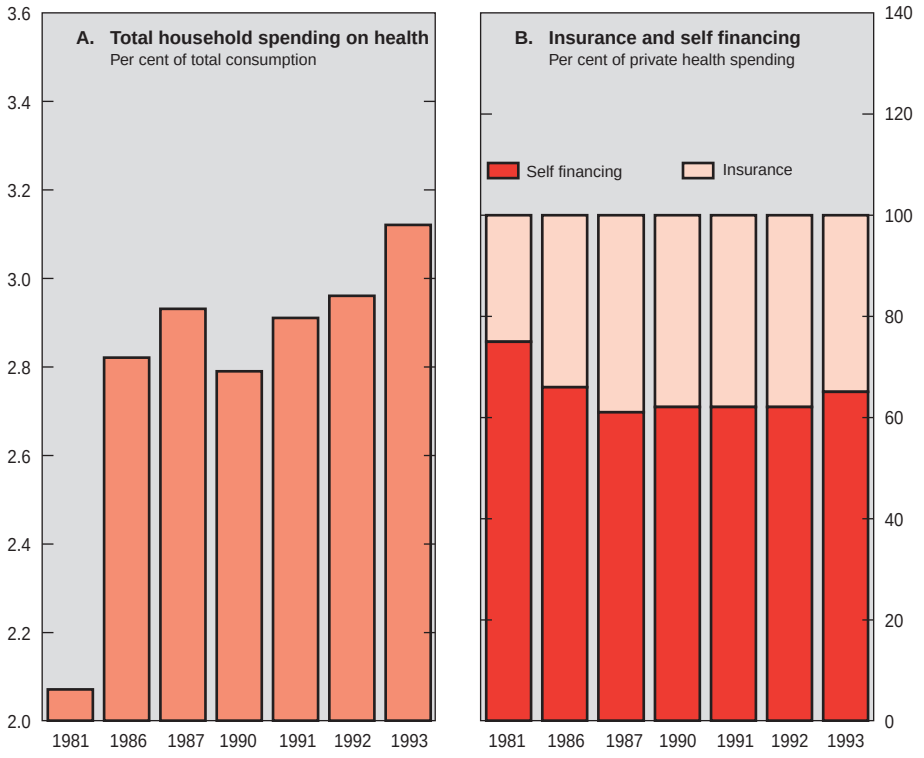
Figure 32. **GROWTH OF HEALTH CARE SPENDING BY PROGRAMME**



Source: Department of Health.

in real terms, private health insurance expenditure rose rapidly (Figure 33). Since 1990, at the same time as public spending began to increase, the growth of private health insurance spending slackened. By 1993, the proportion of private health expenditure covered by the VHI scheme had dropped to 34 per cent. To some extent this reflected a switch to self-insurance, as health insurance stagnated and the growth of households' own expenditure on health rose more rapidly. Such expenditure mainly represents payments for such services as general practitioner care, drugs and dental care. The slowdown in the growth of the number of people insured continued until 1995 (the last date for which data are available),

Figure 33. **PRIVATE HEALTH EXPENDITURE**



Source: Health Expenditure in Ireland, Health Fact Sheet 4/95, Institute of Public Administration, Dublin.

with the expansion being only 1 to 2 per cent per year as the market came close to saturation. However, between 1992 and 1995, VHI claims expenditure rose annually 6 per cent faster than inflation as measured by the CPI.

Hospital infrastructure and occupation

The control over the amount of resources devoted to health care has not resulted in a fall in the availability of hospital health care. The admission rate to Irish hospitals for in-patient care has dropped significantly in contrast to the average position in other OECD countries where, on average, it has remained stable (Table 42). By 1994, only four of the OECD countries for which data were available had a lower hospital admission rate, in contrast to 1980 when Ireland had one of the highest admission rates (Figure 29, Panel A). However, since 1986 the drop in in-patient care has been more than offset by the almost quadrupling of day-case treatment. By 1984, day-care represented 27 per cent of total admissions, up from only 8 per cent in 1986. This rise in day-care treatment has meant that the total number of cases treated annually has been rising by 3 per cent a year since 1987.

Ireland's position in an international perspective

There have been significant improvements in hospital productivity over the past fifteen years in Ireland. The average length of stay has been reduced by 29 per cent, somewhat greater than achieved in other OECD countries. Nevertheless, some scope for further reduction of the length of stay would appear to exist as it is still higher than in some economies (Figure 29, Panel B). Better management of hospital facilities has allowed an increase in the average bed occupancy rate to nearly 85 per cent, one of the highest in OECD area. The combination of reductions in length of stay, higher occupancy and a slight fall in the number of admissions has permitted a major reduction in the number of hospital beds – of nearly 40 per cent – much greater than seen in other OECD countries during the same period. In the Irish case, this has permitted the outright closure of a number of small hospitals.

Comparison of public and private facilities

The provision of private medical care in public hospitals is geographically concentrated, as is the population. Over one-third of the private beds found in

Table 42. **Acute hospital facilities and admissions**
Change in key indicators for selected OECD countries
1980-1993

	Percentage change in:				
	Acute hospital beds/1 000	Admissions as a percentage of population	Average length of stay	Bed-days per capita	Occupancy rate
Australia	–31 [1]	–14 [4]	–34 [1]	–40	18 [1]
Belgium	–13	25 [5]	–20	–18	4
Canada	–26 [1]	–5 [6]	–16 [1]	–13	–2 [8]
Denmark	–27	13	–31	–19	9
Finland	–6 [2]	16 [7]	–35	–21 [7]	–9 [7]
France	–19	19	–36	–22	–3
Germany	–6	13 [8]	–17	–8	3 [2]
Ireland	**–43**	**–13**	**–29**	**–41**	**5** [10]
Italy	–28 [2]	n.a.	n.a.	n.a.	n.a.
Netherlands	–21	–8	–26	–31	–14
New Zealand	–18 [3]	n.a.	n.a.	n.a.	n.a.
Norway	–35 [2]	–7 [1]	–38	–31	–3 [6]
Spain	–3 [4]	10 [9]	2 [9]	–11 [9]	5 [9]
Sweden	–33	10	–35	–31	6
United Kingdom	–28	72	–40 [2]	0	1 [11]
United States	–17 [2]	–18 [2]	–7 [2]	–33	–12 [1]

1. 1980-1991.
2. 1980-1992.
3. 1986-1991.
4. 1984-1991.
5. 1981-1993.
6. 1980-1989.
7. Pre-unification boundaries are used for Germany.
8. 1985-1993.
9. 1980-1990.
10. 1984-1993.
11. 1982-1992.

Source: OECD Health Data, 1995.

public hospitals are located in the Eastern part of the country (essentially the Dublin area) (Table 43). In this area the number of patients discharged from private beds account for about one-third of all discharges from public hospitals. In five of the eight health boards, the occupancy of private beds is low, with private bed-days used accounting for only 10 to 13 per cent of all bed-days compared to one-fifth of total available beds. Such low use of private beds pulls down the overall occupancy rate of private beds in public hospitals to 57 per cent against 92 per cent for public beds. However, the occupancy rate for public beds is biased up, in that hospitals are free to put public patients in private beds for

Table 43. **Public and private health care resources in public hospitals**

1994

Health board	Beds Available[1]				Bed-days Used[1]			
	Public	Private	Non-designated	Total	Public	Private	Non-designated	Total
	Per cent				Per cent			
East	78.0	16.4	5.6	4 922	84.4	11.5	4.1	1 522 974
Midland	80.8	15.4	3.8	480	82.5	14.6	2.9	144 949
Mid-West	63.5	30.6	5.9	759	69.4	28.9	1.7	247 727
North-East	73.7	21.1	5.2	949	78.1	18.1	3.8	251 746
North-West	65.0	21.0	14.0	666	75.7	12.2	12.1	200 074
South-East	76.1	20.1	3.8	1 078	85.2	12.1	2.7	312 377
South	70.8	20.2	9.0	1 771	87.1	10.0	2.9	526 656
West	76.9	19.6	3.5	1 236	85.0	13.0	2.0	377 468

1. Excluding day cases.
Source: Provisional estimates from the Department of Health.

medical reasons or in emergencies. Taking this factor into account, the occupancy rate of private beds in public hospitals was 69 per cent in 1995. The average stay is considerably shorter in private beds than in public ones (5.8 days against 7 days).

This tendency for shorter stays in private beds is also reflected in a greater use of day treatment in the private part of the public sector. For most health

Table 44. **Public and private care in public hospitals**

1994

Regional health board	Inpatients discharged				Day cases			
	Public	Private	Non-designated	Total	Public	Private	Non-designated	Total
	Per cent				Per cent			
East	83.8	13.8	2.4	189 708	73.7	6.6	19.8	123 249
Midland	83.0	15.9	1.1	25 325	69.3	30.7	0.0	5 542
Mid-West	66.9	32.4	0.7	41 125	68.5	27.4	4.1	11 030
North-East	78.8	18.2	3.0	38 357	69.1	23.2	7.7	6 086
North-West	81.2	12.8	6.0	34 228	91.6	0.0	8.4	12 414
South-East	82.4	16.5	1.2	53 543	77.7	22.3	0.0	9 508
South	84.7	13.1	2.2	82 682	73.6	22.1	4.4	17 459
West	84.8	14.3	0.9	57 866	81.4	8.2	10.4	10 042

Source: Provisional estimates from the Department of Health.

boards, the proportion of day cases treated in private beds (between 20 and 30 per cent for five health board areas) is higher than the proportion of in-patients (Table 44). As day cases account for an increasing proportion of total hospital workload in the Eastern Health Board and a large concentration of private hospitals are located in this region, the indications are that the private day case workload may be more concentrated in the private hospitals rather than in private beds in public hospitals.

Problems with the current system

Although the Irish health system has delivered a continuous improvement in health standards, with a growth of expenditure that until recently has been well controlled, there is still scope for further improvements in efficiency that could be achieved through better allocation of resources. Such gains would allow some growth in areas where demand is rising or where (such as cancer and heart disease) death rates have been increasing. In addition, there are some concerns about the extent to which the medical system relies on staff still being trained. The section below examines these and some other problems.

Demand effects

Demographic trends

The Irish population is now entering a phase when the number of old people will start to grow more rapidly than the total population and this may have some effect on health service demands. According to the higher of two official projections,[46] between 1991 and 2011 the number of people aged over 65 may increase by slightly over 20 per cent. This would raise the proportion of old people in the population to 13.6 per cent from 11.4 per cent in 1991. The lower projection would put the proportion at only 12.5 per cent of the total population by the same time.

An ageing population may have an influence on health service requirements. The need for social care services, particularly for the very old, is a case in point. However, since the growth in the numbers of older people, with on the whole worse health than younger people, is still relatively limited – less than 1 per cent annually – its impact on health demand may be modest. It is likely to be

outweighed by the increasing expectations of health care as consumer incomes grow. Moreover, the negative impact on health needs arising from ageing may be offset to a certain extent by the improved average health status of elderly people and by the lower number of births and very young children which will tend to lower health spending.[47] On the other hand, there are other factors which may raise future health spending including the increasing life expectancy of the mentally handicapped, increasing childcare costs stemming form new legislation and the development of new technologies and medicines. As a result, when attempting to project future health service needs, it will be important to consider the relative effects of each of these factors and not just concentrate on the potential negative effect of ageing.

A further demand-side problem may arise in the short term with the introduction of competition in the private health insurance market. With effect from 1 January 1997, BUPA Ireland – a subsidiary of the similarly named UK company – has become a competitor to the government-owned VHI. After initially proposing a series of health insurance products, some of which had age-related premia, the company has reaffirmed its commitment to community rating and will introduce a new range of products that do not conflict with this principle.

Waiting lists

In common with other health systems which have limited capacity and demand that is not constrained by price, an element of rationing by waiting is to be found in the Irish health system. By 1993, waiting lists for surgery in the public system had risen to around 40 000 cases. Though small in relation to the total number of hospital cases treated (700 000 per year), the delays had caused public concern. While low in total, delays for some non-urgent cases were long. Moreover, the fact that better-off people, who were likely to be privately insured, did not experience such delays gave rise to concerns about the equity of the system.

Spending on first level medical care

Spending on the community care programme has been increasing rapidly. This has been due mainly to the rising real cost of providing first-level health care to the poorest third of the population; costs have grown by more than 9 per cent annually in the 1990s driven by higher outlays on drugs. With the numbers

eligible for the free general practitioner services programme falling by ½ per cent per year, real spending per person has been rising even faster, reaching Ir£ 212 per person per year in 1994 of which spending on drugs represented 64 per cent. Doctors participating in the free treatment scheme now receive indicative drug spending targets based on average prescribing costs weighted by age and sex. A portion of the savings made by individual doctors may be available for practice development projects which have the prior approval of the health board. Following the introduction of this scheme, the annual average increase in GMS drug costs has slowed to 6 per cent annually from 10 per cent. Another area where expenditure has risen rapidly is the Drug Cost Subsidisation Scheme. Under this programme, patients who need ongoing medication costing more than Ir£ 32 per month ($50) are given an entitlement to free medicines whatever their income level. Part of the increase in spending is due, on the one hand, to greater awareness amongst the public of the programme and, on the other hand, to the prescription of new, more expensive treatments which may reduce long-term health costs.

Supply constraints

Controls on service provision and utilisation

While, at the time of introduction of charges in the late 1980s, the intention seemed to be to relieve some of the pressures on health expenditure, which were particularly severe in that period, it appears also that there was a rather broad policy objective to use such charges to direct service use. The degree of consistency between the incentives implied by the structuring of the charges and declared health policy objectives may, however, be open to question (Nolan, 1993). One specific example might be cited: the charge for the first visit for outpatient care in accident and emergency centres, without a general practitioner (GP) referral, is Ir£ 12 (slightly below 5 per cent of average weekly earnings). Patients referred by a GP do not incur a charge. The outpatient charge does not reflect the real cost of service provision which will exceed the cost of a similar service provided by a GP. Depending on the area of the country, the outpatient charge is actually often lower than the fee for a GP visit which is currently between Ir£ 15 and Ir£ 20 (about $25 and $30 respectively). This differential suggests that it may be more rational for patients to seek treatment at the hospital department and pay the charge rather than consult the GP.

Controls on the acquisition of new technology

Any investment in new technology by the public hospitals must be approved by the Department of Health. There are currently geographic imbalances in the availability of new technology. The current distribution of high-tech magnetic resonance and induction (MRI) scanners provides a useful illustration of the problems which may arise. In Ireland, there are now five MRI scanners, which translates into approximately 1.4 per million population. While this is a similar rate to that found in the United Kingdom in 1992, provision is higher in the Netherlands and Sweden at 1.8 and 2.6, respectively (Battista, *et al.*, 1995). However in Ireland, only two of these machines are in the public sector with only one located in a public hospital outside of Dublin where two-thirds of the population reside. In order to equalise availability, the government has decided that the next such machine will be installed outside Dublin. The remaining three scanners are located in private hospitals (in Dublin), although the public sector purchases examinations on these machines. For private hospitals, the decisive factor governing the decision to install such machinery is whether the insurer will pay for its use.

While investment in new technology may be an important contributory factor to pressures for increasing expenditure in the health sector, focusing on the control of spending is not sufficient to manage technology (Battista, *et al.*, 1995). If the objectives of efficacy and cost-effectiveness are to be achieved with any investment in new technology, some co-ordination at the national level may be required in addition to operational assessment in the clinical context. In this respect, it may be sensible to draw on the assessments of other countries, in order to minimise administrative cost, as has been done through participation in the EU Technology Assessment Initiative.

Control of pay

While the volume of resources used in the health system has grown by 1½ per cent annually since 1988, the prices paid to purchase or employ the necessary resources have risen more rapidly since 1988. During the period of fiscal consolidation, broadly during the 1980s, the relative price of health expenditure was held down, growing by only ½ per cent annually. This reflected a downward pressure on public-sector wages. Since then, public-sector wages have risen more rapidly and the relative price of health expenditure has risen by

4½ per cent annually. This rapid expansion in pay is continuing. In the 1997 budget, a large part of the increase in spending will be devoted to funding pay increases. Spending on health care is projected to rise by over 9 per cent of which 6 percentage points is for pay. This budget envelope will have to encompass the agreed pay rises of between 5 and 18 per cent for nurses and an eventual settlement for senior hospital doctors (''consultants'') who have been offered significant pay rises in addition to the standard increases under the central incomes policy. The proposed consultants' increase would raise their pay bill by up to 16 per cent, but it will involve a significant change in their employment contracts, with duties becoming better specified. There will also be a better definition of the split of their time between private and public hospitals.

Management of the hospital system

The management of the hospital system seems to be unduly centralised and bureaucratic. Many decisions appear to be taken at the level of the health board. This might be justified if decision-making at a higher level led to a better regional strategy and planning for the provision of resources. This does not however seem to be the case.[48] Budgets for individual hospitals take into account the actual outputs of hospitals only to a limited extent and are generally based on historical spending. Moreover, in a hospital it is rare for budget responsibility to be devolved to operating units. Indeed, there appears to be some tension between the hospital administrations and the consultants, with the former reluctant to share budgetary information and the latter, in some cases, acting as practitioners independent of the hospital team.

Medical staff

The past decade has seen a significant expansion in the number of hospital doctors with their number growing by about 2½ per cent annually. Nearly all of the increase has come in the training grades below consultant level. Such growth was necessary because of a decision to place a limit of 65 hours on the working week of doctors who were below consultant level. As a result of this situation, 41 per cent of sub-consultant posts, excluding interns, are filled by non-EU nationals, with peaks of over 60 per cent in paediatrics and surgery; these proportions are higher in areas outside Dublin. While all non-consultant jobs are supposed to have a major training component, in practice a large number of posts, especially in small and mid-sized hospitals, are unable to offer such a

programme. Such posts are unattractive to Irish medical graduates who leave to finish their training in the United Kingdom or the United States. Many subsequently return and about half of the appointments to consultant posts have trained in these two countries. While such an interchange does provide evidence of the high quality of initial medical training in Ireland, the subsequent fourteen-year training period, which lasts on average until 38, appears unduly long.

Summary

The principal problems facing the health service in Ireland appear to be related to the control of spending and ensuring that resources are used and allocated efficiently. Four major areas of concern can be observed. *First,* there has been a marked rise in the pay of health service personnel and a rise in the overall cost of medicines paid for by the government, which has been reflected in the increased cost of public programmes of first level medical care. *Second,* the management of the hospital system may be rather bureaucratic, with little devolution of budgetary authority and a lack of communication between the managerial and medical staff. *Third,* there are some problems in managing the complicated interface between the public and private provision of medical care. Patients may go to public hospitals for out-patient care in the accident and emergency service because charges are cheaper than going to see a general practitioner where most people have to pay the full costs of a consultation. Within hospitals, there is difficulty in accurately assigning the time that senior medical staff devote to public and private care. *Finally,* there is some concern about the extent to which Irish doctors leave the country to complete their training, reflecting the long training period and inadequate sub-consultant career posts. Over the longer term, there must be some concern about the possible growth in spending on nursing care for the aged. Another area where problems may arise is through the impact of competition in the private insurance market.

Improving the system: reform to date and future strategies

These problems of the health system, outlined above, are not new and, in fact, are often present to much greater extent in many OECD countries. They have existed for many years and have been the subject of a series of official reports and studies, some of which are dealt with below together with principal

proposed solutions. While some of these proposals have been followed by action, progress in implementing other key areas has been slow, even if the direction for change seems clear. These areas are pointed out in the paragraphs below, together with possible future changes in other areas.

Past policy initiatives

The Commission on Health Funding

In response to the fall in public expenditure on health in the mid to late-1980s (see above) the government set up a Commission on Health Funding in 1987. This body was charged with making recommendations on the extent and sources of the future funding and on any changes in administration. In the event, the Commission found that the key problem in the health system was its management rather than its funding. An essential recommendation was that the administration of the health services should be restructured with the Minister and the Health Department fulfilling policy and legislative functions while the overall management of the health services would be transferred to an executive authority. This authority would be expected to operate through a new management structure, while the health boards would be reconstituted as health councils and take on representative rather than executive functions.

The Commission recommended, in 1989, that health service funding should continue to be primarily tax-based. It was at the origin of the generalisation of public-sector hospital care to the better-off (see above). It also supported the continuation of modest user charges for public services by all but the poorest patients. All of these suggestions were implemented by 1991.

The Commission also made a number of important recommendations concerning the interface between the public and private sectors. One conclusion was that tax relief on private insurance contributions should be phased out. As a result, tax relief is now only available at the standard rather than the marginal tax rate, so reducing the relief on health insurance by 44 per cent for those paying the higher rate of income tax, the threshold of which is relatively low (see Chapter III). In 1995-96 such reliefs generated tax expenditure of Ir£ 58 million per year, less than 4 per cent of gross public health spending. The Commission also recommended that there should be clearer differentiation between the public and private practice of hospital consultants. This was pursued to some extent in

the subsequent revision of consultant contracts in which the number of hours per week spent by consultants on public practice was specified and their public salary set at a level that reflected the extent of their commitment.

Shaping a healthier future

In 1994 the Department of Health set out a new strategy in *Shaping a healthier future* with the objective of providing a new policy framework for the development of health services in Ireland. The strategy called for improving the effectiveness of health and personal social services by reshaping the way that services are planned and delivered. Three key areas were identified: *i) services* such as prevention, treatment and care would need to become more focused on improvements in health status and quality of life and so place more emphasis on the provision of the most appropriate care in an equitable fashion; *ii)* the *management* of the system would need to provide for more decision-making and accountability at the regional level, together with the application of better methods of performance measurement; *iii)* finally, *consumers* would need to be treated with greater sensitivity while *providers* would need to be given a greater role in decision-making.

With regard to *services,* a number of criteria were suggested. Uniform rules for eligibility and service charges could be introduced on a geographic basis, waiting times for public services could be reduced and special attention given to disadvantaged groups. If adopted, these proposals would ensure a more equitable delivery of services. There should also be a more critical evaluation of the performance of hospitals through such techniques as clinical audit.

Proposals were made for reorganising the *management* of the system. These involved a redefinition of the role of the Department of Health and the rebirth of the health boards as health authorities. The issue of accountability and definition of the roles of the health board levels were the subject of legislation in 1996. This was concerned with improving financial accountability and expenditure control procedures in health boards; clarifying the respective roles of the members of health boards and their chief executive officers; and starting to remove the Department of Health from detailed involvement in operational matters. Various initiatives to restructure the material purchasing and finance functions in hospitals are being undertaken.

The government has now proposed to reorganise the administration of hospitals in the Dublin area. A key element in the management reforms will be a change in the status of the voluntary hospitals which are funded directly by the Department of Health while other acute hospitals are funded through the local health boards. As a result, the health board covering Dublin has little effective involvement with the provision of acute hospital services in its area. The parallel operation of the provision of acute hospital services through the voluntary hospitals and the health board structure results in some fragmentation of the system, together with differences in efficiencies within the sector, and impedes the proper co-ordination and development of services at the local level.[49] Under the new proposal, the Eastern Health Board will be renamed the Eastern Regional Health Authority and will receive a block grant from the Department of Health. It will then be responsible for managing the allocation of funds to all public hospitals in its geographic area and for delivering, planning and evaluating health and personal social services such as home nursing.

The government has also introduced a series of measures to strengthen the management capacity of health services. This plan aims to improve the human resources of the hospitals by introducing comprehensive performance appraisals for staff, encouraging continuing training and improving the selection, promotion and recruitment procedures. In addition, the senior executive of health authorities and boards will be placed on fixed-term contracts.

In pursuit of greater accountability towards *consumers*, the Strategy suggests that mechanisms must be put in place which ensure that those with decision-making powers are adequately accountable to the funders and consumers of the services. At the same time, the government introduced an initiative to improve consumer satisfaction with the public hospital system by reducing waiting lists for elective surgery. By targeting a relatively small amount of expenditure on certain specialities, the government has been able to reduce the number of people awaiting treatment by 35 per cent in the three and half years to end 1996, from about 40 000 to just under 26 000. In addition, the government set a number of objectives for the *providers* of the health service. It specified national objectives for service development in a wide range of areas including health promotion, primary care services, services for women, children and special needs groups and acute hospital services. It also set national targets for reductions in risk factors associated with premature mortality and for improvements in other indicators of

health status. These outcomes were to be monitored through a system of annual reports for each region covering health status, service provision and measures of patient satisfaction and an annual performance review produced by the Department of Health. The government pledged that it would aim to provide, over the 1994 to 1997 period, the money for the development needs identified in the plan while observing the limits of budgetary policy. The government has recently introduced a cancer programme boosting both hospital and non-hospital spending against this illness.

Reform of budget mechanisms to improve efficiency

Hospital budgets had been generally set on an incremental basis during the 1980s, with some marginal changes depending on the perceived efficiency of hospitals and changing service needs. The first proposals for reforming budgetary mechanisms were made by the Commission in 1989, but progress in implementation has been slow. The Commission recommended that hospitals should receive global budgets for the provision of an agreed service level based on: *i)* an assessment of the activity level implied by the hospital's agreed role and catchment area; and *ii)* the case-mix based cost of meeting this.[50] By 1990, it was established that the case-mix technique could be successfully applied to Irish hospital discharge data and a system that related hospital workload, measured by similar groups of treatment, to spending patterns was developed for acute hospital services.[51] Following a successful attempt[52] in 1991 to measure hospital workload in a manner that is equally meaningful to both clinicians and accountants, through the use of aggregates of clinically similar treatments or illnesses, the basis for the introduction of budgeting based on the case-mix system was in place.

The essentials of the case-mix adjustment may be summarised as follows: hospitals are stratified according to teaching or non-teaching status; activity data from the relevant hospitals are assigned to related-illness or treatment groups and, on the basis of the average cost of treating each illness group, a case-mix adjusted cost is estimated for the individual hospital and hospital group.[53] The actual budget is then determined on the basis of an average or blend of the specific hospital's case-mix adjusted costs and the average cost for the hospital group to which the specific hospital under consideration belongs. The ''blending approach'' was judged necessary because the case-mix adjusted costs estimated

for individual hospitals showed considerable variations and so adoption of an average cost basis would cause large changes in the money available to individual hospitals, with the inefficient losing and the efficient gaining.

A further element in the determination of a hospital budget is the extent of the income generated by the private-sector activity of the hospital. The Department of Health assumes a certain level of income from private treatment in the confines of the public hospital. The only constraint placed on private treatment stems from the agreed limit on the number of private beds in the hospital which can only be altered with the agreement of the Department. Where actual income differs from the assumed level, the hospital can keep the money as there is no retrospective adjustment of the budgeted income. The extent of private care in the hospitals is determined by the senior doctors (consultants) who are in charge of admissions. These doctors, when acting privately outside their contract with the health board (even in public hospitals), are paid by the private patient.

In 1993, the system was made partially operational for the largest acute hospitals. For the first time budgets incorporated a case-mix adjustment within the allocation process. A blend rate of 5 per cent for in-patient costs was adopted in the first year which meant that sector-average costs only had a 5 per cent weight in determining a budget. The remaining 95 per cent was based on its own past costs. The application of the sector average cost on such a limited basis was intended to herald a gradual movement towards lowering costs. By 1996, despite the evidence of considerable variation in efficiency between hospitals, the importance of sector average costs in determining public hospital budgets has only been raised to 15 per cent for in-patients and an adjustment for day cases of 5 per cent blend had only just been introduced. Even so, this level of ''blending'' has resulted in budget losses of up to 1 per cent for a number of major hospitals. Nevertheless, this case-mix budget adjustment is now applied within the budgeting process for nearly all acute hospitals discharging more than 5 000 patients per year. Moreover, the growth in the budget now depends on the expected workload of the hospital in each category of treatment.

Future changes

The budgeting system

The case-mix system for budgeting is being extended and should be applied to all hospitals discharging more than 5 000 patients per year by 1997. As a

result, the most inefficient hospitals should receive some financial penalty, while more efficient hospitals should get a financial reward, though the adjustments would have to be considered marginal in the context of the overall hospital budget. While the application of a more limited adjustment may have been understandable as a preliminary measure within the first application in 1993, the fact that limited progress has been made in addressing the quantified inefficiencies in the interim must raise questions about the overall commitment to this process.

The application of a case-mix adjustment within a budgeting model is intended to facilitate the estimation of resource deployment by hospital workload in a manner which will enable a meaningful assessment and improvement of hospital efficiency levels. As for the financial penalty or reward related to the efficiency assessment, these should provide the incentives necessary to improve productivity. A critical requirement for the successful operation of this system, however, is that hospitals are in a position to understand the source of the inefficiencies observed. Hospitals are informed about their ranking relative to others within their group and generally have key problems regarding their cost profile identified. With the expertise in dealing with information having improved substantially, it would seem that the use of a case-mix concept is an effective mechanism to improve hospital efficiency.

Restructuring medical training

Some improvement in the overall mix of medical staff would seem to be called for, so that patients are less likely to be seen by staff still seeking further qualifications. One proposal is to reduce the overall number of hospital doctors but to increase the proportion of consultants to 50 per cent from the current 33 per cent level.[54] Achieving such a change would require shortening the time spent by doctors in post-qualification training to about six years. However, as workloads would remain the same, such an increase in consultant numbers would require a major change in work practices. For this reason, it might only be possible to introduce a reform following the creation of a junior consultant grade or an permanent career grade below that of consultant. Major changes in the work practices of consultants would be necessary, as they would have to take on lower level work. To be viable, such changes would require a major reduction in the number of sub-consultant staff, otherwise the payroll for hospital doctors

would increase considerably. As yet no definite proposal has been agreed by the government and the organisations representing hospital doctors and consultants.

Competition within the private insurance market

Competition is now allowed in the health insurance market following Irish compliance with EU legislation. The relevant legislation was passed in June 1994 and the orders for the new regulatory structure for the market were implemented in March 1996. As well as allowing competition, the law enshrines the concept of insurance tariffs that do not vary with age or health. The consequences of this legislation will only be felt in the future, as the first competitor to VHI only entered the market at the beginning of 1997. The new regulations maintain the ban on relating insurance premia to age. They rule out the possibility of pre-inscription medical screening and do not allow the insurance company to terminate policies, except for non-payment. As a result of these provisions, each individual will pay the same premium for a given level of health insurance cover, irrespective of age, sex or health status. Health insurers are required to accept all applications for membership from individuals aged under 65. The regulations do, however, provide for maximum waiting periods in respect of pre-existing medical conditions. They also provide for risk-equalisation payments between the competing insurance companies. Registration of insurance companies is mandatory but approval will be based on the minimum of regulation.

These regulations are to be based on equity principles. They ensure the continued availability of the current type of cover and are aimed at preventing individuals from being under-insured due to lack of proper understanding of the restrictions which might apply to some types of policies. In addition, they will maintain the current level of inter-generational solidarity. While the benefit regulations generally prescribe the minimum level of cover which all health insurance contracts must provide, they do not apply to health insurance contracts solely concerned with ancillary health services or the reimbursement of statutory inpatient costs in publicly-funded hospitals.

The regulations governing risk equalisation are intended to ensure that under a community-rated system with open enrolment, no insurer incurs disproportionately heavy claims because of preferred risk selection by other insurers in the market. Under this system, an insurer with a higher than average risk profile will receive a transfer of funds from the system while those with a lower than average

risk profile will pay into the system. While the operation of the risk equalisation mechanism will only be triggered with the development of serious competition in the market, all insurers are required to return data in a specified format on a quarterly basis from 30 June 1996. These data are essentially concerned with the number of settled inpatient claims in respect of covered persons for the quarter, classified by age, sex, length of stay and benefits paid. The data requirements currently encompassed by the risk equalisation legislation do not cover morbidity-related risk which is outside the scope of the scheme. While the government is satisfied that the scheme will work well, representations from the two competing companies have resulted in the government establishing an independent review of the system.

At present, it is difficult to judge the impact that such competition will have on the insurance market and health provision generally. Some of the regulations, such as those relating to minimum coverage, seem unnecessary on consumer protection grounds, as there is now a universal public hospital service, but may be justified by the need to have an adequate spread of risk to allow community rating to work. Others, such as not relating premia to age, may be justified on equity grounds and the need to minimise the extent to which the elderly population use the public system.

Concluding remarks

The Irish health system is based on a mixture of public and private care which has resulted in good provision of health care at relatively low cost to the taxpayer. Two-thirds of first-level, ambulatory medical care – provided by general practitioners – is paid for by consumers, directly. The state only finances such care for the poorest one-third of the population, while medical insurance payments only cover this type of treatment if outlays are high. On the other hand, since 1991, public hospital care has been available to all groups in society, subject to a daily accommodation charge. Nevertheless, about one-third of all hospital beds are financed by voluntary, private medical insurance and even one-fifth of public sector beds are financed by such payments. As a result, the cost of the system to the tax-payer has been restrained to 6.8 per cent of GNP in 1996, helped also by the increasing productivity of the state hospital system in the past ten years, where the number of cases treated has risen by 3 per cent per year while the annual increase in employment has been only 2 per cent. At the same

time, there has been an improvement in the health of the population, especially for those under 45 whose mortality is now below the European average. However, the life expectation for those aged over 65 is still the lowest in Europe.

Further modernisation of management methods in hospitals may be required. Greater pressure should be brought to bear on high-cost hospitals through an expansion of the so-called case-mix system which relates hospital budgets to average costs (adjusted for the type cases dealt with in a given hospital). Within hospitals, there may be a case for devolution of budgets and spending decisions to medical teams, although there may be reluctance to admit that clinical need cannot be completely independent of available finance. The interface between public and private sectors may also need reforming, to a certain extent. Prices are often set in ways that do not appear to conform with the design of the Irish system. Most consumers are expected to be responsible for their own first-level medical care and yet the pricing of public hospitals encourages them to use these facilities as a source of first-level medical care that is cheaper than private general practitioners. Within hospitals, there appears to be some need to define better the distribution of doctors' time between the public and private sectors. In addition, future action will be required to modernise the system of medical training that is long and results in the need to attract doctors to Ireland at the same time as Irish doctors are emigrating. Finally, more thought needs to be given to the way in which the growth in the cost of publicly-provided first-level medical care can be reduced. One way forward might be to include the costs of drugs within the capitation fee paid to doctors for servicing those patients eligible for state-provided medical care. In this way, further incentives could be introduced to limit the increase in drug costs.

List of acronyms

ALMP	Active labour market policies
DSW	Department of Social Welfare
ESO	Employment Services Offices
FAS	National Training Agency
FIS	Family Income Supplement
IDA	Industrial Development Agency
ILO	International Labour Organisation
IFSC	International Financial Services Centre
P2000	Partnership 2000 for Inclusion, Employment and Competitiveness
PCW	Programme for Work and Competitiveness
PSRI	Pay Related Social Insurance
SST	Specific Skills Training
VHI	Voluntary Health Insurance
VTOS	Vocational Training Opportunities Scheme

Notes

1. The EU inflation index for Ireland, which covers only 85 per cent of the products and services in the overall CPI due to the exclusion of mortgage interest, education and all health expenditure, also moderated to 1.7 per cent in the year to February 1997. Irish inflation was then 0.5 percentage points below the EU average by this measure.
2. For the first time, the 1995 Irish national accounts included all the profits of foreign enterprises. Previous estimates had only taken into account the remitted profits of such companies.
3. A further decision to pre-fund an expected liability of Ir£ 60 million to people who had received blood transfusions contaminated by the hepatitis C virus added to the spending overrun on an EBR basis, though not on a general government basis as no actual payment was made during the year.
4. Expenditure of Ir£ 203 million was added to reduce debt related to pension liabilities assumed by the government when the An Post and Telecom Eireann became public corporations and to supplement a reserve fund for the payment of the accumulated interest on postal savings schemes.
5. An advanced VAT payment system was introduced in 1993 to offset the ending, with the inauguration of the Single Market, of VAT on imports from the EU.
6. This increase has been calculated after adjusting the 1996 base by deducting Ir£ 98 million in respect of Rate Support Payments to local authorities. This payment was reduced in 1997 to reflect new financing arrangements for local authorities.
7. The EU version of the 1993 System of National Accounts, adopted by the United Nations Statistical Convention in February 1993, will be implemented in Europe in 1999 as the 1995 European System of National Accounts (ESA95).
8. However, the definition of the general government deficit will change in 1999, so adding to the deficit.
9. In 1989, they accounted for a combined 11 per cent of the workforce compared with 13.8 per cent when the two categories were summed together.
10. Probabilities estimated by the Department of Finance using registered unemployment data, which, as explained in Box 3, are not directly comparable to the Labour Force Survey statistics used in this chapter.
11. Using registered unemployment data, O'Connell and Sexton (1994) calculated that long-term unemployed in 1993 accounted for 48 per cent of the unemployed men in the 25 to 34 age group compared to 61 per cent of men between 45 and 54.

12. The participation rate of women aged 15 to 24 fell from 55 per cent in 1971 to under 40 per cent in 1995.
13. In the long run, though, the unemployment rate is likely to be unrelated to the size of the labour force since real wages should decline as the population increases to create more jobs.
14. Under Irish law, a person can qualify for citizenship if his or her parents or grandparents were born in Ireland.
15. The remaining 25 per cent gave various reasons such as ''to see the world''. The survey covered 1985/86 school leavers who emigrated in 1992.
16. Since its peak in 1980, however, the birth rate has fallen about a third to 13 per thousand, a level close to the European average. The demographic transition is already evident in the reduced share of the population under the age of 15 from 28.1 per cent in 1988 to 24.6 per cent in 1995. Consequently, the natural increase in the working-age population will begin to slow after the turn of the century.
17. To address these problems, the government established an Expert Working Group on the Integration of the Tax and Social Welfare Systems. The Group issued a report in 1996.
18. The exemption in 1996 was set at Ir£ 3 900 for each adult, plus Ir£ 450 for the first two children and Ir£ 650 for each additional child.
19. Since it is only imposed on income above the exemption limit, the marginal relief rate generates a lower tax bill for low-income workers than the 27 per cent rate, which is imposed on all income above the personal allowance. The marginal rate is phased out when the standard 27 per cent rate results in a lower tax bill. According to the Expert Working Group, the marginal relief system is an effective way of targeting tax relief on low-paid workers.
20. In 1995, an Ir£ 50 exemption had been introduced.
21. The average social insurance tax rate for ten continental EU countries was 37 per cent in 1995 (Department of Social Welfare, 1996).
22. Early retirement and disability benefit schemes are not widely used in Ireland compared to other countries, perhaps due to the availability of a permanent income from unemployment assistance.
23. In contrast, *unemployment benefits* increased only 9 per cent in real terms over the same period. Consequently, the level of unemployment assistance increased from 58 per cent of unemployment benefits in 1977 to 83 per cent in 1994. With the elimination of the pay-related component, payments under the two system were equal in 1995.
24. According to Martin (p. 102, 1996), Greece, Italy, Japan, Portugal, Spain, Sweden and Switzerland do not pay benefits to those unemployed more than three years. This is also the case in Korea.
25. This correlation does not hold at replacement rates above 100 per cent. Persons in this category, however, only account for 1 per cent of the labour force.
26. This is consistent with Secretariat estimates showing that most employees face net replacement rates of between 41 and 60 per cent, OECD, *Employment Outlook,* 1996.
27. This allowance is in addition to the Child Benefit, a universal payment made to all children regardless of the level of family income, employment or marital status.

28. Other secondary benefits include the Back-to-School Allowances, Butter Vouchers, Fuel Allowances and Christmas Bonus Payments.

29. A stricter means test is imposed on part-time workers claiming unemployment benefits than for those receiving unemployment assistance. The benefit is withdrawn after one day's work for an employee earning the average wage.

30. In the public sector, a local bargaining clause was included in the PCW to deal with cases that had not been finalised under the 1991 to 1994 national agreement. However, in most cases, the bulk of the increases will not be paid until 1997.

31. In 1994, 25 550 days were lost due to industrial disputes, before increasing to 130 300 in 1995, largely as a result of a national retail chain store strike.

32. A typical negative feature of centralised wage bargaining is that it does not allow sufficient flexibility in relative wages compared to enterprise-based negotiations, which allow wages to adjust to divergent labour demand developments across different groups, such as low-skilled and young workers. This does not appear to have been the case in Ireland.

33. This section draws on *The OECD Jobs Strategy: Enhancing the Effectiveness of Active Labour Market Policies.*

34. Since April 1996, participation in this programme qualifies a worker for unemployment insurance benefits after the part-time job has ended.

35. See Sabel (1996) for a discussion of the role of local partnerships.

36. This survey shows that public employment is less successful than other ALMPs. In order to say anything more specific about the effectiveness of a scheme, including its deadweight loss and substitution effects, it is necessary to compare the post-programme experience of participants in public employment with a control group. However, there are few, if any, rigorous evaluations of ALMPs in Ireland.

37. *Company Training in Ireland in 1993,* published in 1995.

38. Estimates of the size of the underground economy range from 5 to 10 per cent of GNP (Expert Working Group, 1996).

39. The 1997 budget also introduced a gradual withdrawal of the Adult Dependence Allowance for a spouse that earns more than Ir£ 60 a week in order to alleviate the poverty trap that affects such families when the earnings of the spouse increase.

40. *Report of the Steering Committee on the Future Developments of Higher Education,* HEA, June 1995.

41. According to the Expert Working Group, raising personal allowances to the tax exemption limit for married persons would decrease tax revenue by about Ir£ 290 million a year (0.7 per cent of GDP). In addition, the child exemption limits would need to be phased out as child benefits were increased.

42. This would probably require a legal change to make clear that the child benefit is considered to be the income of the parent. If the benefit was not taxable, it would cost Ir£ 400 million.

43. Stationery Office (1984).

44. There are three main occupational schemes covering the police force, the prison service and the national electricity company employees and their dependents, representing around 3 per cent of the population.
45. See Fahey and Murray (1994).
46. Central Statistics Office (1995).
47. For more discussion of this subject, see National Council for the Elderly (1995).
48. See Report 36 of the Review Body on Higher Remuneration in the Public Sector (1997).
49. See Wiley and Fetter (1990), Stationery Office (1991) and Department of Health (1994), p. 33.
50. A discussion of the data sources used for this analysis may in found in *OECD Health Systems, 1993, Vol. II.*
51. Wiley and Fetter (1990).
52. The National Case Mix project.
53. For more details of the case-mix system see Wiley (1995).
54. See the ''The report on medical manpower'' by Dr. N. Tierney, Department of Health (1993).

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STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	Average 1986-1995	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
A. Percentage changes from previous year											
Private consumption [1]	3.6	2.0	3.3	4.4	6.2	1.1	2.3	4.4	1.7	6.8	3.7
Gross fixed capital formation [1]	2.7	–2.8	–1.1	0.4	13.0	10.5	–7.4	–2.5	–2.1	8.7	10.1
GDP at market prices [1]	4.8	–0.4	4.7	4.3	6.1	7.8	2.1	4.0	3.1	6.5	10.3
GDP price deflator	2.7	6.5	2.2	3.4	5.4	–0.8	1.8	2.2	4.2	1.2	0.5
Industrial production	8.7	2.2	8.9	10.7	11.6	4.7	3.3	9.1	5.6	11.9	18.9
Employment	1.5	0.2	1.0	–0.1	1.2	3.3	–0.2	0.4	1.8	3.5	4.0
Compensation of employees (current prices)	6.8	6.8	5.2	6.0	7.3	8.1	6.6	6.8	8.2	6.4	7.0
Productivity (GDP/employment)	3.3	–0.7	3.7	4.3	4.9	4.3	2.3	3.6	1.2	2.9	6.1
Unit labour costs (Compensation/GDP)	2.0	7.3	0.5	1.6	1.1	0.3	4.4	2.7	4.9	–0.1	–3.0
B. Percentage ratios											
Gross fixed capital formation as % of GDP at constant prices	16.2	18.0	17.0	16.4	17.5	17.9	16.3	15.2	14.5	14.8	14.7
Stockbuilding as % of GDP at constant prices	0.5	1.2	0.0	–0.7	0.6	2.6	2.3	–0.3	–0.4	–0.9	0.1
Foreign balance as % of GDP at current prices	6.7	1.2	4.5	6.4	5.8	4.7	4.9	7.6	10.7	9.7	11.8
Compensation of employees as % of GDP at current prices	49.7	51.7	50.9	50.0	48.0	48.5	49.7	50.0	50.3	49.7	47.9
Direct taxes as percent of household income	14.7	14.7	15.3	16.0	14.0	14.2	14.2	14.1	14.6	14.9	14.6
Household saving as percent of disposable income	11.5	11.7	12.5	10.6	9.3	11.2	12.1	11.6	13.8	10.6	11.7
Unemployment as percent of total labour force	15.0	17.0	16.8	16.1	14.6	13.2	14.7	15.5	15.6	14.1	12.2
C. Other indicator											
Current balance (billion dollars)	0.4	–0.8	–0.1	0.0	–0.6	–0.4	0.3	0.5	1.8	1.4	1.4

1. At constant 1990 prices.
Source: OECD.

Table B. **Expenditure on gross national product, current prices**

Ir£ million

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Private consumption	12 431	13 155	14 282	15 774	16 280	17 147	18 358	19 012	20 836	22 055
Public consumption	3 532	3 565	3 530	3 672	4 067	4 481	4 844	5 248	5 581	5 883
Gross fixed investment	3 385	3 422	3 620	4 320	4 881	4 664	4 688	4 734	5 249	6 031
Final domestic demand	19 348	20 142	21 432	23 767	25 228	26 292	27 890	28 994	31 666	33 969
Stockbuilding	142	11	–150	176	720	621	–95	–138	–208	94
Total domestic demand	19 490	20 153	21 282	23 942	25 947	26 913	27 796	28 856	31 458	34 063
Exports	10 416	11 899	13 684	16 196	16 175	16 984	18 881	22 033	25 170	30 016
Imports	10 186	10 959	12 230	14 732	14 891	15 587	16 597	18 573	21 796	25 463
Foreign balance	229	940	1 453	1 464	1 284	1 397	2 284	3 461	3 374	4 553
GDP (market prices)	19 734	21 109	22 754	25 459	27 231	28 310	30 080	32 316	34 833	38 616
Net factor income from abroad	–1 861	–1 946	–2 486	–3 039	–2 921	–2 796	–3 209	–3 521	–3 575	–4 815
GNP (market prices)	17 873	19 163	20 268	22 420	24 311	25 513	26 870	28 796	31 258	33 801
Memorandum:										
Composition of stockbuilding										
Agriculture	. .	. .	. .	. .	80	35	83	–1	58	79
Non-agriculture (incl. EEC intervention stocks)	. .	. .	. .	. .	640	587	–178	–137	–265	14

Source: CSO, *National Income and Expenditure*, 1996.

Table C. **Expenditure on gross national product, constant 1990 prices**

Ir£ million

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Private consumption	14 043	14 510	15 154	16 095	16 280	16 658	17 395	17 685	18 889	19 595
Public consumption	4 310	4 101	3 897	3 860	4 067	4 187	4 290	4 339	4 486	4 622
Gross fixed investment	3 936	3 891	3 908	4 416	4 881	4 520	4 408	4 314	4 687	5 160
Final domestic demand	22 288	22 502	22 958	24 371	25 228	25 366	26 092	26 337	28 062	29 377
Stockbuilding	256	3	–158	152	720	641	–78	–130	–279	52
Total domestic demand	22 544	22 505	22 800	24 523	25 947	26 007	26 014	26 207	27 783	29 429
Exports	10 902	12 398	13 496	14 888	16 175	17 033	19 334	21 208	24 097	28 205
Imports	11 228	11 921	12 506	14 189	14 891	15 235	16 433	17 611	20 132	22 616
Foreign balance	–326	476	990	698	1 284	1 798	2 901	3 597	3 965	5 589
GDP (market prices)	21 812	22 830	23 803	25 259	27 231	27 805	28 915	29 804	31 748	35 018
Net factor income from abroad	–1 893	–1 949	–2 440	–2 777	–2 921	–2 804	–3 287	–3 389	–3 423	–4 524
GNP (market prices)	19 919	20 880	21 364	22 482	24 311	25 000	25 629	26 415	28 325	30 494
Memorandum:										
Composition of stockbuilding										
Agriculture	. .	. .	. .	. .	80	39	88	–4	54	74
Non-agriculture (incl. EEC intervention stocks)	. .	. .	. .	. .	640	602	–165	–61	–333	–21

Sources: CSO, *National Income and Expenditure*, 1996.

Table D. **Agricultural output and income**

Ir£ million

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Gross agricultural output [A]	2 739	2 721	2 872	3 157	3 377	3 227	3 136	3 354	3 431	3 401	3 573
Inputs [B]	1 262	1 289	1 179	1 229	1 348	1 351	1 368	1 380	1 441	1 563	1 653
Gross agricultural product[1] [C = A – B]	1 477	1 432	1 693	1 928	2 029	1 876	1 768	1973	1990	1 838	1 920
Other expenses less subsidies [D][2]	258	290	304	287	303	173	230	168	150	–99	–149
INCOME (self-employed) [= C – D]	1 219	1 142	1 389	1 641	1 726	1 703	1 539	1 806	1 839	1 937	2 069
Annual changes	–9.5	–6.3	21.6	18.1	5.2	–1.3	–9.7	17.3	1.9	5.3	6.8
Volume changes:											
Gross agricultural output	–1.5	–1.2	1.2	1.6	2.2	7.4	0.3	5.2	–3.0	–1.1	3.4
Inputs	1.4	6.3	–3.7	0.9	5.5	1.8	1.1	1.6	4.3	7.4	3.6
Gross agricultural product	–3.4	–7.7	5.9	2.3	–1.0	12.9	–0.3	7.8	–8.0	–7.8	3.3
Price changes:											
Gross agricultural output	–2.0	0.5	4.3	8.2	4.4	–11.0	–3.1	1.7	5.5	0.2	1.7
Inputs	2.2	–4.0	–5.2	3.2	4.0	–0.7	0.1	–0.7	0.1	1.0	2.1
Gross agricultural product	–5.5	5.0	11.7	11.4	5.7	–18.5	–5.4	3.5	9.6	0.2	1.1

1. At market prices. Gross agricultural product measures value added.
2. Depreciation plus wages and salaries plus land annuities plus agricultural levies less subsidies.

Source: CSO, *Statistical Abstract.*

Table E. **Prices and wages**

Index 1990 = 100

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Agricultural prices	97.2	107.4	112.8	100.0	96.4	97.8	104.2	105.8	108.1	102.8
Livestock price index	101.1	113.7	112.3	100.0	95.5	94.1	99.7	102.6	101.0	93.2
Consumer prices	91.1	93.0	96.9	100.0	103.2	106.5	108.0	110.5	113.3	115.1
Wholesale prices [1]	93.6	97.4	102.9	100.0	101.2	102.1	106.9	108.0	110.3	. .
Industrial prices [2]	93.1	96.9	101.6	100.0	100.8	102.5	107.2	108.4	111.1	111.9
Average hourly earnings in manufacturing	88.3	92.1	95.7	100.0	105.6	110.5	116.9	119.0	121.7	126.0

1. General wholesale price index.
2. Output of manufacturing industry.

Source: CSO, *Economic Series.*

Table F. **Household appropriation account**[1]

Ir£ million

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
Total personal income	15 930	16 857	18 464	19 767	20 958	23 021	24 516	26 108	27 955	29 593
of which:										
Agricultural	1 402	1 332	1 586	1 870	1976	1 966	1 824	2 112	2 168	2 282
Non-agricultural employee compensation	9 465	10 113	10 679	11 309	12 104	13 028	13 883	14 833	16 059	17 096
Transfers	3 129	3 419	3 791	3 917	3 862	4 095	4 558	4 848	5 213	5 480
Direct taxation	3 332	3 625	3 999	4 464	4 325	4 659	4 981	5 330	5 880	6 284
(as percentage of income)	20.9	21.5	21.7	22.6	20.6	20.2	20.3	20.4	21.0	21.2
Personal disposable income	12 598	13 232	14 466	15 303	16 633	18 362	19 536	20 778	22 075	23 309
Savings ratio (percentage)	15.9	14.6	11.2	9.7	7.5	11.3	12.2	11.6	13.9	10.6
Personal consumption	10 598	11 306	12 845	13 811	15 378	16 280	17 147	18 358	19 012	20 836
Memorandum:										
Real disposable income[2]	14 891	14 952	15 956	16 238	16 973	18 362	18 979	19 688	20 534	21 131
(percentage change)	1.7	0.4	6.7	1.8	4.5	8.2	3.4	3.7	4.3	2.9

1. Break in series in 1990.
2. Deflated by personal consumption deflator.

Source: CSO, *National Income and Expenditure*, OECD estimates.

Table G. **General government revenue and spending**

	1970	1980	1990	1994	1995	1996
	Percentage of GNP					
A. General government[1]						
Current receipts	34.7	38.8	40.7	42.8	40.7	41.8
Non-interest expenditure	29.6	38.5	36.0	39.9	38.8	38.2
Primary budget balance	5.1	0.3	6.1	4.6	3.5	4.7
Net property income	–4.0	–6.6	–8.7	–6.6	–5.9	–5.7
General government budget balance	1.1	–6.3	–2.6	–2.0	–2.4	–1.1
B. The structure of expenditure and taxation						
General government primary expenditure	40.0	52.9	44.4	44.8	43.3	43.0
Current consumption	15.3	20.9	18.4	19.5	18.9	18.5
Transfers to persons	9.0	12.9	16.0	17.5	17.2	17.2
Subsidies	4.8	3.8	1.3	1.2	1.3	1.6
Capital formation (gross)	4.3	5.9	2.3	2.7	2.7	2.8
General government expenditure by function						
Education	5.1	6.0	5.6	6.3	. .	. .
Transportation	3.4	4.7	2.3	2.9	. .	. .
Health	4.5	8.0	6.3	7.1	. .	. .
Defence	–	–	1.5	1.4	. .	. .
Agriculture	–	–	2.0	2.0	. .	. .
Total tax revenue	30.9	34.4	39.8	42.0	40.4	40.9
Income tax	8.9	12.6	14.4	16.6	15.6	16.5
of which:						
Personal	6.2	11.0	12.5	13.0	12.2	12.5
Corporate	2.7	1.5	2.0	3.6	3.4	3.9
Social security	2.5	4.9	5.8	6.0	5.8	5.5
Consumption tax	19.1	16.8	17.0	16.5	16.5	16.7
Tax rates (per cent)						
Top rate of income tax	35[2]	60	53	48	48	48
Lower rate of income tax	–	25	30	27	27	27
Corporate tax rate (full rate)	5[3]	45	43	40	38	38
Consumption tax rate (standard rate)	25[4]	25	23	21	21	21

1. National accounts basis
2. Standard rate. A variable surtax was also applied.
3. Turnover tax.
4. Wholesale tax.

Source: Department of Finance, CSO, *National Income and Expenditure*, OECD.

Table H. **Public sector investment**

Ir£ million

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Total public investment	1 335	1 391	1 653	1 687	1 842	2 084	2 230	2 633	3 126	3 425
(as % of GNP)	*6.7*	*6.3*	*6.8*	*6.6*	*6.9*	*7.2*	*7.1*	*7.8*	*8.6*	*8.9*
State-owned companies	534	616	733	722	843	902	961	1 124	1 398	1 514
(as % of GNP)	*2.7*	*2.8*	*3.0*	*2.8*	*3.1*	*3.1*	*3.1*	3.3	*3.9*	*3.9*
Banks	131	188	188	195	256	232	271	502	576	546
Telecommunications	135	154	167	152	159	165	185	188	261	290
Electricity	96	80	92	110	138	171	193	146	202	256
Land and sea transport	17	19	35	38	57	73	130	138	155	175
Air transport	96	112	182	159	100	49	61	36	90	111
Gas	34	21	23	41	112	177	62	49	46	82
Postal services	8	13	8	11	13	21	22	15	24	24
Peat	3	5	12	10	6	7	8	8	12	14
Television	5	6	7	5	6	6	17	19	17	12
Oil	0	0	0	0	0	0	11	23	16	4
Irish Steel	1	0.9	0.7	1.3	7.4	0.6	1.8	0	0	0
Siuicre Eireann	8	17	18	0	0	0	0	0	0	0
Aids to private sector	353	318	363	388	384	406	455	503	643	697
(as % of GNP)	*1.8*	*1.4*	*1.5*	*1.5*	*1.4*	*1.4*	*1.5*	*1.5*	*1.8*	*1.8*
Industry and services	155	161	193	223	205	228	189	219	273	300
Agriculture, fisheries, forestry	58	69	111	120	118	104	175	164	222	244
Housing	140	88	58	45	61	74	91	120	148	153
Government investment	449	457	557	577	615	776	813	1 006	1 085	1 214
(as % of GNP)	*2.2*	*2.1*	*2.3*	*2.3*	*2.3*	*2.7*	*2.6*	*3.0*	*3.0*	*3.2*
Roads	121	164	189	207	234	305	248	290	312	367
Water and sewage	63	65	73	75	75	121	125	109	122	151
Housing	57	48	64	79	82	127	205	195	210	234
Education and health	105	102	113	104	117	126	164	203	197	197
Other	103	78	119	112	107	97	72	209	241	264

Source: Budget 1994 and earlier issues; Public Capital Programme, Summary version 1996, 1997.

Table I. **The relationship between the Exchequer Borrowing Requirement and the Maastricht definition of government borrowing**

Ir£ million

	1992	1993	1994	1995	1996
Exchequer Borrowing Requirement	–713	–690	–672	–627	–437
Excluding: Loan disbursements	104	242	282	146	45
Loan repayments	150	337	301	401	–127
Net Exchequer Borrowing Requirement	–759	–785	–691	–882	–519
Plus: accruals adjustment, expenditure	–43	–52	–117	–103	159
accruals adjustment, revenue	18	–67	17	–27	4
Net Exchequer Borrowing, accruals basis	–784	–904	–791	–1 012	–356
Plus transactions of off-budget funds:					
Interest paid by exchequer vs Central Government interest	52	49	8	199	75
Gross trading income of semi-state bodies	–8	–18	–31	–20	4
Other transactions	–14	6	75	–22	–60
Net central government borrowing (–)	–754	–867	–739	–855	–337
Net local government borrowing (–) / lending (+)	–5	56	121	33	–40
Net social security fund borrowing (–) / lending (+)	5	1	7	6	–16
Net general government borrowing (Maastricht target variable)	–754	–810	–611	–816	–394
Memorandum item:					
As per cent of nominal GNP					
Exchequer Borrowing Requirement	–2.7	–2.4	–2.1	–1.9	–1.2
Net general government borrowing	–2.8	–2.8	–2.0	–2.4	–1.1
As per cent of nominal GDP					
Exchequer Borrowing Requirement	–2.3	–2.1	–1.9	–1.6	–1.0
Net general government borrowing	–2.5	–2.4	–1.7	–2.0	–0.9

Note: Borrowing is shown as a negative item.
Source: Ministry of Finance.

Table J. **Public sector balance sheet**

Ir£ million

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996[1]
Liabilities										
General government debt[2]	24 491	25 738	26 257	26 422	27 343	28 274	31 370	31 690	32 755	32 002
Per cent of GDP (ESA79)	116.2	113.6	103.4	95.8	95.0	92.0	94.5	87.9	81.6	72.8
Small Investor savings	1 212	1 567	1 854	2 019	2 258	2 491	2 843	3 404	3 973	4 554
National government debt (gross)										
Total	24 450	25 361	25 857	25 933	26 793	27 128	29 932	30 059	31 186	30 702
Domestic	14 757	15 862	16 689	17 085	17 665	16 273	17 658	18 541	20 280	21 984
of which:										
Held by non-residents	1 758	2 665	3 667	3 859	4 161	2 321	4 047	3 679	4 352	5 370
External	9 693	9 498	9 167	8 848	9 128	10 856	12 275	11 518	10 906	8 718
Public enterprises										
Total	4 783	4 789	4 827	5 133	5 174	5 213	5 549	5 691	. .	. .
Domestic	2 512	2 750	2 718	2 865	2 969	3 051	3 619	3 644	. .	. .
External	2 271	2 039	2 109	2 268	2 205	2 162	1 930	2 047	. .	. .
Assets										
Official reserves	2 821	3 161	2 521	2 892	3 256	2 113	4 278	4 041	5 473	4 960
Domestic liquid assets	756	749	954	850	1 146	51	686	291	634	790
Foreign liquid assets	0	0	45	0	269	734	889	540	343	0
Government loans	4 015	1 319	1 098	973	954	934	940	737	514	. .
Net worth of public enterprises	1 127	1 272	2 005	2 523	2 707	2 651	2 665	2 839	2 886	. .
Total of above	8 719	6 501	6 623	7 238	8 332	6 483	9 458	8 588	9 979	. .

1. Provisional.
2. Figures prior to 1990 contain an element of estimation.

Source: European Commission; Central Bank; Department of Finance; CSO, *Statistical Abstract*, budget 1994.

Table K. **Balance of payments**

OECD basis; US$ million

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Exports, fob	12 297	15 536	18 378	20 347	23 303	23 607	28 082	28 479	33 483	43 949
Imports, fob	11 158	12 934	14 550	16 332	19 342	19 293	21 048	20 371	24 187	30 621
Trade balance	1 138	2 602	3 829	4 015	3 961	4 315	7 034	8 108	9 295	13 328
Services, net	–3 236	–3 999	–5 278	–6 079	–6 667	–6 565	–8 609	–8 196	–9 596	–13 536
Balance on goods and services	–2 098	–1 397	–1 449	–2 064	–2 706	–2 250	–1 575	–88	–301	–208
Private transfers, net	1 622	1 682	1 810	1 882	2 062	1999	1 608	1 443	1 463	1 518
Official transfers, net	–367	–354	–362	–381	273	588	511	472	263	262
Current balance	–842	–68	–2	–564	–371	337	545	1 827	1 425	1 571
Long-term capital (excluding special transactions)	1 061	849	–347	–1 212	–2 913	–2 513	–3 968	553	–4 224	. .
a) Private[1]	1 432	–834	271	–799	–2 676	–2 299	–5 108	1 155	–2 725	. .
b) Official	–371	1 683	–618	–412	–237	–214	1 140	–602	–1 499	. .
Basic balance	243	753	–264	–1 701	–2 850	–1 029	–1 493	3 678	–1 169	. .
Non-monetary short-term official capital	0	0	0	–3	0	–3	1 251	–1 183	–93	. .
Errors and omissions	–1 078	349	347	1 135	2 487	2 177	511	1 341	799	. .
Balance on non-monetary transactions	–835	1 102	83	–569	–363	1 145	269	3 836	–463	. .
Private monetary institutions' short-term capital	735	–213	441	–333	1 212	–695	–2 298	–1 270	314	. .
Balance on official settlements	–92	895	537	–907	733	451	–2 044	2 569	–152	. .

1. Includes non-monetary short-term private capital.

Source: Direct communication to the OECD.

Table L. **Foreign trade and payments**

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Imports, Ir£ million	9 428	8 621	9 155	10 215	12 284	12 469	12 851	13 195	14 885	17 283	20 420
Exports, Ir£ million	9 743	9 374	10 724	12 305	14 597	14 337	15 019	16 744	19 830	22 753	27 726
Trade balance, Ir£ million	315	753	1 568	2 090	2 313	1 868	2 168	3 549	4 945	5 470	7 306
Volume of imports (1990 = 100)	72.3	74.4	79.1	82.8	93.6	100	100.8	105.6	113.0	127.9	144.8
Volume of exports (1990 = 100)	65.2	67.6	77.4	82.9	92.2	100	105.6	121.1	133.4	153.2	183.4
Import unit values (1990 = 100)	104.5	92.9	92.8	98.9	105.3	100	102.3	100.2	105.4	108.1	112.7
Export unit values (1990 = 100)	104.3	96.7	96.6	103.6	110.5	100	99.3	96.6	103.9	103.8	105.7
Terms of trade	99.8	104.1	104.1	104.8	104.9	100	97.1	96.4	98.6	96.0	93.8
Official external reserves, Ir£ million [1]	2 272	2 205	2 821	3 161	2 521	2 892	3 256	2 113	4 278	4 041	5 473

1. At end of year.

Source: CSO, *Economic Series;* Central Bank of Ireland, *Quarterly Bulletin.*

Table M. **Foreign trade by commodities**

$ million

	1987	1988	1989	1990	1991	1992	1993	1994	1995
	Exports, f.o.b.								
SITC Section									
0. Food and live animals	3 986	4 412	4 544	4 740	4 892	6 246	5 660	6 334	7 705
1. Beverages and tobacco	353	383	422	542	572	642	639	624	747
2. Crude materials, inedible, except fuels	630	802	854	819	758	797	673	813	888
3. Mineral fuels, lubricants and related materials	115	103	103	156	151	172	173	167	194
4. Animal and vegetable oils and fats	15	17	18	18	19	24	26	29	38
5. Chemicals	1 910	2 435	2 932	3 748	4 253	5 440	5 531	7 070	8 070
Of which:									
Pharmaceuticals	377	560	696	960	1 150	1 532	1 392	1 837	2 106
6. Manufactured goods, classified by materials	1 400	1 590	1 638	1 922	1 956	2 150	1 680	1 874	2 109
7. Machinery and transport equipment	5 011	5 846	6 593	7 447	7 124	7 637	8 327	10 230	15 128
Of which:									
Office equipment	3 340	3 446	4 114	4 580	4 010	4 446	5 281	6 081	9 347
8. Miscellaneous manufactured articles	1 943	2 439	2 778	3 379	3 615	4 358	4 025	5 058	5 901
9. Commodities and transactions not classified elsewhere	608	710	802	1 026	889	865	2 082	1 871	3 011
Total	15 970	18 736	20 685	23 796	24 229	28 331	28 814	34 071	43 790

Table M. **Foreign trade by commodities** *(cont.)*

$ million

	1987	1988	1989	1990	1991	1992	1993	1994	1995
	Imports, c.i.f.								
STIC Section									
0. Food and live animals	1 466	1 625	1 636	1 862	1985	2 180	1 792	2 088	2 299
1. Beverages and tobacco	155	192	198	256	295	352	278	311	303
2. Crude materials, inedible, except fuels	383	429	467	545	506	485	468	578	613
3. Mineral fuels, lubricants and related materials	1 007	870	963	1 341	1 227	1 177	1 042	999	1 071
4. Animal and vegetable oils and fats	56	66	69	66	66	70	78	97	122
5. Chemicals	1 676	1 958	2 149	2 563	2 744	2 901	2 656	3 312	4 119
Of which:									
Pharmaceuticals	260	317	342	425	504	538	442	580	724
6. Manufactured goods, classified by materials	2 147	2 528	2 653	3 231	3 148	3 322	2 550	2 979	3 684
7. Machinery and transport equipment	4 561	5 357	6 591	7 442	7 202	8 010	7 998	9 920	13 652
Of which:									
Office equipment	1 497	1 493	1 822	1 951	1 882	2 247	2 614	3 344	5 596
8. Miscellaneous manufactured articles	1 745	2 047	2 197	2 826	2 999	3 343	2 876	3 134	3 492
9. Commodities and transactions not Classified elsewhere	419	486	492	583	579	630	1 930	2 268	2 951
Total	13 614	15 558	17 416	20 716	20 751	22 469	21 667	25 686	32 307

Source: OECD, *Foreign Trade Statistics.*

Table N. **Production structure and performance indicators**

	Per cent share of GDP at constant factor cost			
	1970	1980	1990	1995
A. Production structure				
Agriculture and fishing	16.9	11.6	9.7	7.9
Industry	36.2	37.3	35.6	41.1
of which:				
Selected modern industries [1]	. .	. .	12.4	18.2
Distribution, transport and communication	19.0	17.7	19.4	17.2
Public administration and defense	5.8	6.8	5.6	4.8
Other domestic	26.6	31.2	33.8	33.7
Adjustment for financial services	–2.5	–4.5	–4.1	–4.7
	Per cent share of total employment			
	1971	1981	1991	1996
B. Employment structure				
Agriculture, forestry and fishing	25.9	16.6	13.8	10.6
Mining and quarrying	1.0	1.0	0.5	0.4
Manufacturing	20.3	20.9	19.0	19.1
Electricity, gas and water	1.3	1.3	1.0	1.1
Construction	8.0	9.0	6.7	6.7
Services	43.5	51.3	59.0	62.1
of which:				
Transport, storage and communication	5.7	6.1	6.0	6.2
Commerce, insurance and finance	16.3	18.3	20.5	21.2
Community, social and personal services	21.5	26.9	32.4	34.6
	Per cent share of gross domestic fixed capital formation at constant prices			
	1977	1980	1990	1995
C. Investment structure				
By Industry:				
Agriculture, forestry and fishing	15.0	11.3	10.7	8.7
Fuel and power products	6.0	7.1	3.2	4.5
Manufacturing	20.5	22.7	19.3	15.7
Building and construction	3.8	4.9	2.4	2.0
Market services	47.5	47.6	57.8	62.3
Dwelling	–	–	23.6	32.5
Other market services	–	–	34.2	29.9
Non-market services	7.0	6.4	6.6	6.7
By Asset:				
Dwelling	24.3	21.3	23.6	32.5
Roads	2.0	2.1	4.3	5.4
Other building and construction	23.3	28.4	27.7	25.6
Transport and equipment	13.3	11.5	19.1	16.4
Agricultural machinary	5.3	2.6	2.0	2.2
Other machinary and equipment	31.7	34.1	23.3	17.9

1. Chemicals, computers and instrument engineering and electrical engineering.

Source: CSO, *National Income and Expenditure, Labour Force Survey, Statistical Abstract.*

Table O. **Labour market indicators**

	1991	1992	1993	1994	1995	1996
A. Labour market performance						
Standardised unemployment rate[1]	14.7	15.1	15.7	14.8	12.2	11.9
Unemployment rate: Male	14.2	15.0	15.6	14.7	12.1	11.9
Female	15.5	15.2	15.9	14.8	12.2	11.9
Youth	21.7	23.1	25.1	23.2	19.1	18.2
Share of long-term unemployment in total unemployment[2]	59.4	56.3	56.9	60.7	58.2	57.6
	1971	1981	1985	1990	1993	1996
B. Structural or institutional characteristics						
Participation rate:[3] Total	60.9	60.2	59.7	60.2	60.9	62.3
Male	89.0	84.9	83.0	77.7	76.1	75.7
Female	32.2	34.8	35.9	42.6	45.5	48.8
Employment/population (15-64 years)	56.7	53.7	49.1	52.3	51.1	54.8
Average hours worked (in manufacturing)[4]	42.7	41.3	41.2	43.0	41.9	..
	1983	1986	1991	1992	1993	1996
Part-time work (as per cent of employment)	5.0	6.1	8.3	9.0	10.8	11.5
Employees (percentage of total employment)	79.3**	78.5	78.6	77.6	78.1	79.1
Non wage labour costs[5] (percentage of total compensation)	–	–	6.4	6.5	6.7	6.5

** 1981.
1. Based on ILO definitions using the Labour Force Survey, April.
2. People looking for a job for one year or more.
3. Participation rate = Labour force (15-64 years) / population group (15-64 years).
4. 1971 figure is estimated on basis of the change in hours in transportable goods.
5. Employers' contributions to social security and pensions funds.

Source: Department of Finance; CSO, *Labour Force Survey*.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Greece
Population											
Total	Thousands	1994	17 840	8 031	10 124	29 251	5 206	5 088	57 960	81 407	10 430
Inhabitants per sq. km	Number	1994	2	96	332	3	121	15	106	228	79
Net average annual increase over previous 10 years	%	1994	1.4	0.6	0.3	1.6	0.2	0.4	0.5	2.9	0.5
Employment											
Total civilian employment (TCE)[2]	Thousands	1994	7 680 (93)	3 737	3 724 (92)	13 292	2 508	2 015	21 781 (93)	35 894	3 790
of which: Agriculture	% of TCE	1994	5.3 (93)	7.2	2.6 (92)	4.1	5.1	8.3	5.1 (93)	3.3	20.8
Industry	% of TCE	1994	23.7 (93)	33.2	27.7 (92)	22.6	26.8	26.8	27.7 (93)	37.6	23.6
Services	% of TCE	1994	71 (93)	59.6	69.7 (92)	73.3	68.1	64.9	67.2 (93)	59.1	55.5
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1994	331.6	198.1	227.9	544	146.7	97.2	1 328.5	1 832.3	73.1 (93)
Per capita	US$	1994	18 588	24 670	22 515	18 598	28 181	19 106	22 944	27 826	7 051 (93)
At current prices using current PPP's[3]	Bill. US$	1994	327.9	162.3	204.2	596.7	107	82.5	1 111.8	1 601.7	118
Per capita	US$	1994	18 382	20 210	20 166	20 401	20 546	16 208	19 201	24 325	1 450
Average annual volume growth over previous 5 years	%	1994	2.2	2.5	1.6	1.1	1.9	–1.6	1.1	2.6	1.4 (93)
Gross fixed capital formation (GFCF)	% of GDP	1994	21.4	24.8	17.4	18.7	14.8	14.3	18.1	18.5	17.4 (93)
of which: Machinery and equipment	% of GDP	1993	9.8	8.7	7.8	6.2	7.2	5.90	8.10	7.5	7.8
Residential construction	% of GDP	1993	5.2	6.3	4.6	6.1	2.9	3.7	4.9	6.5	3.6
Average annual volume growth over previous 5 years	%	1994	0.8	3.7	0.4	–0.1	–2.8	–12.9	–1	0.8	2.7 (93)
Gross saving ratio[4]	% of GDP	1994	16.8	25.3	22	16	17	16.6	19	21	15.5 (93)
General government											
Current expenditure on goods and services	% of GDP	1994	17.5	18.8	15	20.2	25.3	22.4	19.6	17.7	19.1 (93)
Current disbursements[5]	% of GDP	1993	36.9	48.4	55.3	49	61.1	58.9	51.5	45.6	51.2
Current receipts	% of GDP	1993	33.5	48.6	50.1	43	58.3	52.5	46.8	45.7	40.2
Net official development assistance	% of GNP	1993	0.34	0.31	0.41	0.46	1.03	0.76	0.66	0.44	..
Indicators of living standards											
Private consumption per capita using current PPP's[3]	US$	1993	10 803	10 546	12 090	11 863	10 042	8 814	11 395	10 733	6 367
Passenger cars, per 1 000 inhabitants	Number	1990	430	382	387	469	311	386	413	480[8]	169
Telephones, per 1 000 inhabitants	Number	1991	464	432	410	586	577	544	511	420[8]	413
Television sets, per 1 000 inhabitants	Number	1991	480	478	451	639	536	501	407	556[8]	197
Doctors, per 1 000 inhabitants	Number	1993	2.2 (91)	2.3	3.7	2.2	2.8 (92)	2.6 (92)	2.8	3.2 (92)	3.8 (92)
Infant mortality per 1 000 live births	Number		6.1	6.5	8	6.8	5.4	4.4	6.5	5.8	8.5
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1994	3	5.5	3.7	3.3	3.5	4.8	3.5	5.2	14.6
Consumer prices	%	1994	3	3.4	2.8	2.8	2.1	3.3	2.5	3.3	16.2
Foreign trade											
Exports of goods, fob*	Mill. US$	1994	47 363	44 881	137 259[7]	165 358	41 850	29 514	235 337	422 243	8 958
As % of GDP	%	1994	14.3	22.7	60.2	30.4	28.5	30.4	17.7	23	11.5 (93)
Average annual increase over previous 5 years	%	1994	5	6.7	6.5	7.1	8.3	4.9	5.6	4.4	3.4
Imports of goods, cif*	Mill. US$	1994	49 731	55 071	126 006[7]	148 297	35 932	23 091	220 508	376 566	21 111
As % of GDP	%	1994	15	27.8	55.3	27.3	24.5	23.8	16.6	20.6	30.1 (93)
Average annual increase over previous 5 years	%	1994	4	7.2	5	5.4	6.1	–1.3	3.5	6.9	5.4
Total official reserves[6]	Mill. SDRs	1994	7 730	11 523	9 505[7]	8 416	6 203	7 304	17 986	52 994	9 924
As ratio of average monthly imports of goods	Ratio	1994	1.9	2.5	0.9	0.7	2.1	3.8	1	1.7	5.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Iceland	Ireland	Italy	Japan	Luxembourg	Mexico	Netherlands	New Zealand
Population										
Total	Thousands	1994	267	3 571	57 190	124 960	398	93 010	15 382	3 526
Inhabitants per sq. km	Number	1994	3	51	190	331	153	47	377	13
Net average annual increase over previous 10 years	%	1994	1.1	0.1	0	0.4	0.8	2	0.6	0.8
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	138	1 168 (93)	20 152 (93)	64 530	162 (91)	32 439	6 631	1 560
of which: Agriculture	% of TCE	1994	9.4	12.7 (93)	7.5 (93)	5.8	3.7 (91)	25.8	4	10.4
Industry	% of TCE	1994	26.1	27.7 (93)	33 (93)	34	31.5 (91)	22.2	23	25
Services	% of TCE	1994	65.2	59.7 (93)	59.6 (93)	60.2	64.8 (91)	52.1	73	64.6
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	6.2	52	1 017.8	4 590	10.6 (92)	371.2	334.3	51.2
Per capita	US$	1994	23 199	14 550	17 796	36 732	27 073 (92)	3 991	21 733	14 513
At current prices using current PPP's[3]	Bill. US$	1994	5.1	54.3	1 068.4	2 593.7	11.7	673.3	285.9	57.3
Per capita	US$	1994	19 271	15 212	18 681	20 756	29 454	7 239	18 589	16 248
Average annual volume growth over previous 5 years	%	1994	0.6	4.7	1	2.1	4.1 (92)	3	2.3	2.5
Gross fixed capital formation (GFCF)	% of GDP	1994	15.2	15.1	16.4	28.6	20.4 (93)	20.7	19.3	20
of which: Machinery and equipment	% of GDP	1993	3.9	6.3	7.4	11.5	..	9.4	8.6	9.3
Residential construction	% of GDP	1993	4.4	4.1	5.3	5.4	..	4.9	5.1	4.9
Average annual volume growth over previous 5 years	%	1994	–4	1	–2.3	1.4	6.5 (92)	7.7	0.4	5.8
Gross saving ratio[4]	% of GDP	1994	16.9	19.5	18.8	31.2	60.2 (92)	15.1	24.4	20.7
General government										
Current expenditure on goods and services	% of GDP	1994	20.6	16	17.1	9.8	17.1 (92)	11.8[9]	14.2	14.7
Current disbursements[5]	% of GDP	1993	34.9	..	53.2	26.9	..	..	55.4	..
Current receipts	% of GDP	1993	35.9	..	47.1	32.9	..	..	54.5	..
Net official development assistance	% of GNP	1993	..	0.15	0.42	0.27	0.34 (92)	..	0.88	0.22
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	11 546	7 750	11 029	11 791	15 545	4 853	10 726	9 266
Passenger cars, per 1 000 inhabitants	Number	1990	464	228	478	282	470	85	356	440
Telephones, per 1 000 inhabitants	Number	1991	527	300	400	454	511	70	477	436
Television sets, per 1 000 inhabitants	Number	1991	319	276	421	613	267	148	485	443
Doctors, per 1 000 inhabitants	Number	1993	3	1.7 (92)	1.7 (91)	1.7 (92)	2.1 (92)	1	2.5 (90)	2
Infant mortality per 1 000 live births	Number		4.8	5.9	7.3	4.5 (92)	8.5 (92)	18	6.3	7.3
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	..	4.6	5.9	2.4	..	5.3	3.2	2.1
Consumer prices	%	1994	6.3	2.7	5.2	2	3.1	16.1	2.8	2.5
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	1 628	34 125	189 802	396 149	..	60 882	155 084	12 169
As % of GDP	%	1994	26.3	65.7	18.6	8.6	..	16.4	46.4	23.8
Average annual increase over previous 5 years	%	1994	2.7	10.5	6.2	7.6	..	21.7	7.6	6.5
Imports of goods, cif*	Mill. US$	1994	1 464	25 812	167 690	274 916	..	79 346	139 800	11 859
As % of GDP	%	1994	23.6	49.7	16.5	6	..	21.4	41.8	23.2
Average annual increase over previous 5 years	%	1994	0.7	8.1	1.9	5.5	..	25.5	6.1	6.1
Total official reserves[6]	Mill. SDRs	1994	201	4 189	22 102	86 214	..	4 301	23 655	2 540
As ratio of average monthly imports of goods	Ratio	1994	1.6	1.9	1.6	3.8	..	0.7	2	2.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population										
Total	Thousands	1994	4 337	9 900	39 150	8 781	6 994	60 573	58 375	260 651
Inhabitants per sq. km	Number	1994	13	107	78	20	169	78	238	28
Net average annual increase over previous 10 years	%	1994	0.5	–0.1	0.2	0.5	0.7	2.1	0.3	1
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	1970 (93)	4 372	11 760	3 926	3 772	19 664	25 044 (93)	123 060
of which: Agriculture	% of TCE	1994	5.6 (93)	11.5	9.8	3.4	4	44.8	2.2 (93)	2.9
Industry	% of TCE	1994	23.1 (93)	32.8	30.1	25	28.8	22.2	26.2 (93)	24
Services	% of TCE	1994	71.3 (93)	55.7	60.2	71.6	67.2	33	71.6 (93)	73.1
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	103.4 (93)	87	482.4	196.6	257.3	130.7	1 019.5	6 649.8
Per capita	US$	1994	23 984 (93)	8 792	12 321	22 389	36 790	2 157	17 468	25 512
At current prices using current PPP's[3]	Bill. US$	1994	95.3	122	531.7	153	167.4	319.3	1 030.2	6 649.8
Per capita	US$	1994	21 968	12 335	13 581	17 422	23 942	5 271	17 650	25 512
Average annual volume growth over previous 5 years	%	1994	2.1 (93)	1.4	1.5	–0.3	0.5	3.6	0.8	2.1
Gross fixed capital formation (GFCF)	% of GDP	1994	22 (93)	25.7	19.8	13.7	22.8	24.5	15	17.2
of which: Machinery and equipment	% of GDP	1993	..	..	5.7	5.7	7.5	10.3	7.3	7.7
Residential construction	% of GDP	1993	..	..	4.1	4	14.9[10]	9.1	3.1	4
Average annual volume growth over previous 5 years	%	1994	–3.93	2.7	–1.2	–7.6	–0.4	5.1	–2.1	4.6
Gross saving ratio[4]	% of GDP	1994	21.9 (93)	24.2	18.8	13.7	29.3	22.5	13.5	16.2
General government										
Current expenditure on goods and services	% of GDP	1994	22.1 (93)	17.2	16.9	27.3	14.1	11.7	21.6	16.4
Current disbursements[5]	% of GDP	1993	..	..	43.7	67.3	36.7	..	42.7	35.8
Current disbursements[5]	% of GDP	1993	..	..	40.1	59	36	..	36.8	31.7
Current receipts	% of GDP	1993	1.23	0.36	0.32	1.33	0.49	..	0.34	0.19
Net official development assistance	% of GNP	1993	9 826	7 780	8 412	9 240	13 730	3 617	10 942	16 444
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	378	260	307	418	441	29	361	568
Passenger cars, per 1 000 inhabitants	Number	1990	515	273	340	687	603	143	445	553
Telephones, per 1 000 inhabitants	Number	1991	423	187	400	468	406	175	434	814
Television sets, per 1 000 inhabitants	Number	1991	3.2 (92)	2.9	4.1	3	3	0.9	1.5 (92)	2.3 (92)
Doctors, per 1 000 inhabitants	Number	1993	5	8.7	7.6	4.8	5.6	52.6	6.6	8.5 (92)
Infant mortality per 1 000 live births	Number		4	..	7.2	5.4	..	..	6.7	2.8
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	2.7	9	5.6	5.7	3.9	73	4.6	3.6
Consumer prices	%	1994	34 645	17 072	73 129	61 122	70 467	18 456	205 170	512 627
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	30.9 (93)	19.6	15.2	31.1	27.4	14.1	20.1	7.7
As % of GDP	%	1994	5	6.1	10.5	3.4	6.4	9.5	6.1	7.1
Average annual increase over previous 5 years	%	1994	27 345	25 967	92 182	51 730	68 126	22 976	227 026	663 256
Imports of goods, cif*	Mill. US$	1994	23.3 (93)	29.9	19.1	26.3	26.5	17.6	22.3	10
As % of GDP	%	1994	2.9	6.6	5.2	1	3.2	37.9	2.8	7
Average annual increase over previous 5 years	%	1994	13 033	10 627	28 475	15 929	23 790	4 911	28 094	43 350
Total official reserves[6]	Mill. SDRs	1994	5.7	4.9	3.7	3.7	4.2	2.6	1.5	0.8

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

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